MATHEMATICS II

AMATH2E/S
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Preface

In this module, we study differentiation and integration, and their applications to solving various mathematical problems. We also study differential equations, matrices and the use of matrix methods to solving systems of simultaneous linear equations.

In the first unit we study differentiation continuing from where it was left in Mathematics 1. Specifically, we look at derivatives of inverse trigonometric functions, hyperbolic functions, inverse hyperbolic functions and parametric equations. We also discuss application of differentiation to solving maxima and minima problems. We conclude unit 1 by discussing partial derivatives and applications to rates of change and small increments. In unit 2 we explore various techniques of integration with special emphasis to integration by parts, use of partial fractions, inverse functions, trigonometric substitutions and the t-formula. In unit 3 we consider applications of integration to calculating areas between curves, mean values, root-mean-square values, length of curves and volumes of solids of revolution.

In unit 4 we solve simple first order ordinary differential equations. In particular we solve equations of variable separable type and linear equations. We also briefly discuss solutions of second order homogeneous linear odes with constant coefficients. In Unit 5 we deal with operations with matrices including finding determinants and inverses. In unit 6 we apply the knowledge of determinants and matrix inverses to solving systems of simultaneous linear equations. Elementary row operation methods are also discussed as an extra optional.

In each unit we incorporate several worked examples, class exercises and homework exercises. A thorough study of the learning material presented, as well as the class exercises should enable a student to work through the tutorial and homework questions with little difficulty. The learning material is written in such a way that it is not difficult for students to follow the material even on their own. Students are further encouraged to make use of their lecturer and tutorial assistants should they have any queries arising from their studies.

It is my hope that you will enjoy going through this module.

L. Rundora 2012

Basic Information: Algebra and Geometry

Arithmetic Operations

$$a(b+c) = ab + ac \qquad \frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$$

$$\frac{a+c}{b} = \frac{a}{b} + \frac{c}{b} \qquad \frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

Laws of Exponents

$$x^m x^n = x^{m+n} \qquad \frac{x^m}{x^n} = x^{m-n}$$

$$(x^m)^n = x^{mn} \qquad x^{-n} = \frac{1}{x^n}$$

$$(xy)^n = x^n y^n \qquad \left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}$$

$$x^{\frac{1}{n}} = \sqrt[n]{x} \qquad x^{\frac{m}{n}} = \sqrt[n]{x^m} = (\sqrt[n]{x})^m$$

$$\sqrt[n]{xy} = \sqrt[n]{x}\,\sqrt[n]{y} \qquad \sqrt[n]{\frac{x}{y}} = \frac{\sqrt[n]{x}}{\sqrt[n]{y}}$$

Factoring Special Polynomials

$$x^2 - y^2 = (x+y)(x-y) \qquad x^3 + y^3 = (x+y)(x^2 - xy + y^2) \qquad x^3 - y^3 = (x-y)(x^2 + xy + y^2)$$

Binomial Theorem

$$(x+y)^2 = x^2 + 2xy + y^2 \qquad (x-y)^2 = x^2 - 2xy + y^2$$

$$(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3 \qquad (x-y)^3 = x^3 - 3x^2y + 3xy^2 - y^3$$

$$(x+y)^n = x^n + nx^{n-1}y + \frac{n(n-1)}{2!}x^{n-2}y^2 + \frac{n(n-1)(n-2)}{3!}x^{n-3}y^3 + \cdots + nxy^{n-1} + y^n$$

Quadratic Formula

$$\text{If} \quad ax^2 + bx + c = 0, \quad \text{then} \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Inequalities and absolute value

If $a < b$ and $b < c$, then $a < c$

If $a < b$, then $a + c < b + c$

If $a < b$ and $c > 0$, then $ca < cb$

If $a < b$ and $c < 0$, then $ca > cb$

If $a > 0$, then $|x| = a$ means $x = a$ or $x = -a$, $|x| < a$ means $-a < x < a$ and $|x| > a$ means $x > a$ or $x < -a$.

Geometric Formulae

Area of a Triangle: $A = \frac{1}{2}bh = \frac{1}{2}ab\sin\theta$

Area of a circle: $A = \pi r^2$ **Circumference:** $C = 2\pi r$

Area of sector: $A = \frac{1}{2}r^2\theta$ **arc length:** $s = r\theta$ (θ in radians)

Volume of a sphere: $V = \frac{4}{3}\pi r^3$ **surface area of a sphere:** $A = 4\pi r^2$

Volume of a cylinder: $V = \pi r^2 h$

Volume of a right circular cone: $V = \frac{1}{3}\pi r^2 h$
surface area: $A = \pi r\sqrt{r^2 + h^2}$

Distance and Midpoint Formulae
Distance between $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Midpoint between P_1 and P_2: $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

Equation of a straight line: $y = mx + c$ m is the slope or gradient and c is the y-intercept.

Equation of a circle: $(x - h)^2 + (y - k)^2 = r^2$ (h, k) is the centre and r is the radius of the circle.

Angle measurement

$$\pi \text{ radians} = 180^\circ \qquad 1^\circ = \frac{\pi}{180} \text{ rad} \qquad 1 \text{ rad} = \frac{180^\circ}{\pi}$$

Right Angle Trigonometry

$$\sin\theta = \frac{\text{opp}}{\text{hyp}} \qquad \csc\theta = \frac{\text{hyp}}{\text{opp}}$$

$$\cos\theta = \frac{\text{adj}}{\text{hyp}} \qquad \sec\theta = \frac{\text{hyp}}{\text{adj}}$$

$$\tan\theta = \frac{\text{opp}}{\text{adj}} \qquad \cot\theta = \frac{\text{adj}}{\text{opp}}$$

Trigonometric ratios of special angles

θ	radians	$\sin\theta$	$\cos\theta$	$\tan\theta$
$0°$	0	0	1	0
$30°$	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
$45°$	$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
$60°$	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$90°$	$\frac{\pi}{2}$	1	0	$-$
$180°$	π	0	-1	0
$270°$	$\frac{3\pi}{2}$	-1	0	$-$
$360°$	2π	0	1	0

Sine rule

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Cosine rule

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$b^2 = a^2 + c^2 - 2ac\cos B$$

$$c^2 = a^2 + b^2 - 2ab\cos C$$

Special Functions

Inverse Trigonometric Functions

$$\text{arcsin}x = \sin^{-1} x = y \iff \sin y = x \quad \text{and} \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

$$\text{arccos}x = \cos^{-1} x = y \iff \cos y = x \quad \text{and} \quad 0 \leq y \leq \pi$$

$$\text{arctan}x = \tan^{-1} x = y \iff \tan y = x \quad \text{and} \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

Exponential and Logarithmic Functions

$$\log_a x = y \iff a^y = x$$

$$\ln x = \log_e x, \quad \text{where} \quad \ln e = 1$$

$$\ln x = y \iff e^y = x$$

Cancelation Equations

$$\log_a(a^x) = x \qquad a^{\log_a x} = x \qquad \ln(e^x) = x \qquad e^{\ln x} = x$$

Laws of Logarithms

$$\log_a(xy) = \log_a x + \log_a y$$
$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$
$$\log_a(x^r) = r\log_a x$$

Hyperbolic Functions

$$\sinh x = \frac{e^x - e^{-x}}{2} \qquad \text{cosech } x = \frac{1}{\sinh x}$$
$$\cosh x = \frac{e^x + e^{-x}}{2} \qquad \text{sech } x = \frac{1}{\cosh x}$$
$$\tanh x = \frac{\sinh x}{\cosh x} \qquad \coth x = \frac{\cosh x}{\sinh x}$$

Inverse Hyperbolic Functions

$$y = \sinh^{-1} x \iff \sinh y = x \qquad \sinh^{-1} x = \ln(x + \sqrt{x^2+1})$$
$$y = \cosh^{-1} x \iff \cosh y = x \qquad \cosh^{-1} x = \ln(x + \sqrt{x^2-1})$$
$$y = \tanh^{-1} x \iff \tanh y = x \qquad \tanh^{-1} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

Table of Integrals

1. $\int f'(x)[f(x)]^n dx = \frac{[f(x)]^{n+1}}{n+1} + c; n \neq -1$

2. $\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + c$

3. $\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$

4. $\int f'(x) e^{f(x)} dx = e^{f(x)} + c$

5. $\int f'(x) \cos f(x) dx = \sin f(x) + c$

6. $\int f'(x) \sin f(x) dx = -\cos f(x) + c$

7. $\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$

8. $\int f'(x) \mathrm{cosec}^2 f(x) dx = -\cot f(x) + c$

9. $\int f'(x) \sec f(x) \tan f(x) dx = \sec f(x) + c$

10. $\int f'(x) \mathrm{cosec} f(x) \cot f(x) dx = -\mathrm{cosec} f(x) + c$

11. $\int f'(x) \tan f(x) dx = \ln|\sec f(x)| + c$

12. $\int f'(x) \cot f(x) dx = \ln|\sin f(x)| + c$

13. $\int f'(x) \sec f(x) dx = \ln|\sec f(x) + \tan f(x)| + c$

14. $\int f'(x) \mathrm{cosec} f(x) dx = \ln|\mathrm{cosec} f(x) - \cot f(x)| + c$

15. $\int f'(x) \cosh f(x) dx = \sinh f(x) + c$

16. $\int f'(x) \sinh f(x) dx = \cosh f(x) + c$

17. $\int f'(x)\text{sech}^2 f(x)dx = \tanh f(x) + c$

18. $\int f'(x)\text{cosech}^2 f(x)dx = -\coth f(x) + c$

19. $\int f'(x)\text{sech} f(x)\tanh f(x)dx = -\text{sech} f(x) + c$

20. $\int f'(x)\text{cosech} f(x)\coth f(x)dx = -\text{cosech} f(x) + c$

21. $\int f'(x)\tanh f(x)dx = \ln|\cosh f(x)| + c$

22. $\int f'(x)\coth f(x)dx = \ln|\sinh f(x)| + c$

23. $\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}}dx = \sin^{-1}\frac{f(x)}{a} + c \quad \text{or} \quad -\cos^{-1}\frac{f(x)}{a} + c$

24. $\int \frac{f'(x)}{\sqrt{[f(x)]^2 + a^2}}dx = \sinh^{-1}\frac{f(x)}{a} + c \quad \text{or} \quad \ln\left[\frac{f(x)}{a} + \sqrt{\left(\frac{f(x)}{a}\right)^2 + 1}\right] + c$

25. $\int \frac{f'(x)}{\sqrt{[f(x)]^2 - a^2}}dx = \cosh^{-1}\frac{f(x)}{a} + c \quad \text{or} \quad \ln\left[\frac{f(x)}{a} + \sqrt{\left(\frac{f(x)}{a}\right)^2 - 1}\right] + c$

26. $\int \frac{f'(x)}{[f(x)]^2 + a^2}dx = \frac{1}{a}\tan^{-1}\frac{f(x)}{a} + c \quad \text{or} \quad -\frac{1}{a}\cot^{-1}\frac{f(x)}{a} + c$

27. $\int \frac{f'(x)}{a^2 - [f(x)]^2}dx = \frac{1}{a}\tanh^{-1}\frac{f(x)}{a} + c \quad \text{or} \quad \frac{1}{2a}\ln\left|\frac{a + f(x)}{a - f(x)}\right| + c$

28. $\int \frac{f'(x)}{[f(x)]^2 - a^2}dx = -\frac{1}{a}\coth^{-1}\frac{f(x)}{a} + c \quad \text{or} \quad \frac{1}{2a}\ln\left|\frac{f(x) - a}{f(x) + a}\right| + c$

29. $\int f'(x)\sqrt{a^2 - [f(x)]^2}dx = \frac{1}{2}\left[f(x)\sqrt{a^2 - [f(x)]^2}\right] + a^2\sin^{-1}\frac{f(x)}{a} + c$

30. $\int f'(x)\sqrt{[f(x)]^2 + a^2}dx = \frac{1}{2}\left[f(x)\sqrt{[f(x)]^2 + a^2}\right] + a^2\sinh^{-1}\frac{f(x)}{a} + c$

31. $\int f'(x)\sqrt{[f(x)]^2 - a^2}dx = \frac{1}{2}\left[f(x)\sqrt{[f(x)]^2 - a^2}\right] - a^2\cosh^{-1}\frac{f(x)}{a} + c$

32. $\int udv = uv - \int vdu$

TRIGONOMETRIC IDENTITIES

33. $\sin^2 A + \cos^2 A = 1$

34. $\sec^2 A = 1 + \tan^2 A$

35. $\text{cosec}^2 A = 1 + \cot^2 A$

36. $\sin 2A = 2 \sin A \cos A$

37. $\cos 2A = \cos^2 A - \sin^2 A$

38. $\sin^2 A = \frac{1}{2}(1 - \cos 2A)$

39. $\cos^2 A = \frac{1}{2}(1 + \cos 2A)$

40. $\sin A \cos B = \frac{1}{2}[\sin(A - B) + \sin(A + B)]$

41. $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$

42. $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$

Unit 1
Differentiation and Partial Differentiation

1.1 What this Unit is all About

In this unit we discuss differentiation of inverse trigonometric functions, hyperbolic functions, inverse hyperbolic functions and parametric equations. We also discuss application of differentiation to solving maxima and minima problems. We end the unit by discussing partial derivatives up to second order derivatives and application to rates of change and small changes.

By the end of this unit you should be able to:
1. Differentiate inverse trigonometric functions.
2. Differentiate hyperbolic functions and inverse hyperbolic functions.
3. Differentiate parametric equations.
4. Solve maxima/minima problems.
5. Perform partial differentiation.
6. Solve problems involving rates of change and small changes.

1.2 Differentiation

The concept of differentiation was introduced in AMATH1E/S and there you dealt with problems of finding derivatives of various functions. In this unit we deal with differentiation as a continuation from where you left it in AMATH1E/S. Suffice to say that the techniques and differentiation formulae that you have learnt remain the same, the only new thing is that this time around we will be applying these to different type of functions. In particular we focus on inverse trigonometric functions, hyperbolic functions, inverse hyperbolic functions and parametric equations.

Perhaps it will be worthwhile to start by recalling the techniques/formulae that you have used before.

Differentiation from first principles

$$\frac{dy}{dx} = \lim_{h \to o} \frac{f(x+h) - f(x)}{h}$$

Derivative of a constant
If k is a constant, $\dfrac{d}{dx}(k) = 0$

The power rule for differentiating polynomials

$\dfrac{d}{dx}(ax^n) = anx^{n-1}$, a is a constant.

Product Rule for differentiating products

$$\frac{d}{dx}(f(x) \cdot g(x)) = f(x) \cdot g'(x) + g(x) \cdot f'(x)$$

Quotient Rule for differentiating rational functions

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{(g(x))^2}$$

Chain Rule or function of a function rule

$\dfrac{d}{dx}[f(x)]^n = n[f(x)]^{n-1} \cdot f'(x)$ and if $y = f(u(x))$, then $\dfrac{dy}{dx} = \dfrac{dy}{du} \cdot \dfrac{du}{dx}$

You are also urged to recall implicit differentiation for finding $\dfrac{dy}{dx}$ when y is an implicit function of x. This, as well as logarithmic differentiation, will also be encountered in the foregoing.

1.2.1 Inverse trigonometric functions

Recall that the only functions for which inverse functions are defined are one-to-one functions. Trigonometric functions, however, are not one-to-one and thus they do not have inverse functions. However, if we restrict their domains, they will be one-to-one in the restricted domains and thus we can find inverses of such restricted trigonometric functions. In particular the sine function $y = \sin x, \quad -\infty < x < \infty$ is not one-to-one, whereas $y = \sin x, \quad -\dfrac{\pi}{2} \leq x \leq \dfrac{\pi}{2}$ is one-to-one and so the later has an inverse. See Figures 1.1 and 1.2.

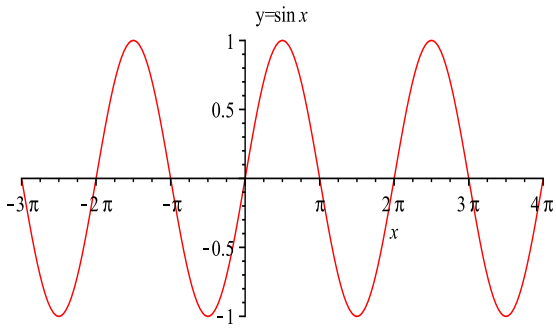

Figure 1.1

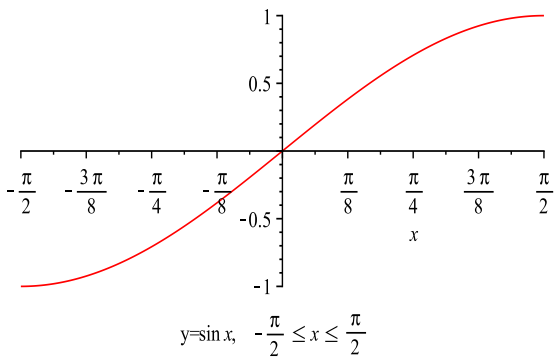

Figure 1.2

This inverse is denoted by $\sin^{-1}$ or arcsin. Now, in general $f^{-1}(x) = y \iff f(y) = x$. In this way

$$\sin^{-1} x = y \iff \sin y = x \quad \text{and} \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}.$$

Please note that

$$\sin^{-1} x \neq \frac{1}{\sin x}$$

and this applies to all other inverse functions. If $-1 \leq x \leq 1$, $\sin^{-1} x$ is the number between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ whose sine is x. Observe that

$$\sin^{-1}(\sin x) = x \quad \text{for} \quad -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

and

$$\sin(\sin^{-1} x) = x \quad \text{for} \quad -1 \leq x \leq 1$$

These cancellation equations correspondingly apply to all other inverse functions as well.

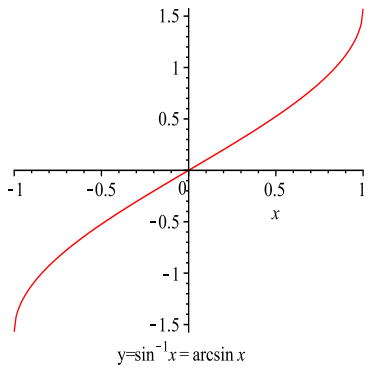

Figure 1.3

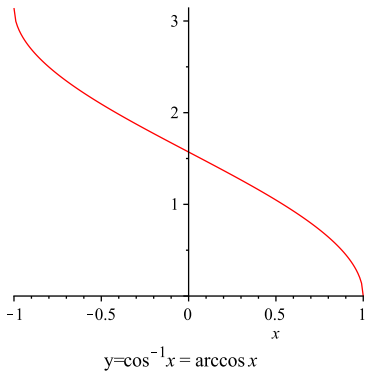

Figure 1.4

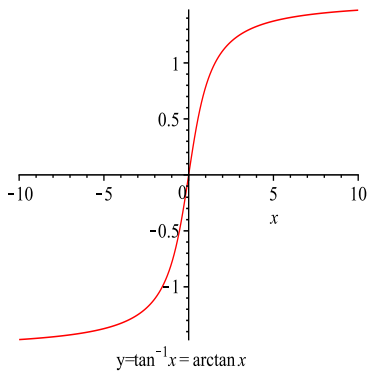

Figure 1.5

Activity 1.1 *Find the exact value of each expression:*

(i) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$ *(ii)* $\cos^{-1}(-1)$ *(iii)* $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(iv) $\sec^{-1} 2$ *(v)* $\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$ *(vi)* $\cos\left(\arcsin\frac{1}{2}\right)$

(vii) $\tan\left(\sin^{-1}\left(\frac{2}{3}\right)\right)$ *(viii)* $\operatorname{cosec}\left(\arccos\frac{3}{5}\right)$ *(ix)* $\sin\left(2\tan^{-1}\sqrt{2}\right)$

Now, since the inverse sine function $\sin^{-1} x$ is continuous in the interval in which it is defined, it is differentiable. We use implicit differentiation to find its derivative.

Let $y = \sin^{-1} x$, then $\sin y = x \implies \cos y \dfrac{dy}{dx} = 1$ and $\dfrac{dy}{dx} = \dfrac{1}{\cos y}$. Now, since $\cos y = \sqrt{1-\sin^2 y} = \sqrt{1-x^2}$ then

$$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-x^2}}$$

and so

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}} \qquad -1 < x < 1$$

In general if $y = \sin^{-1}(f(x))$, then $\sin y = f(x)$ and $\cos y \dfrac{dy}{dx} = f'(x)$, so that

$$\frac{dy}{dx} = \frac{f'(x)}{\cos y} = \frac{f'(x)}{\sqrt{1-[f(x)]^2}}$$

This formula, together with formulae for other inverse functions appear in the table of integrals. Thus, as an alternative to using implicit differentiation, derivatives of inverse trigonometric functions can be found by direct use of formulae.

Example 1.1 *If* $y = \arcsin\left(\frac{3x^2-1}{2}\right)$ *find* $\dfrac{dy}{dx}$ *and simplify the answer.*

Solution
Here $f(x) = \dfrac{3x^2-1}{2} \implies f'(x) = 3x$ and therefore

$$\frac{dx}{dx} = \frac{3x}{\sqrt{1-\left(\frac{3x^2-1}{2}\right)^2}} = \frac{3x}{\sqrt{\frac{4-(3x^2-1)^2}{4}}} = \frac{6x}{\sqrt{3-9x^4+6x^2}}$$

Example 1.2 *Differentiate* $y = \operatorname{arcsec}(2x^2)$

Solution
First note that $\operatorname{arcsec}(2x^2) = \arccos\left(\dfrac{1}{2x^2}\right)$. Thus $f(x) = \dfrac{1}{2x^2} \implies f'(x) = -\dfrac{1}{x^3}$ and

$$\frac{dy}{dx} = \frac{-\left(-\frac{1}{x^3}\right)}{\sqrt{1-\frac{1}{4x^4}}} = \frac{\frac{1}{x^3}}{\frac{\sqrt{4x^4-1}}{2x^2}} = \frac{2}{x\sqrt{4x^4-1}}$$

Example 1.3 *Find* $\dfrac{d}{dx}(\cot^{-1}(\operatorname{cosec} x))$

Solution

$$\frac{d}{dx}(\cot^{-1}(\text{cosec } x)) = \frac{d}{dx}\left(\tan^{-1}\left(\frac{1}{\text{cosec } x}\right)\right) = \frac{d}{dx}(\tan^{-1}(\sin x)) = \frac{\cos x}{1+\sin^2 x}$$

Activity 1.2 *1. Differentiate the following and simplify where possible*

(a) $y = \arctan\left(\frac{x+1}{2}\right)$

(b) $y = (\tan^{-1} x)^2$

(c) $y = \tan^{-1} x^2$

(d) $y = \dfrac{1}{\tan^{-1} x}$

(e) $y = \cos^{-1}(\sin^{-1} x)$

(f) $y = \arctan\sqrt{\frac{1-x}{1+x}}$

(g) $y = \arccos\left(\frac{b+a\cos x}{a+b\cos x}\right)$

(h) $y = \arccos\left(\frac{1+x^2}{1-x^2}\right)$

(i) $y = \arctan\left(\frac{x}{16-x^2}\right)$

(j) $y = \cos^{-1}\left(e^{2x}\right)$

(k) $y = e^{\text{arccot } 2x}$

2. If $y = e^{\arctan x}$*, show that* $(1+x^2)\dfrac{d^2y}{dx^2} + (2x-1)\dfrac{dy}{dx} = 0$

3. If $y = e^{2\arcsin x}$*, prove that* $(1-x^2)\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} - 4y = 0$

1.2.2 Hyperbolic functions

Recall that trigonometric functions are defined in terms of the cosine and the sine functions. For instance $\sec x = \dfrac{1}{\cos x}$, $\cot x = \dfrac{1}{\tan x} = \dfrac{\cos x}{\sin x}$, etc,etc. Correspondingly, hyperbolic functions are all defined from the hyperbolic cosine function (cosh) and the hyperbolic sine function (sinh).

Definition 1.1 *1.* $\cosh x = \dfrac{e^x + e^{-x}}{2}$ 2. $\sinh x = \dfrac{e^x - e^{-x}}{2}$

From these two, we can define $\tanh x = \dfrac{\sinh x}{\cosh x} = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$,
$\coth x = \dfrac{\cosh x}{\sinh x} = \dfrac{e^x + e^{-x}}{e^x - e^{-x}}$, $\text{sech } x = \dfrac{1}{\cosh x} = \dfrac{2}{e^x + e^{-x}}$,
$\text{cosech } x = \dfrac{1}{\sinh x} = \dfrac{2}{e^x - e^{-x}}$, etc,etc.
Hyperbolic functions also play an important role in integral calculus.

To each trigonometric identity there is a corresponding hyperbolic identity. To obtain the hyperbolic identity from the trigonometric identity, simply replace the trigonometric function with the corresponding hyperbolic function and in front of a product or an implied product of sine put a negative sign. For example, from $\sin^2 x + \cos^2 x = 1$ we obtain $\cosh^2 x - \sinh^2 x = 1$ and from $1 + \tan^2 x = \sec^2 x$ we obtain $1 - \tanh^2 x = \text{sech}^2 x$.

Figure 1.6

Figure 1.7

Activity 1.3 *Complete the following table*

Trigonometric Identity	Hyperbolic Identity
$\sin^2 A + \cos^2 A = 1$	$\cosh^2 A - \sinh^2 A = 1$
$1 + \tan^2 A = \sec^2 A$	$1 - \tanh^2 A = \text{sech}^2 A$
$\text{cosec}^2 A = 1 + \cot^2 A$	
$\sin(-A) = -\sin A$	
$\cos(-A) = \cos A$	
$\tan(-A) = -\tan A$	
$\sin(A+B) = \sin A \cos B + \cos A \sin B$	
$\cos(A+B) = \cos A \cos B - \sin A \sin B$	
$\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$	
$\sin 2A = 2 \sin A \cos A$	
$\cos 2A = \cos^2 A - \sin^2 A$	
$\sin^2 A = \dfrac{1}{2}(1 - \cos 2A)$	
$\cos^2 A = \dfrac{1}{2}(1 + \cos 2A)$	
$\sin A \cos B = \dfrac{1}{2}[\sin(A-B) + \sin(A+B)]$	
$\sin A \sin B = \dfrac{1}{2}[\cos(A-B) - \cos(A+B)]$	
$\cos A \cos B = \dfrac{1}{2}[\cos(A-B) + \cos(A+B)]$	

Now, $\dfrac{d}{dx}(\cosh x) = \dfrac{d}{dx}\left(\dfrac{e^x + e^{-x}}{2}\right) = \dfrac{e^x - e^{-x}}{2} = \sinh x$

and $\dfrac{d}{dx}(\sinh x) = \dfrac{d}{dx}\left(\dfrac{e^x - e^{-x}}{2}\right) = \dfrac{e^x + e^{-x}}{2} = \cosh x.$

From these two derivatives, and in conjunction with the laws or techniques of differentiation, we can obtain the derivative of any given hyperbolic function.

Example 1.4 *If* $y = \cosh^3 x$, *find* $\dfrac{dy}{dx}$.

Solution

The chain rule gives $\dfrac{dy}{dx} = 3\cosh^2 x \sinh x$

Example 1.5 *If* $y = \sinh^2 x \cosh x$, *find* $\dfrac{dy}{dx}$.

Solution

The chain rule and product rule give

$\dfrac{dy}{dx} = 2\sinh x \cosh x \cosh x + \sinh^2 x \sinh x = 2\sinh x \cosh^2 x + \sinh^3 x$

Example 1.6 *If* $y = x^3 e^{\tanh x}$, *find* $\dfrac{dy}{dx}$.

Solution
$\dfrac{dy}{dx} = 3x^2 e^{\tanh x} + x^3 \text{sech}^2 x e^{\tanh x}$

Example 1.7 *Differentiate* $y = \sin^{-1}(\cosh x) + \cosh(\sin^{-1} x)$

Solution
$$\frac{dy}{dx} = \frac{\sinh x}{\sqrt{1-\cosh^2 x}} + \sinh(\sin^{-1} x) \cdot \frac{1}{\sqrt{1-x^2}} = \frac{\sinh x}{\sqrt{1-\cosh^2 x}} + \frac{\sinh(\sin^{-1} x)}{\sqrt{1-x^2}}$$

Activity 1.4 *Differentiate the following:*

1. $y = \cosh \dfrac{2}{x}$
2. $y = \sinh^2 \dfrac{1}{x}$
3. $y = \tanh \sqrt[3]{x^2}$
4. $y = \ln\left(\frac{1+\tanh x}{1-\tanh x}\right)$
5. $y = \sinh(\cos^{-1} x) + \sin^{-1}(\text{sech } x)$
6. $y = \sinh(\cos^{-1} x) + \cos^{-1}(\sinh x)$
7. $y = \sqrt{x}\text{cosech}\sqrt{x}$
8. $y = x^{\sinh 3x}$
9. $y = \text{sech}^3 (2^x)$
10. $y = \text{sech}(\sinh 2x - \sin 3x)$

1.2.3 Inverse Hyperbolic Functions

In this section we look at inverse hyperbolic functions. Since the discussion here follows more or less the discussion under inverse trigonometric functions, similar details will be omitted. If
$$y = \cosh^{-1} x$$
then
$$\cosh y = x$$
and similarly
$$y = \text{arcsinh } x \Longrightarrow \quad \sinh y = x.$$

Now, since hyperbolic functions are defined in terms of the exponential function, it is reasonable to expect inverse hyperbolic functions to be defined in terms of the natural logarithmic function.

Let $y = \cosh^{-1} x$. Then $\cosh y = x \Longrightarrow$

$$\begin{aligned}
\frac{e^y + e^{-y}}{2} &= x \\
\Longrightarrow \quad e^y + e^{-y} &= 2x \\
\Longrightarrow \quad e^{2y} - 2xe^y + 1 &= 0 \\
\Longrightarrow \quad z^2 - 2xz + 1 = 0, \quad z &= e^y \\
\Longrightarrow \quad z &= \frac{2x \pm \sqrt{4x^2 - 4}}{2} \\
\Longrightarrow \quad z &= x \pm \sqrt{x^2 - 1} \\
\Longrightarrow \quad e^y &= x \pm \sqrt{x^2 - 1} \\
\Longrightarrow \quad y &= \ln(x \pm \sqrt{x^2 - 1}) \\
\Longrightarrow \quad \cosh^{-1} x &= \ln(x + \sqrt{x^2 - 1}), \quad x \geq 1
\end{aligned}$$

Activity 1.5 *Show that*

$$\text{arcsinh } x = \ln(x + \sqrt{x^2 + 1}) \qquad -\infty < x < \infty$$

and

$$\text{arctanh } x = \frac{1}{2} \ln\left(\frac{1 + x}{1 - x}\right) \qquad -1 < x < 1$$

The process of obtaining derivatives of inverse hyperbolic functions is as explained under inverse trigonometric functions. In general if

$$y = \sinh^{-1} f(x),$$

then

$$\sinh y = f(x).$$

Differentiating implicitly gives

$$\cosh y \frac{dy}{dx} = f'(x) \quad \Longrightarrow \quad \frac{dy}{dx} = \frac{f'(x)}{\cosh y}.$$

From the hyperbolic identity

$$cosh^2 y - \sinh^2 y = 1,$$

we have

$$\cosh y = \sqrt{1 + \sinh^2 y} = \sqrt{1 + [f(x)]^2},$$

so that

$$\frac{d}{dx}(\sinh^{-1} f(x)) = \frac{f'(x)}{\sqrt{1 + [f(x)]^2}}.$$

Similar formulae can be deduced for other inverse hyperbolic functions.

Example 1.8 *If* $y = \text{arctanh } x^2$, *find* $\dfrac{dy}{dx}$

Solution
Here, $f(x) = x^2$ and $f'(x) = 2x$ so that

$$\frac{dy}{dx} = \frac{2x}{1 - x^4}$$

Example 1.9 *If* $\ln y = \cosh^{-1} x$, *find* $\dfrac{dy}{dx}$

Solution
$\ln y = \cosh^{-1} x \quad \Longrightarrow \quad \dfrac{1}{y}\dfrac{dy}{dx} = \dfrac{1}{\sqrt{x^2 - 1}}$ and so

$$\frac{dy}{dx} = \frac{y}{\sqrt{x^2 - 1}} = \frac{e^{\cosh^{-1} x}}{\sqrt{x^2 - 1}}$$

Example 1.10 *Differentiate* $y = \text{cosech}^{-1}(\text{sech } x^3)$

Solution
$y = \text{cosech}^{-1}(\text{sech } x^3) = \sinh^{-1}\left(\frac{1}{\text{sech } x^3}\right) = \sinh^{-1}(\cosh x^3)$. Thus,

$$\frac{dy}{dx} = \frac{3x^2 \sinh x^3}{\sqrt{1 + \cosh^2 x^3}}$$

Activity 1.6 *1. Differentiate the following:*

(a) $y = \dfrac{\cot(\tanh x)}{\sinh(\cosh^{-1} x)}$

(b) $y = \tanh^{-1}(\cosh x) \cdot \text{sech}(\sinh x)$

(c) $y = \coth^{-1}(\tan^{-1} 2x)$

(d) $y = \text{sech}^{-1}\left(\frac{3}{x^2}\right)$

(e) $y = \ln(\text{arccosh } x)^{\text{arcsinh } x}$

(f) $y = \sinh^{-1}(\sin x) + \sin(\text{arcsinh } x)$

2. Determine $\dfrac{dy}{dx}$ *if:*

(a) $(\cosh x)^{\sinh^{-1} y} = (\tanh^{-1} y)^{\coth x}$

(b) $(\cot^{-1} y)^{\ln y} = x^{\cosh^{-1} x} e^{\frac{x}{2}}$

(c) $\text{arccoth}\left(\frac{x}{y}\right) = \ln x + \ln y$

3. If $\ln y = \cosh^{-1} x$, *find the value of* $(1 + x^2)\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} - y$

1.2.4 Parametric Equations

Sometimes it is not possible to express y explicitly in terms of x, i.e. it may not be possible to have $y = f(x)$. An alternative to this representation is to relate the variables x and y through another independent variable, say θ, i.e. $x = x(\theta), \quad y = y(\theta)$. This third variable θ is called a parameter and

$$x = x(\theta), \quad y = y(\theta)$$

are called parametric equations. Now, given such parametric equations, our objective here is to find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$.

Let $x = x(\theta), \quad y = y(\theta)$. Chain rule gives $\dfrac{dy}{dx} = \dfrac{dy}{d\theta} \cdot \dfrac{d\theta}{dx}$. Since $\dfrac{d\theta}{dx} = \dfrac{1}{\frac{dx}{d\theta}}$, we have that

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}.$$

Further $\dfrac{d^2y}{dx^2} = \dfrac{d}{dx}\left(\dfrac{dy}{dx}\right) = \dfrac{d}{d\theta}\left(\dfrac{dy}{dx}\right) \cdot \dfrac{d\theta}{dx}$, which gives

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{d\theta}\left(\frac{dy}{dx}\right)}{\frac{dx}{d\theta}}.$$

Example 1.11 *Given* $x = 2\cos 2t, \quad y = 2\sin t$, *determine* $\dfrac{dy}{dx}$ *and* $\dfrac{d^2y}{dx^2}$.

solution
$\dfrac{dx}{dt} = -4\sin 2t, \quad \dfrac{dy}{dt} = 2\cos t$ and

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2\cos t}{-4\sin 2t} = -\frac{\cos t}{2\sin 2t} = -\frac{\operatorname{cosec} t}{4}$$

Now, $\dfrac{d}{dt}\left(\dfrac{dy}{dx}\right) = \dfrac{\operatorname{cosec} t \cot t}{4}$ and

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}} = -\frac{\operatorname{cosec} t \cot t}{16\sin 2t} = -\frac{\operatorname{cosec}^3 t}{32}$$

Example 1.12 *Find an equation of the tangent line to the parametric curve*

$$x = 2\sin 2t, \quad y = 2\sin t$$

at the point $(\sqrt{3}, 1)$.

solution
$\dfrac{dx}{dt} = 4\cos 2t, \quad \dfrac{dy}{dt} = 2\cos t$. The slope is

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2\cos t}{4\cos 2t} = \frac{\cos t}{2\cos 2t}$$

The point $(\sqrt{3}, 1)$ corresponds to the parameter value $t = \dfrac{\pi}{6}$, so that the slope of the tangent at this point is

$$\frac{dy}{dx}|_{t=\frac{\pi}{6}} = \frac{\cos\left(\frac{\pi}{6}\right)}{2\cos\left(\frac{\pi}{3}\right)} = \frac{\sqrt{3}}{2}$$

Thus, the equation of the tangent line is $y - 1 = \dfrac{\sqrt{3}}{2}(x - \sqrt{3}) \implies y = \dfrac{\sqrt{3}}{2}x - \dfrac{1}{2}$.

Activity 1.7 *1. Find an equation of the tangent line to the curve $x = t^4 + 1, \quad y = t^3 + t$ at the point $t = -1$.*

2. A curve C is defined by the parametric equations $x = t^2, \quad y = t^3 - 3t$. Show that C has two tangents at the point $(3, 0)$ and find their equations.

3. Find an equation of the tangent to the cycloid $x = r(\theta - \sin\theta), \quad y = r(1 - \cos\theta)$ at the point where $\theta = \dfrac{\pi}{3}$.

4. Find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$ for each of the following:

(a) $y = \sinh t, \quad x = t + \cosh t$

(b) $x = \cos^{-1}(\ln t), \quad y = -2\sin^{-1}(3^t)$

(c) $y = \sin\theta + \theta\cos\theta, \quad x = \cos\theta - \theta\sin\theta$

(d) $y = 3\sinh\theta, \quad x = \cosh^3\theta, \quad$ *at* $\theta = \dfrac{\pi}{6}$

(e) $x = e^t\cos t, \quad y = e^t\sin t$

(f) $x = \cos^3 2\theta, \quad y = \sin^3 2\theta$

(g) $x = a\cos 2\theta, \quad y = a\sin\theta$, *a-constant.*

1.2.5 Maxima/Minima problems

One important application of differential calculus is in the area of optimization where real life problems are reduced into problems of finding maximum and minimum values of functions. The techniques and/or theory that we are going to use here have already been introduced earlier in AMATH1E/S when problems of curve sketching where discussed. An example of a real life optimization problem

is a situation where a businessman is faced with the problem of finding a strategy to minimise costs and maximise profits. Interesting problems in engineering and technology include maximising or minimising areas, volumes, distances, forces, angles, times, and many other quantities.

The challenge in solving such practical problems often lies in converting the word problem into a mathematical optimization problem by setting up the function that is to be minimised or maximised. It is important that the problem is read carefully and understood in terms of identifying the unknown variables, the given quantities as well as the given conditions. In most problems it is useful to sketch a diagram and identify the given and required quantities on the diagram.

Example 1.13 *A farmer has* $2400m$ *of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?*

solution
Here we wish to maximise the area A of a rectangle. It is important to note that there is no fence on the side where there is a river. Let x and y be the length and breadth of the rectangle in metres. Then

$$A = xy$$

We need to eliminate one of the variables. It is given that the total length of the fence is $2400m$, i.e.

$$2x + y = 2400, \Longrightarrow \quad y = 2400 - 2x$$

and thus

$$A = x(2400 - 2x) = 2400x - 2x^2$$

We maximise this function.

$$\frac{dA}{dx} = 2400 - 4x$$

The critical value(s) of x are found when $\dfrac{dA}{dx} = 0$, that is

$$2400 - 4x = 0 \Longrightarrow \; x = 600$$

Now, we use the second derivative test to check whether this value of x gives the maximum area:

$$\frac{d^2A}{dx^2} = -4 < 0$$

for all x, and this proves that A is always concave down and so the local maximum at $x = 600$ must be the global maximum. The rectangular field must be $600m$ wide and $2400 - 2 \times 600 = 1200m$ long. $A_{\max} = A(600) = 720000m^2$.

In general, to solve optimization problems, formulate the function $f(x)$ and solve $f'(x) = 0$ to get critical values and use $f''(x)$ to determine the nature of the critical values.

Activity 1.8 *1. Find two numbers whose sum is 23 and whose product is a maximum.*

2. *Find two numbers whose difference is 100 and whose product is a minimum.*

3. *The sum of two positive numbers is 16. What is the smallest possible value of the sum of their squares?*

4. *Find the dimensions of a rectangle with perimeter* $100m$ *whose area is as large as possible.*

5. *A model used for the yield* Y *of an agricultural crop as a function of the nitrogen level* N *in the soil (measured in appropriate units) is*

$$Y = \frac{KN}{1 + N^2}$$

where K *is a positive constant. What nitrogen level gives the best yield?*

6. *The rate (in mg carbon/*m^3/h*) at which photosynthesis takes place for a species of phytoplankton is modeled by the function*

$$P = \frac{100I}{I^2 + I + 4}$$

where I *is the light intensity (measured in appropriate units). For what light intensity is* P *a maximum?*

7. *A box with an open top is to be constructed from a square piece of cardboard,* $3m$ *wide, by cutting out a square from each of the four corners and bending up the sides. Find the largest volume that such a box can have.*

8. *If* $1200cm^2$ *of material is available to make a box with a square base and an open top, find the largest possible volume of the box.*

9. *A box with a square base and open top must have a volume of* $32000cm^3$*. Find the dimensions of the box that minimise the amount of material used.*

10. *A piece of wire* $10m$ *long is cut into two pieces. One piece is bent into a square and the other is bent into an equilateral triangle. How should the wire be cut so that the total area enclosed is (a) a maximum? (b) a minimum?*

11. *A cylindrical can without a top is made to contain* Vcm^3 *of a liquid. Find the dimensions that will minimise the cost of the material used to manufacture the can.*

1.3 Partial Derivatives

1.3.1 First partial derivatives

Recall that if $y = f(x)$, then $\dfrac{dy}{dx} = f'(x)$ can be obtained from first principles using the definition

$$\frac{dy}{dx} = \lim_{h \to o} \frac{f(x+h) - f(x)}{h}.$$

In this section we begin by adapting this definition to the case of a function of two variables $f(x, y)$.

Definition 1.2 *If f is a function of two variables, its* **partial derivatives** *are the functions f_x and f_y defined by*

$$f_x(x, y) = \lim_{h \to o} \frac{f(x+h, y) - f(x, y)}{h}$$

and

$$f_y(x, y) = \lim_{h \to o} \frac{f(x, y+h) - f(x, y)}{h}$$

Notations for partial derivatives

If $z = f(x, y)$, we write

$$f_x(x, y) = f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x, y) = \frac{\partial z}{\partial x}$$

and

$$f_y(x, y) = f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x, y) = \frac{\partial z}{\partial y}$$

Rule for finding partial derivatives of $z = f(x, y)$

To find $\dfrac{\partial z}{\partial x}$, regard y as a constant and differentiate $f(x, y)$ with respect to x. To find $\dfrac{\partial z}{\partial y}$, regard x as a constant and differentiate $f(x, y)$ with respect to y. All the usual rules/laws and properties of differentiation apply.

Example 1.14 *If $f(x, y) = x^3 - x^2y^3 + 2y^2$, find $f_x(2, -1)$ and $f_y(2, -1)$*

Solution
$f_x(x, y) = 3x^2 - 2xy^3$ and therefore $f_x(2, -1) = 3 \cdot 2^2 - 2 \cdot 2 \cdot (-1)^3 = 16$
$f_y(x, y) = -3x^2y^2 + 4y$ and so $f - y(2, -1) = -3 \cdot 2^2 \cdot (-1)^2 + 4 \cdot (-1) = -16$

Example 1.15 *If $f(x, y) = \cos\left(\frac{x}{1+y}\right)$, calculate $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$.*

Solution

$$\frac{\partial f}{\partial x} = -\sin\left(\frac{x}{1+y}\right)\frac{\partial}{\partial x}\left(\frac{x}{1+y}\right) = -\sin\left(\frac{x}{1+y}\right)\cdot\frac{1}{1+y} = -\frac{1}{1+y}\sin\left(\frac{x}{1+y}\right)$$

and

$$\frac{\partial f}{\partial y} = -\sin\left(\frac{x}{1+y}\right)\frac{\partial}{\partial y}\left(\frac{x}{1+y}\right) = -\sin\left(\frac{x}{1+y}\right)\cdot\frac{-x}{(1+y)^2} = \frac{x}{(1+y)^2}\sin\left(\frac{x}{1+y}\right)$$

Example 1.16 *Find* $\frac{\partial z}{\partial x}$ *and* $\frac{\partial z}{\partial y}$ *if* z *is defined implicitly as a function of* x *and* y *by the equation* $x^2 + y^2 + z^2 + 3xyz = 1$

Solution

$$\begin{aligned} 2x + 2z\frac{\partial z}{\partial x} + 3yz + 3xy\frac{\partial z}{\partial x} &= 0 \\ \implies \quad (2z + 3xy)\frac{\partial z}{\partial x} &= -2x - 3yz \\ \implies \quad \frac{\partial z}{\partial x} &= -\frac{2x + 3yz}{2z + 3xy} \end{aligned}$$

Similarly,

$$\begin{aligned} 2y + 2z\frac{\partial z}{\partial y} + 3xz + 3xy\frac{\partial z}{\partial y} &= 0 \\ \implies \quad \frac{\partial z}{\partial y} &= -\frac{2y + 3xz}{2z + 3xy} \end{aligned}$$

Example 1.17 *If* $f(x_1, x_2, x_3) = x_1 \sin(2x_2 - x_3)$, *find the first partial derivatives* $\frac{\partial f}{\partial x_1}$, $\frac{\partial f}{\partial x_2}$ *and* $\frac{\partial f}{\partial x_3}$.

Solution

$$\frac{\partial f}{\partial x_1} = \sin(2x_2 - x_3)$$

$$\frac{\partial f}{\partial x_2} = 2x_1 \cos(2x_2 - x_3)$$

and

$$\frac{\partial f}{\partial x_3} = -x_1 \cos(2x_2 - x_3)$$

1.3.2 Second partial derivatives

If f is a function of two variables, then we have seen that its partial derivatives

$$f_x = \frac{\partial f}{\partial x}$$

and

$$f_y = \frac{\partial f}{\partial y}$$

are also functions of two variables. We can thus consider their partial derivatives

$$(f_x)_x = \frac{\partial^2 f}{\partial x^2},$$

$$(f_x)_y = \frac{\partial^2 f}{\partial y \partial x},$$

$$(f_y)_x = \frac{\partial^2 f}{\partial x \partial y}$$

and

$$(f_y)_y = \frac{\partial^2 f}{\partial y^2}$$

which are called **second partial derivatives** of f. The notation used, and indeed which also provides a way of calculating second partial derivatives, is the following:

$$(f_x)_x = f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)$$

$$(f_x)_y = f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$$

$$(f_y)_x = f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$$

$$(f_y)_y = f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)$$

Example 1.18 *Given* $z = \dfrac{xy}{x-y}$, *find all the second partial derivatives.*

Solution

$$\frac{\partial z}{\partial x} = \frac{y(x-y) - xy}{(x-y)^2} = -\frac{y^2}{(x-y)^2}$$

$$\frac{\partial z}{\partial y} = \frac{x(x-y) + xy}{(x-y)^2} = \frac{x^2}{(x-y)^2}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial x}\left(-\frac{y^2}{(x-y)^2}\right) = -\frac{2y^2}{(x-y)^3}$$

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial y}\left(-\frac{y^2}{(x-y)^2}\right) = \frac{-2y(x-y)^2 - 2y^2(x-y)}{(x-y)^4} = \frac{2xy^2 - 2x^2y}{(x-y)^4}$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial x}\left(\frac{x^2}{(x-y)^2}\right) = \frac{2x(x-y)^2 - 2x^2(x-y)}{(x-y)^4} = \frac{2xy^2 - 2x^2y}{(x-y)^4}$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{x^2}{(x-y)^2}\right) = -\frac{2x^2}{(x-y)^3}$$

Activity 1.9 *1. Find the first partial derivatives of the following functions:*

(a) $f(x,y) = y^5 - 3xy$

(b) $f(x,y) = \dfrac{x-y}{x+y}$

(c) $z = \tan(xy)$

(d) $w = \ln(x + 2y + 3z)$

(e) $u = xy\sin^{-1}(yz)$

(f) $w = ze^{xyz}$

2. Find $\dfrac{\partial z}{\partial x}$ *and* $\dfrac{\partial z}{\partial y}$

(a) $z = \tanh\sqrt{xy} - \ln\sqrt{\frac{x^3}{y}}$

(b) $x^2 + y^2 + z^2 = 3xyz$

(c) $x - z = \arctan(yz)$

(d) $\sin(xyz) = x + 2y + 3z$

(e) $yz = \ln(x + z)$

(f) $z = f(x + y)$

(g) $z = f(x)g(y)$

(h) $z = f(xy)$

(i) $z = f(\frac{x}{y})$

3. Find all the second partial derivatives

(a) $z = \tan^{-1}\left(\dfrac{x+y}{1-xy}\right)$

(b) $z = \dfrac{1}{\sqrt{x^2+y^2}}$

(c) $z = \cos xy$

(d) $z = \ln\sqrt{x^2+y^2}$

4. *If* $u = \dfrac{x+y+z}{\sqrt{x^2+y^2+z^2}}$, *show that* $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z} = 0$

5. *If* $z = \ln(e^x + e^y)$ *show that* $\dfrac{\partial z}{\partial x} + \dfrac{\partial z}{\partial y} = 1$

6. *Show that the equation* $\dfrac{\partial^2 z}{\partial x^2} + \dfrac{\partial^2 z}{\partial y^2} = 0$ *is satisfied by* $z = \ln\sqrt{x^2+y^2} + \dfrac{1}{2}\tan^{-1}\left(\dfrac{y}{x}\right)$

1.3.3 Applications: Differentials

For a function of one variable $y = f(x)$, the differential of y is defined as $dy = f'(x)dx$. For a differentiable function of two variables, $z = f(x, y)$, the differential dz, also called the total differential, is defined by

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$$

This expression is the basis for solving problems involving small increments and rates of change. From

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$$

we get

$$\Delta z \approx \frac{\partial z}{\partial x}\Delta x + \frac{\partial z}{\partial y}\Delta y$$

which gives an approximation of the small change in z, Δz as a result of small changes in x and y, Δx and Δy respectively. We also have

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt},$$

where t is time. This expression gives a calculation of the rate of change of z, $\dfrac{dz}{dt}$ given the rates of changes $\dfrac{dx}{dt}$ and $\dfrac{dy}{dt}$ of x and y respectively. If z is a function of only one variable or a function of more than two variables, the formulae are adapted accordingly.

Small increments

Example 1.19 *The dimensions of a rectangular box are measured to be* $70cm$, $60cm$ *and* $50cm$, *and each measurement is correct to within* $0.2cm$. *Estimate the largest possible error when the volume of the box is calculated from these measurements.*

Solution
If the dimensions of the box are x, y and h, then its volume is $V = xyh$

$$\Delta V \approx \frac{\partial V}{\partial x}\Delta x + \frac{\partial V}{\partial y}\Delta y + \frac{\partial V}{\partial h}\Delta h = yh\Delta x + xh\Delta y + xy\Delta h$$

Here $|\Delta x| \leq 0.2$, $|\Delta y| \leq 0.2$ and $|\Delta h| \leq 0.2$. Thus

$$\Delta V \approx (60)(50)(0.2) + (70)(50)(0.2) + (70)(60)(0.2) = 2140$$

Thus, an error of only $0.2cm$ in measuring the dimensions leads to an error of $2140cm^2$, i.e. about 1% of the volume.

Example 1.20 *The lift, L, of a body in a fluid of density ρ is given by*

$$L = \frac{1}{2}C\rho V^2 A$$

where C is the lift coefficient, V is the free stream velocity and A is the area. If C can be measured to within 1%, ρ to within 0.5%, V to within 0.6% and A to within 0.1%, find the largest percentage error in the value of L.

Solution

$$\begin{aligned}
\Delta L &\approx \frac{\partial L}{\partial C}\Delta C + \frac{\partial L}{\partial \rho}\Delta \rho + \frac{\partial L}{\partial V}\Delta V + \frac{\partial L}{\partial A}\Delta A \\
&= \frac{1}{2}\rho V^2 A(\pm 0.01C) + \frac{1}{2}CV^2 A(\pm 0.005\rho) + C\rho V A(\pm 0.006V) + \frac{1}{2}C\rho V^2(\pm 0.001A) \\
&= \frac{1}{2}C\rho V^2 A(\pm 0.01) + \frac{1}{2}C\rho V^2 A(\pm 0.005) + \frac{1}{2}C\rho V^2 A(\pm 0.012) + \frac{1}{2}C\rho V^2 A(\pm 0.001) \\
&= \frac{1}{2}C\rho V^2 A(\pm 0.01 \pm 0.005 \pm 0.012 \pm 0.001) \\
&= L(\pm 0.028)
\end{aligned}$$

and the largest percentage error in the value of L is 2.8%

Rates of change

Example 1.21 *The altitude of a right circular cone is $20cm$ and is increasing at $0.1cms^{-1}$. The radius of the base is $10cm$ and is decreasing at $0.2cms^{-1}$. How fast is the volume of the cone changing?*

Solution
The volume of the right circular cone is given by $V = \frac{1}{3}\pi r^2 h$ where r is the radius

and h is the altitude. $\dfrac{dh}{dt} = 0.1$ and $\dfrac{dr}{dt} = -0.2$

$$\begin{aligned}\frac{dV}{dt} &= \frac{\partial V}{\partial r}\frac{dr}{dt} + \frac{\partial V}{\partial h}\frac{dh}{dt}\\ &= \frac{2}{3}\pi r h\frac{dr}{dt} + \frac{1}{2}\pi r^2\frac{dh}{dt}\\ &= \frac{2}{3}\pi(10)(20)(-0.2) + \frac{1}{3}\pi(100)(0.1)\\ &= -73.3\end{aligned}$$

and the volume is decreasing at a rate of $73.3cm^3s^{-1}$.

Activity 1.10 *1. The radius of a cylinder increases at the rate of $0.3cm/s$ while the height decreases at the rate of $0.4cm/s$. Find the rate at which the volume is changing at the instant when $r = 0.06m$ and $h = 14cm$.*

2. The frequency, f, of an LC circuit is given by

$$f = \frac{1}{2\pi\sqrt{LC}}$$

where L represents an inductance and C a capacitance. If L is decreased by 1.5% and C is decreased by 0.5%, find the approximate percentage change in f.

3. A square pyramid reservoir has a base of side $12m$ and height $18m$. It contains oil that is flowing out at a rate of $2m^3s^{-1}$. At the moment when the depth of oil is $8m$, find the rate at which the depth is decreasing.
Hint: *The volume of a square pyramid with a base of side a and height h is given by $V = \frac{1}{3}a^2h$.*

4. The perimeter of a rectangle is $25m$ and one side decreases at a rate of $0.02m/s$. At the moment when the side is $25cm$, at what rate does the area change if the perimeter remains constant?

5. The height of a cone of volume $20cm^3$ decreases at a rate of $0.15ms^{-1}$ while the volume remains constant. At what rate does the radius increase when the altitude is $2.4cm$?

6. The moment of inertia, I, of a body is given by

$$I = \frac{1}{12}bd^3$$

where b and d are dimensions of the body. If $b = 5m$, $d = 0.8m$ and the change in b is $5cm$ and the change in d is $0.8cm$, find the approximate change in I.

Unit 2
Integration

2.1 What this Unit is all About

In this unit we define the definite integral, the indefinite integral and state the fundamental theorem of calculus. We study the various techniques of integration, among which are the method of substitution, use of partial fractions, integration by parts, integration of trigonometrical functions, integration of inverse trigonometrical functions, integration of hyperbolic functions, integration of inverse hyperbolic functions, and use of the technique of completing the square.

By the end of this unit you should be able to:
1. Understand integration as the inverse process to differentiation.
2. State the fundamental theorem of calculus.
3. Distinguish between a definite and an indefinite integral.
4. Carry out integration by the method of substitution.
5. Integrate exponential and logarithmic forms.
6. Integrate by parts, by use of partial fractions and by use of completing the square.
7. Integrate any given trigonometric and/or inverse trigonometric function.
8. Understand and use the V-formula.

2.2 Integration as a reverse process of differentiation

2.2.1 The Indefinite integral

In AMATH1E/S and in the previous unit you dealt with problems of finding derivatives of given functions. Integration involves the inverse problem of finding the function whose derivative is given. Specifically, if $y = f(x)$, differentiating this results in $y' = f'(x)$.
Now, given the latter, integrating gives the former.

Example 2.1 *Given $f'(x) = 3x^2$, find $f(x)$.*

We immediately see that $f(x) = x^3$, since $\dfrac{d}{dx}(x^3) = 3x^2$.

This operation of determining the original function from its derivative is the inverse operation of differentiation. It is called **antidifferentiation**.

Definition 2.1 *A function F is an* **antiderivative** *of a function f if for every x in the domain of f, $F'(x) = f(x)$.*

Remark 2.1 *If $F(x)$ is an antiderivative of $f(x)$, then $F(x) + C$, where C is a constant, is also an antiderivative of $f(x)$.*

Example 2.2 *$F(x) = x^3$, $G(x) = x^3 - 5$, $H(x) = x^3 + 7$, etc are all antiderivatives of $3x^2$ because $\frac{d}{dx}(F(x)) = 3x^2$, $\frac{d}{dx}(G(x)) = 3x^2$, $\frac{d}{dx}(H(x)) = 3x^2$.*

Thus the process of anti-differentiation does not determine a single function, but rather a family of functions, each differing from the others by a constant.
Notation
The process of finding antiderivatives (The anti-differentiation process) is called **Integration**.

- $\int$- is the integral sign.
- $\int f(x)dx$- is the **indefinite integral** of $f(x)$. It denotes the family of antiderivatives of $f(x)$.
- If $F'(x) = f(x)\ \ \forall\ x$, we write $\int f(x)dx = F(x) + C$.
- $dx \Longrightarrow$ with respect to x.
- C is a constant of integration.

Example 2.3 • $\int 2dx = 2x + C$

- $\int x^2 dx = \frac{x^3}{3} + C$
- $\int 3x^2 dx = x^3 + C$
- $\int 4tdx = 2t^2 + C$
- $\int f'(x)dx = f(x) + C$

2.2.2 Basic rules of integration

1. $\int kdx = kx + C$, $k,\ C$ are constants.
2. $\int kf(x)dx = k\int f(x)dx$.

3. $\int [f(x) + g(x)]dx = \int f(x)dx + \int g(x)dx.$

4. $\int [f(x) - g(x)]dx = \int f(x)dx - \int g(x)dx.$

5. $\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1.$

Example 2.4 *1.* $\int 6xdx = 6\int xdx = 6\int x^1 dx = 6\left(\frac{x^2}{2}\right) + C = 3x^2 + C.$

2. $\int \frac{1}{x^4}dx = \int x^{-4}dx = \frac{x^{-3}}{-3} + C = -\frac{1}{3x^3} + C.$

3. $\int \sqrt{x}dx = \int x^{\frac{1}{2}}dx = \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{2}{3}x^{\frac{3}{2}} + C.$

4.

$$\begin{aligned}\int (2x^{\frac{4}{3}} + 3x - 1)dx &= 2\int x^{\frac{4}{3}}dx + 3\int xdx - \int dx \\ &= \frac{6}{7}x^{\frac{7}{3}} + \frac{3}{2}x^2 - x + C\end{aligned}$$

5.

$$\begin{aligned}\int (x-1)(6x-5)dx &= \int (6x^2 - 11x + 5)dx \\ &= 2x^3 - \frac{11}{2}x^2 + 5x + C\end{aligned}$$

6.

$$\begin{aligned}\int \frac{2x-1}{x^{\frac{4}{3}}}dx &= \int \left(\frac{2x}{x^{\frac{4}{3}}} - \frac{1}{x^{\frac{4}{3}}}\right)dx \\ &= \int (2x^{\frac{-1}{3}} - x^{\frac{-4}{3}})dx \\ &= \frac{2}{\frac{2}{3}}x^{\frac{2}{3}} - \frac{1}{\frac{-1}{3}}x^{\frac{-1}{3}} + C \\ &= 3x^{\frac{2}{3}} + 3x^{\frac{-1}{3}} + C\end{aligned}$$

NB: If in doubt, **check** your answer by differentiation. For instance, in (3) in the above example:

$$\frac{d}{dx}\left[\frac{2}{3}x^{\frac{3}{2}} + C\right] = \frac{d}{dx}\left(\frac{2}{3}x^{\frac{3}{2}}\right) + \frac{d}{dx}(C) = \frac{2}{3}\cdot\frac{3}{2}x^{\frac{1}{2}} + 0 = x^{\frac{1}{2}} = \sqrt{x}.$$

Example 2.5 *Find the particular solution $y = f(x)$ that satisfies the differential equation $f'(x) = 3\sqrt{x} + 3$ and the initial condition $f(1) = 0$.*

Solution

$$\begin{aligned} f(x) &= 3 \cdot \frac{2}{3} x^{\frac{3}{2}} + 3x + C \\ &= 2x^{\frac{3}{2}} + 3x + C \\ f(1) &= 2 + 3 + C \\ 4 &= 5 + C \\ C &= -1 \end{aligned}$$

Hence

$$y = 2x^{\frac{3}{2}} + 3x - 1.$$

Exercises

1. Find the indefinite integral and check your result by differentiation.
   (a) $\int (x^3 + 2)dx$ (b) $\int \left(\sqrt{x} + \frac{1}{2\sqrt{x}}\right) dx$ (c) $\int \sqrt[3]{x^2} dx$
   (d) $\int (\sqrt[4]{x^3} + 1)dx$ (e) $\int \frac{t^2 + t}{t^2} dt$ (f) $\int (2x + x^{-\frac{1}{2}})dx$
   (g) $\int u(3u^2 + 1)du$ (h) $\int (2t^2 - 1)^2 dt$ (i) $\int y^2 \sqrt{y} dy$
   (j) $\int (1 + 3t)t^2 dt.$

2. Find the particular solution $y = f(x)$ that satisfies the differential equation and the given initial condition.

   (a) $f'(x) = 3\sqrt{x} + 3, \quad f(1) = 4$

   (b) $f'(x) = \frac{1}{5}x - 2, \quad f(10) = -10$

   (c) $f'(x) = 6x(x - 1), \quad f(1) = -1$

   (d) $f'(x) = (2x - 3)(2x + 3), \quad f(3) = 0$

   (e) $f'(x) = \frac{2 - x}{x^3}, x > 0, \quad f(2) = \frac{3}{4}$

   (f) $f'(x) = \frac{-5}{(x - 2)^2}, \quad x \neq 2, \quad f(7) = 1.$

2.3 The definite integral

Figure 2.1

Consider the area bounded by the curve $y = f(x)$, the x-axis and the ordinates $x = a, \quad x = b$ (see figure). Subdivide the interval $a \leq x \leq b$ into n subintervals by means of the points $x_1, x_2, \cdots, x_{n-1}$ chosen arbitrarily. Form the sum

$$f(x_1)(x_1 - x_0) + f(x_2)(x_2 - x_1) + \cdots + f(x_n)(x_n - x_{n-1}).$$

Let $\Delta x_k = (x_k - x_{k-1})$, then the sum above can be written as

$$\sum_{k=1}^{n} f(x_k)(x_k - x_{k-1}) = \sum_{k=1}^{n} f(x_k)\Delta x_k.$$

This sum represents the total area of all rectangles in the figure. Now, let the number of subdivisions n increase in such a way that $\Delta x_k \longrightarrow 0$. In this way we define **the definite integral of** $f(x)$ **between** $x = a$ **and** $x = b$ as

$$\int_a^b f(x)dx = \lim_{\Delta x_k \longrightarrow 0} \sum_{k=1}^{n} f(x_k)\Delta x_k.$$

This definite integral $\displaystyle\int_a^b f(x)dx$ is equal to the area bounded by the curve between $x = a$ and $x = b$. a is called **the lower limit of integration** and b is **the upper limit of integration**.

Example 2.6 *Evaluate the definite integral* $\displaystyle\int_0^2 2xdx$.

Solution

$\displaystyle\int_0^2 2xdx = [x^2]_0^2 = 2^2 - 0^2 = 4 - 0 = 4.$

This value represents the area of the region bounded by the graph of $f(x) = 2x$, the x-axis, and the line $x = 2$.

Theorem 2.1 *The Fundamental Theorem of Calculus*
If f is nonnegative and continuous on the closed interval $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

where F is any function such that $F'(x) = f(x)\ \forall\ x \in [a, b]$.

Remark 2.2

1. *The Fundamental Theorem of Calculus describes a way of evaluating a definite integral, not a procedure for finding antiderivatives.*
2. *The constant of integration C drops because*

$$\begin{aligned}\int_a^b f(x)dx &= [F(x) + C]_a^b \\ &= [F(b) + C] - [F(a) + C] \\ &= F(b) - F(a) + C - C \\ &= F(b) - F(a).\end{aligned}$$

3. *Definite integrals do not necessarily represent areas and can be negative, zero or positive.*

2.3.1 Properties of the definite integral

If $f(x)$ and $g(x)$ are integrable in $[a, b]$ then

1. $\displaystyle\int_a^b kf(x)dx = k\int_a^b f(x)dx$, k is a constant.
2. $\displaystyle\int_a^b (f(x) \pm g(x))dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$
3. $\displaystyle\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$, provided $f(x)$ is integrable in $[a, c]$ and $[c, b]$ and $a \leq c \leq b$.
4. $\displaystyle\int_a^b f(x)dx = -\int_b^a f(x)dx$
5. $\displaystyle\int_a^a f(x)dx = 0$
6. If $m \leq f(x) \leq M$ in $[a, b]$, m, M-constants, then

$$m(b - a) \leq \int_a^b f(x)dx \leq M(b - a).$$

7. If $f(x) \leq g(x)$ in $[a, b]$, then $\int_a^b f(x)dx \leq \int_a^b g(x)dx$

8. $\left|\int_a^b f(x)dx\right| \leq \int_a^b |f(x)|dx$, if $a < b$.

2.4 Exponential and logarithmic integrals

Recall that $\frac{d}{dx}(e^x) = e^x$.
$\Longrightarrow \int e^x dx = e^x + C$ and in a more general sense

$$\int e^{f(x)} f'(x)dx = e^{f(x)} + C.$$

Example 2.7 *1.* $\int 2e^x dx = 2\int e^x dx = 2e^x + C.$

2. $\int 2e^{2x}dx = \int e^{2x}(2)dx = e^{2x} + C$

3. $\int (e^x + x)dx = \int e^x dx + \int x dx = e^x + \frac{1}{2}x^2 + C.$

4. $\int e^{3x+1}dx = \frac{1}{3}\int e^{3x+1}(3)dx = \frac{1}{3}e^{3x+1} + C.$

5. $\int 5xe^{-x^2}dx = \int \left(\frac{-5}{2}\right) e^{-x^2}(-2x)dx = \frac{-5}{2}\int e^{-x^2}(-2x)dx = \frac{-5}{2}e^{-x^2} + C$

Remark 2.3 *You cannot introduce a missing variable in the integrand. For instance, in finding* $\int e^{x^2}dx$*, it is illegal to multiply and divide by* $2x$*. i.e.*

$$\int e^{x^2}dx \neq \frac{1}{2x}\int e^{x^2}(2x)dx.$$

Recall that $\frac{d}{dx}(\ln x)dx = \frac{1}{x}$.
$\Longrightarrow \int \frac{1}{x}dx = \ln|x| + C$ and in a more general sense

$$\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + C$$

Example 2.8 *1.* $\int \frac{3}{x}dx = 3\int \frac{1}{x}dx = 3\ln|x| + C.$

2. $\displaystyle\int \frac{2x}{x^2}dx = \int \frac{f'(x)}{f(x)} = \ln x^2 + C = 2\ln|x| + C.$

3. $\displaystyle\int \frac{4}{4x+1}dx = \ln|4x+1| + C.$

4. $\displaystyle\int \frac{1}{3x+1}dx = \frac{1}{3}\int \frac{3}{3x+1}dx = \frac{1}{3}\ln|3x+1| + C.$

In general, to find

$$\int \frac{a}{bx+k}dx,$$

write

$$\int \frac{a}{bx+k}dx$$

as

$$\frac{a}{b}\int \frac{b}{bx+k}dx.$$

The last integral is in the form

$$\frac{a}{b}\int \frac{f'(x)}{f(x)}dx$$

so that

$$\int \frac{a}{bx+k}dx = \frac{a}{b}\ln|bx+k| + C.$$

Exercises

1. Find the integrals.
   (a) $\displaystyle\int e^{4x}dx$ (b) $\displaystyle\int -3e^{-3x}dx$ (c) $\displaystyle\int 5x^2e^{x^3}dx$
   (d) $\displaystyle\int (2x+1)e^{x^2+x}dx$ (e) $\displaystyle\int (x^2+2x)e^{x^3+3x^2-1}dx$ (f) $\displaystyle\int \frac{1}{x^2}e^{\frac{2}{x}}dx$
   (g) $\displaystyle\int \frac{1}{x^3}e^{\frac{1}{4x^2}}dx$ (h) $\displaystyle\int (e^x - e^{-x})^2dx$

2. Integrate the following:
   (a) $\displaystyle\int \frac{1}{x+1}dx$ (b) $\displaystyle\int \frac{1}{x-5}dx$ (c) $\displaystyle\int \frac{1}{3-2x}dx$
   (d) $\displaystyle\int \frac{x^2}{3-x^3}dx$ (e) $\displaystyle\int \frac{x}{x^2+4}dx$ (f) $\displaystyle\int \frac{x^2+2x+3}{x^3+3x^2+9x+1}dx$
   (g) $\displaystyle\int \frac{e^x}{1+e^x}dx$ (h) $\displaystyle\int \frac{-e^{3x}}{2-e^{3x}}dx$

3. Evaluate the following
   (a) $\displaystyle\int \frac{e^{2x}+2e^x+1}{e^x}dx$ (b) $\displaystyle\int e^x\sqrt{1-e^x}dx$ (c) $\displaystyle\int \frac{1}{(x-1)^2}dx$

(d) $\displaystyle\int \frac{x^2-4}{x}dx$ (e) $\displaystyle\int \frac{2}{1+e^{-x}}dx$ (f) $\displaystyle\int \frac{x^2+2x+5}{x-1}dx$
(g) $\displaystyle\int \frac{x^3+4x^2+3x}{x^2-x}dx$ (h) $\displaystyle\int \frac{1+e^{-x}}{1+xe^{-x}}dx$

2.5 Trigonometric functions

Recall that $\dfrac{d}{dx}(\sin x) = \cos x$ and $\dfrac{d}{dx}(\cos x) = -\sin x$

$$\Longrightarrow \int \sin x dx = -\cos x + C$$

and

$$\int \cos x dx = \sin x + C.$$

using these integrals we can find the integral of any trigonometric function. This will be seen later.

2.6 Some Techniques of Integration

2.6.1 The general power rule

This has already been explored in detail. We state it here for completeness only.

$$\int u^n \frac{du}{dx}dx = \frac{u^{n+1}}{n+1} + C, \quad n \neq -1$$

2.6.2 Change of variable (Substitution)

If a determination of $\displaystyle\int f(x)dx$ is not immediately obvious in terms of elementary functions, useful results may be obtained by changing the variable, say from x to t, according to the transformation $x = g(t)$. In this way we have

$$\int f(x)dx = \int f(g(t))g'(t)dt$$

Example 2.9 *Find* $\displaystyle\int \frac{x}{(x+1)^2}dx$.

Solution

Let $u = x + 1 \Longrightarrow \dfrac{du}{dx} = 1 \Longrightarrow dx = du$ and $x = u - 1$. Thus

$$\begin{aligned}
\int \frac{x}{(x+1)^2} dx &= \int \frac{u-1}{u^2} du \\
&= \int \frac{1}{u} du - \int \frac{1}{u^2} du \\
&= \ln|u| + \frac{1}{u} + C \\
&= \ln|x+1| + \frac{1}{x+1} + C.
\end{aligned}$$

Example 2.10 *Find* $I = \displaystyle\int (x+2)\sin(x^2+4x-6)dx$.

Solution
Let $x^2 + 4x - 6 = u \Longrightarrow (2x+4)dx = du \Longrightarrow (x+2)dx = \dfrac{1}{2}du$.
So

$$I = \frac{1}{2}\int \sin u du = -\frac{1}{2}\cos u + C = -\frac{1}{2}\cos(x^2+4x-6) + C.$$

Example 2.11 *Show that* $\displaystyle\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C$.

Solution
Let $x = \sin y \Longrightarrow dx = \cos y dy$ and $y = \sin^{-1} x$. Thus

$$\begin{aligned}
\int \frac{1}{\sqrt{1-x^2}} dx &= \int \frac{\cos y}{\sqrt{1-\sin^2 y}} dy \\
&= \int \frac{\cos y dy}{\cos y} \\
&= \int dy \\
&= y + C \\
&= \sin^{-1} x + C.
\end{aligned}$$

Example 2.12 *Find* $I = \displaystyle\int \frac{dx}{\sqrt{a^2-x^2}}$.

Solution
Let $x = a\sin y \Longrightarrow dx = a\cos y dy$ and $y = \sin^{-1}\frac{x}{a}$. Hence

$$\begin{aligned}
I &= \int \frac{a\cos y dy}{\sqrt{a^2 - a^2\sin^2 y}} \\
&= \int \frac{a\cos y dy}{a\sqrt{1-\sin^2 y}}
\end{aligned}$$

$$\begin{aligned} &= \int dy \\ &= y + C \\ &= \sin^{-1}\frac{x}{a} + C. \end{aligned}$$

Activity 2.1 *Show that*

1. $\displaystyle\int \frac{1}{1+x^2}dx = \tan^{-1} x + C.$
2. $\displaystyle\int \frac{1}{\sqrt{1-a^2x^2}}dx = \frac{1}{a}\sin^{-1}(ax) + C.$
3. $\displaystyle\int \frac{1}{1+a^2x^2}dx = \frac{1}{a}\tan^{-1}(ax) + C.$

Example 2.13 *Calculate*

$$\int \sin x \cos x dx$$

and check the result by differentiation.

Solution
Let $u = \sin x \Longrightarrow du = \cos x dx$ so that

$$\int \sin x \cos x dx = \int u du = \frac{1}{2}u^2 + C = \frac{1}{2}\sin^2 x + C.$$

Checking: $\displaystyle\frac{d}{dx}\left(\frac{1}{2}\sin^2 x + C\right) = \frac{1}{2}(2)\sin x \cos x = \sin x \cos x.$

Example 2.14 *Calculate* $\displaystyle\int x^2\sqrt{4+x^3}dx.$

Solution
Set $u = 4 + x^3, \Longrightarrow du = 3x^2dx$. Then $x^2\sqrt{4+x^3}dx = \dfrac{1}{3}u^{\frac{1}{2}}du$. Hence

$$\int x^2\sqrt{4+x^3}dx = \frac{1}{3}\int u^{\frac{1}{2}}du = \frac{2}{9}u^{\frac{3}{2}} + C = \frac{2}{9}(4+x^3)^{\frac{3}{2}} + C.$$

Activity 2.2 *Find* $\displaystyle\int \sec^3 x \tan x dx.$
Hint: *Write*

$$\sec^3 x \tan x = \sec^2 x \sec x \tan x$$

and choose proper substitution.

Example 2.15 *Let* $a^x = e^z$. *Then* $\ln a^x = \ln e^z, \Longrightarrow x \ln a = z \ln e$.
$\Longrightarrow z = x \ln a$ *and* $a^x = e^{x \ln a}$. *Thus*

$$\int a^x dx = \int e^{x \ln a} dx = \frac{1}{\ln a} e^{x \ln a} = \frac{a^x}{\ln a} + C.$$

The result above is useful in evaluating integrals such as $\int 3^x dx$.

Example 2.16 *Find* $I = \int_{-1}^{1} \frac{dx}{\sqrt{(x+2)(3-x)}}$.

Solution

$$\begin{aligned} I &= \int_{-1}^{1} \frac{dx}{\sqrt{(x+2)(3-x)}} \\ &= \int_{-1}^{1} \frac{dx}{\sqrt{6+x-x^2}} \\ &= \int_{-1}^{1} \frac{dx}{\sqrt{\frac{25}{4} - (x - \frac{1}{2})}}. \end{aligned}$$

Now let $(x - \frac{1}{2}) = \frac{5}{2} \sin y, \Longrightarrow dx = \frac{5}{2} \cos y dy$ and $y = \sin^{-1} \frac{2}{5}(x - \frac{1}{2})$.

x	y
-1	$\sin^{-1}(\frac{-3}{5})$
1	$\sin^{-1}(\frac{1}{5})$

$$\begin{aligned} I &= \int_{\sin^{-1}(\frac{-3}{5})}^{\sin^{-1}(\frac{1}{5})} \frac{\frac{5}{2} \cos y dy}{\frac{5}{2}\sqrt{1 - \sin^2 y}} \\ &= \int_{\sin^{-1}(\frac{-3}{5})}^{\sin^{-1}(\frac{1}{5})} dy \\ &= [y]_{\sin^{-1}(\frac{-3}{5})}^{\sin^{-1}(\frac{1}{5})} \\ &= \sin^{-1}\left(\frac{1}{5}\right) - \sin^{-1}\left(\frac{-3}{5}\right). \end{aligned}$$

Example 2.17 *Find* $I = \int_0^{\frac{1}{\sqrt{2}}} \frac{x \sin^{-1} x^2}{\sqrt{1 - x^4}} dx$

Solution
Let $\sin^{-1} x^2 = u, \Longrightarrow \sin u = x^2$ and $\cos u du = 2x dx$

$\Longrightarrow \frac{x dx}{\cos u} = \frac{1}{2} du, \Longrightarrow \frac{x dx}{\sqrt{1 - x^4}} = \frac{1}{2} du.$

x	u
0	0
$\frac{1}{\sqrt{2}}$	$\sin^{-1}(\frac{1}{2})$

Therefore $I = \frac{1}{2} \int_0^{\sin^{-1}(\frac{1}{2}} u du = \frac{1}{2} \left[\frac{u^2}{2}\right]_0^{\sin^{-1}(\frac{1}{2}} = \frac{\pi^2}{144}$.

2.6.3 t-formula

In this section we consider integrals of the form $\int \frac{1}{a+b\sin x+c\cos x}dx$ and $\int \frac{1}{a+b\sin^2 x+c\cos^2 x}dx$. In the former, we use the substitution $t=\tan\frac{1}{2}x$. This leads to $x=2\tan^{-1}t \implies dx=\frac{2dt}{1+t^2}$. Use of trigonometric identities result in

$$\begin{aligned}
\cos x &= \cos(\frac{1}{2}x+\frac{1}{2}x)\\
&= \cos^2(\frac{1}{2}x)-\sin^2(\frac{1}{2}x)\\
&= \frac{1}{1+t^2}-\frac{t^2}{1+t^2}\\
&= \frac{1-t^2}{1+t^2}\\
\sin x &= \sin(\frac{1}{2}x+\frac{1}{2}x)\\
&= 2\cos(\frac{1}{2}x)\sin(\frac{1}{2}x)\\
&= \frac{2t}{1+t^2}
\end{aligned}$$

In the later, we use the substitution $t=\tan x$ and similarly obtain $x=\tan^{-1}t$ $\implies dx=\frac{1}{1+t^2}dt$, $\cos x=\frac{1}{\sqrt{1+t^2}}$ and $\sin x=\frac{t}{\sqrt{1+t^2}}$.

Example 2.18 *Determine* $\int \frac{dx}{5+3\cos x}$.

Solution

Let $t=\tan(\frac{1}{2}x), \implies x=2\tan^{-1}t$ and $dx=\frac{2dt}{1+t^2}$. $\sin(\frac{1}{2}x)=\frac{t}{\sqrt{1+t^2}}$, $\cos(\frac{1}{2}x)=\frac{1}{\sqrt{1+t^2}}$ and

$$\begin{aligned}
\cos x &= \frac{1-t^2}{1+t^2}\\
\sin x &= \frac{2t}{1+t^2}
\end{aligned}$$

Hence,

$$\begin{aligned}
\int \frac{dx}{5+3\cos x} &= \int \frac{2}{5+3\left(\frac{1-t^2}{1+t^2}\right)}\cdot\frac{dt}{1+t^2}\\
&= \int \frac{2dt}{8+2t^2}
\end{aligned}$$

$$\begin{aligned} &= \int \frac{dt}{4+t^2} \\ &= \frac{1}{4}\int \frac{dt}{1+(\frac{t}{2})^2} \\ &= \frac{2}{4}\tan^{-1}(\frac{t}{2}) + C \\ &= \frac{1}{2}\tan^{-1}\left(\frac{1}{2}\tan(\frac{1}{2}x)\right) + C \end{aligned}$$

Example 2.19 *Determine* $\int \frac{dx}{1+2\sin^2 x}$.

Solution
Let $t = \tan x$, $\implies$ $x = \tan^{-1} t$ and $dx = \frac{1}{1+t^2}dt$, $\sin x = \frac{t}{\sqrt{1+t^2}}$. This gives $\int \frac{dx}{1+2\sin^2 x} = \int \frac{1}{1+\frac{2t^2}{1+t^2}} \cdot \frac{1}{1+t^2}dt = \int \frac{dt}{1+3t^2} = \frac{1}{\sqrt{3}}\int \frac{\sqrt{3}}{1+(\sqrt{3}t)^2}dt = \frac{1}{\sqrt{3}}\tan^{-1}\sqrt{3}t + C = \frac{1}{\sqrt{3}}\tan^{-1}(\sqrt{3}\tan x) + C$

Activity 2.3 *Determine the following integrals:*
(i) $\int \frac{1}{1+\sin x - \cos x}dx$ *(ii)* $\int \frac{1}{3+\cos^2 x}dx$

2.6.4 Integration by parts

Recall the product rule of differentiation:
$\frac{d}{dx}(uv) = uv' + vu'$. Integrating this on both sides

$$\int \frac{d}{dx}(uv)dx = \int u\frac{dv}{dx}dx + \int v\frac{du}{dx}dx$$

$\implies uv = \int udv + \int vdu$. Rearranging this gives the **integration by parts formula**

$$\int udv = uv - \int vdu$$

Example 2.20 *Find* $\int xe^x dx$.

Solution
Let $u = x$ and $dv = e^x \implies du = dx$ and $v = e^x$. Therefore substituting into the formula gives

$$\int xe^x dx = xe^x - \int e^x dx$$

$$\begin{aligned} &= xe^x - e^x + C \\ &= e^x(x-1) + C \end{aligned}$$

Example 2.21 *Find* $\int x^2 \ln x dx$.

Solution
Let $u = \ln x$ and $dv = x^2 \Longrightarrow du = \frac{1}{x}dx$ and $v = \frac{1}{3}x^3$. Substituting into the formula:

$$\begin{aligned} \int x^2 \ln x dx &= \frac{1}{3}x^3 \ln x - \frac{1}{3}\int x^2 dx \\ &= \frac{1}{3}x^3 \ln x - \frac{1}{9}x^3 + C \\ &= \frac{x^3}{3}(\ln x - \frac{1}{3}) + C \end{aligned}$$

Example 2.22 *Find* $\int 3^{\sqrt{2x+1}} dx$.

Solution
Let $y = \sqrt{2x+1}, \Longrightarrow y^2 = 2x+1, \Longrightarrow 2ydy = 2dx, \Longrightarrow ydy = dx$.
Thus

$$\int 3^{\sqrt{2x+1}} dx = \int 3^y y dy$$

Using integration by parts, we now let $u = y$, $dv = 3^y \Longrightarrow du = dy$ and $dv = 3^y = e^{\ln 3^y} = e^{y \ln 3}, \Longrightarrow v = \frac{1}{\ln 3} e^{y \ln 3} = \frac{3^y}{\ln 3}$. Now

$$\begin{aligned} \int 3^{\sqrt{2x+1}} dx &= \int 3^y y dy \\ &= \frac{y}{\ln 3} 3^y - \frac{1}{\ln 3}\int 3^y dy \\ &= \frac{y}{\ln 3} 3^y - \frac{3^y}{(\ln 3)^2} + C \\ &= \frac{3^y}{\ln 3}\left(y - \frac{1}{\ln 3}\right) + C \\ &= \frac{3^{\sqrt{2x+1}}}{\ln 3}\left(\sqrt{2x+1} - \frac{1}{\ln 3}\right) + C \end{aligned}$$

Exercises

1. Find the integrals.
   (a) $\int x\sqrt{x^2+4}dx$ (b) $\int \frac{t}{\sqrt{1-t^2}}dt$

(c) $\displaystyle\int \frac{\cot(\ln x)}{x}dx$ (d) $\displaystyle\int \frac{dx}{2+2\cos x-\sin x}$ (e) $\displaystyle\int \frac{dx}{1-\sin x}$

(f) $\displaystyle\int \frac{x^2}{(1-x^3)^{\frac{2}{3}}}dx$ (g) $\displaystyle\int \frac{4x+6}{\sqrt{x^2+3x+1}}dx$

2. Show that $\displaystyle\int_1^2 \frac{dx}{(x^2-2x+4)^{\frac{3}{2}}} = \frac{1}{6}$

3. Evaluate the integral by the indicated u-substitution:

(a) $\displaystyle\int x\sqrt{x+1}dx$ set $u = x+1$ (b) $\displaystyle\int x\sqrt{2x-1}dx$ set $u = 2x-1$

(c) $\displaystyle\int_0^1 \frac{x+3}{\sqrt{x+1}}dx$ set $u = x+1$

4. Calculate the indefinite integral:

(a) $\displaystyle\int \cos^4 x \sin x dx$ (b) $\displaystyle\int x\sec^2 x^2 dx$ (c) $\displaystyle\int \frac{\sin\sqrt{x}}{\sqrt{x}}dx$

(d) $\displaystyle\int (1+\tan^2 x)\sec^2 x dx$ (e) $\displaystyle\int x\sin^3 x^2 \cos x^2 dx$ (f) $\displaystyle\int \frac{\sec^2 x}{\sqrt{1+\tan x}}dx.$

5. Evaluate the following

(a) $\displaystyle\int_0^1 x\ln(x+3)dx$ (b) $\displaystyle\int \ln x dx$ (c) $\displaystyle\int x^2 e^{2x} dx dx$

(d) $\displaystyle\int \frac{(\ln x)^2}{x}dx$ (e) $\displaystyle\int_0^1 3^{\sqrt{x}}dx$ (f) $\displaystyle\int_0^e x^5 \ln x dx$

(g) $\displaystyle\int x^2 e^{-x^3}dx$ (h) $\displaystyle\int e^{3x}\cos 2x dx$ (i) $\displaystyle\int x^2(2x-1)^{-7}dx$

(j) $\displaystyle\int \frac{dx}{x(\ln x)^3}$

2.6.5 Use of partial fractions

Every rational fraction, no matter how complicated, can be rewritten as a sum of simpler fractions which may be easier to handle. For instance, it may be easier to integrate the component simpler fractions than the original complicated fraction. We discuss use of partial fractions by means of examples.

1. **Two distinct linear factors in the denominator**

Example 2.23 *Find* $\displaystyle\int \frac{5x-3}{x^2-2x-3}dx.$

Solution
We first express the integrand as partial fractions.Now, note that

$$\frac{5x-3}{x^2-2x-3} = \frac{5x-3}{(x+1)(x-3)}.$$

We find the constants A and B such that

$$\frac{5x-3}{(x+1)(x-3)} = \frac{A}{x+1} + \frac{B}{x-3}$$

Observe that we can clear fractions in this equation to obtain
$5x-3 = A(x-3) + B(x+1) = (A+B)x - 3A + B$.
This will be an identity in x if and only if the coefficients of like powers of x on the two sides are equal:
$A+B=5, \quad -3A+B=-3$. Solving these two equations simultaneously gives $A=2$ and $B=3$. Hence

$$\frac{5x-3}{x^2-2x-3} = \frac{5x-3}{(x+1)(x-3)} = \frac{2}{x+1} + \frac{3}{x-3}.$$

Now,

$$\begin{aligned}\int \frac{5x-3}{x^2-2x-3}dx &= \int \frac{5x-3}{(x+1)(x-3)}dx \\ &= \int \frac{2}{x+1}dx + \int \frac{3}{x-3}dx \\ &= 2\ln|x+1| + 3\ln|x-3| + C.\end{aligned}$$

2. **A Repeated linear factor in the denominator**

Example 2.24 *Find* $\displaystyle\int \frac{6x+7}{(x+2)^2}dx$.

Solution
First express $\dfrac{6x+7}{(x+2)^2}$ as a sum of partial fractions. Since the denominator has a repeated linear factor, $(x+2)^2$, we must express the fraction in the form

$$\frac{6x+7}{(x+2)^2} = \frac{A}{x+2} + \frac{B}{(x+2)^2}$$

After clearing fractions we get
$6x+7 = A(x+2) + B = Ax + (2A+B)$.
Matching coefficients of like terms gives $A=6$ and $7 = 2A+B = 12+B \Longrightarrow B=-5$. Hence,

$$\frac{6x+7}{(x+2)^2} = \frac{6}{x+2} - \frac{5}{(x+2)^2}$$

and

$$\int \frac{6x+7}{(x+2)^2}dx = \int \frac{6}{x+2}dx - \int \frac{5}{(x+2)^2}dx = 6\ln|x+2| + \frac{5}{x+2} + C$$

3. **An improper fraction**

Example 2.25 *Find* $\displaystyle\int \frac{2x^3-4x^2-x-3}{x^2-2x-3}dx.$

Solution
First express $\dfrac{2x^3-4x^2-x-3}{x^2-2x-3}$ as a sum of partial fractions. Divide the denominator into the numerator to get a polynomial plus a proper fraction. Then we write the proper fraction as a sum of partial fractions.

$$\begin{array}{r|l} & \quad 2x \\ \hline x^2-2x-3 & 2x^3-4x^2-x-3 \\ & \underline{2x^3-4x^2-6x} \\ & \qquad\qquad 5x-3 \end{array}$$

Hence,

$$\frac{2x^3-4x^2-x-3}{x^2-2x-3} = 2x + \frac{5x-3}{x^2-2x-3} = 2x + \frac{2}{x+1} + \frac{3}{x-3}$$

as found earlier. So

$$\begin{aligned} \int \frac{2x^3-4x^2-x-3}{x^2-2x-3}dx &= \int 2xdx + \int \frac{5x-3}{x^2-2x-3}dx \\ &= \int 2xdx + \int \frac{2}{x+1}dx + \int \frac{3}{x-3}dx \\ &= x^2 + 2\ln|x+1| + 3\ln|x-3| + C \end{aligned}$$

4. An irreducible quadratic factor in the denominator

Example 2.26 *Determine* $\displaystyle\int \frac{-2x+4}{(x^2+1)(x-1)^2}dx.$

Solution
First express $\dfrac{-2x+4}{(x^2+1)(x-1)^2}$ as a sum of partial fractions. The denominator has an irreducible quadratic factor as well as a repeated linear factor, so we write

$$\frac{-2x+4}{(x^2+1)(x-1)^2} = \frac{Ax+B}{x^2+1} + \frac{C}{x-1} + \frac{D}{(x-1)^2}$$

Notice the numerator over x^2+1 : In general, for quadratic factors, we use first degree numerators, not constant numerators. Removing fractions we get

$$\begin{aligned} -2x+4 &= (Ax+B)(x-1)^2 + C(x-1)(x^2+1) + D(x^2+1) \\ &= (A+C)x^3 + (-2A+B-C+D)x^2 + (A-2B+C)x + (B-C+D) \end{aligned}$$

Equating coefficients of like terms gives
Coefficients of x^3: $0 = A + C$.
Coefficients of x^2: $0 = -2A + B - C + D$.
Coefficients of x: $-2 = A - 2B + C$.
Constant terms (Coefficients of x^0): $4 = B - C + D$.
We solve these equations simultaneously to find the values of A, B, C and D. We get $A = 2,\ B = 1,\ C = -2\ D = 1$. We substitute these values to get

$$\frac{-2x+4}{(x^2+1)(x-1)^2} = \frac{2x+1}{x^2+1} - \frac{2}{x-1} + \frac{1}{(x-1)^2}$$

Hence,

$$\begin{aligned}\int \frac{-2x+4}{(x^2+1)(x-1)^2}dx &= \int \frac{2x+1}{x^2+1}dx - \int \frac{2}{x-1}dx + \int \frac{1}{(x-1)^2}dx \\ &= \int \frac{2x}{x^2+1}dx + \int \frac{1}{x^2+1}dx - \int \frac{2}{x-1}dx + \int \frac{1}{(x-1)^2}dx \\ &= \ln(x^2+1) + \tan^{-1} x - 2\ln|x-1| - \frac{1}{x-1} + C.\end{aligned}$$

Activity 2.4 *Express the following as a sum of partial fractions.*
(i) $\dfrac{5x-13}{(x-3)(x-2)}$ *(ii)* $\dfrac{5x-7}{x^2-3x+2}$ *(iii)* $\dfrac{x+4}{(x+1)^2}$
(iv) $\dfrac{2x+2}{x^2-2x+1}$ *(v)* $\dfrac{x+1}{x^2(x-1)}$ *(vi)* $\dfrac{z}{z^3-z^2-6z}$
(vii) $\dfrac{x^2+8}{x^2-5x+6}$ *(viii)* $\dfrac{x^4+9}{x^4+9x^2}$

Activity 2.5 *Determine*

1. $\displaystyle\int \frac{x+7}{x^2-x-6}dx$
2. $\displaystyle\int \frac{5x^2+20x+6}{x^3+2x^2+x}dx$
3. $\displaystyle\int \frac{x^5+x-1}{x^4-x^3}dx$

2.6.6 Products and powers of trigonometric functions

The form $\displaystyle\int \sin^m x \cos^n x$

If $n = 1$, we have $\displaystyle\int \sin^m x \cos x dx = \frac{1}{m+1}\sin^{m+1} x + C, \quad m \neq -1.$

Example 2.27 $\displaystyle\int \sin^4 x \cos x dx = \frac{1}{5}\sin^5 x + C.$

Example 2.28

$$\begin{aligned}\int \sin^2 x \cos^5 x dx &= \int \sin^2 x \cos^4 x \cos x dx \\ &= \int \sin^2 x (1-\sin^2 x)^2 \cos x dx \\ &= \int (\sin^2 x - 2\sin^4 x + \sin^6 x) \cos x dx \\ &= \frac{1}{3}\sin^3 x - \frac{2}{5}\sin^5 x + \frac{1}{7}\sin^7 x + C\end{aligned}$$

Example 2.29

$$\begin{aligned}\int \sin^5 x dx &= \int \sin^4 x \sin x dx \\ &= \int (1-\cos^2 x)^2 \sin x dx \\ &= \int (1 - 2\cos^2 x + \cos^4 x) \sin x dx \\ &= \int \sin x dx - 2\int \cos^2 x \sin x dx + \int \cos^4 x \sin x dx \\ &= -\cos x + \frac{2}{3}\cos^3 x - \frac{1}{5}\cos^5 x + C\end{aligned}$$

Activity 2.6 *Find* $\int \cos^2 x dx$.

Example 2.30

$$\begin{aligned}\int \sin^2 x \cos^2 x dx &= \int (\sin x \cos x)^2 dx \\ &= \int \left(\frac{1}{2}\sin 2x\right)^2 dx \\ &= \frac{1}{4}\int \sin^2 2x dx \\ &= \frac{1}{4}\int \left(\frac{1}{2} - \frac{1}{2}\cos 4x\right) dx \\ &= \frac{1}{8}\int dx - \frac{1}{8}\int \cos 4x dx \\ &= \frac{1}{8}x - \frac{1}{32}\sin 4x + C\end{aligned}$$

Activity 2.7 $\int \sin^4 x dx$.

Reduction formulae for $\int \sin^n x dx$ and $\int \cos^n x dx$

Write $\int \sin^n x dx = \int \sin^{n-1} x \sin x dx$ and let $f(x) = \sin^{n-1} x$, $g'(x) = \sin x$. Then $f'(x) = (n-1)\sin^{n-2} x \cos x$ and $g(x) = -\cos x$. So

$$\begin{aligned}
I_n &= \int \sin^n x dx \\
&= -\sin^{n-1} x \cos x + (n-1)\int \sin^{n-2} x \cos^2 x dx \\
&= -\sin^{n-1} x \cos x + (n-1)\int \sin^{n-2} x(1-\sin^2 x) dx \\
&= -\sin^{n-1} x \cos x + (n-1)\int \sin^{n-2} x dx - (n-1)\int \sin^n x dx \\
\Longrightarrow \int \sin^n x dx + (n-1)\int \sin^n x dx &= -\sin^{n-1} x \cos x + (n-1)\int \sin^{n-2} x dx \\
\Longrightarrow n\int \sin^n x dx &= -\sin^{n-1} x \cos x + (n-1)\int \sin^{n-2} x dx \\
\Longrightarrow \int \sin^n x dx &= -\frac{1}{n}\sin^{n-1} x \cos x + \frac{(n-1)}{n}\int \sin^{n-2} x dx
\end{aligned}$$

Thus

$$I_n = -\frac{1}{n}\sin^{n-1} x \cos x + \frac{n-1}{n} I_{n-2}.$$

Example 2.31 *Use the reduction formula above to find* $\int \sin^5 x dx$.

Solution

$$\begin{aligned}
\int \sin^5 x dx &= -\frac{1}{5}\sin^4 x \cos x + \frac{4}{5}\int \sin^3 x dx \\
&= -\frac{1}{5}\sin^4 x \cos x + \frac{4}{5}\left[-\frac{1}{3}\sin^2 x \cos x + \frac{2}{3}\int \sin x dx\right] \\
&= -\frac{1}{5}\sin^4 x \cos x - \frac{4}{15}\sin^2 x \cos x - \frac{8}{15}\cos x + C
\end{aligned}$$

Activity 2.8 *Using the reduction formula for* $I_n = \int \cos^n x dx$, *show that*

$$I_n = \frac{1}{n}\cos^{n-1} x \sin x + \frac{n-1}{n} I_{n-2}.$$

Example 2.32

$$\int \cos^5 x dx = \frac{1}{5}\cos^4 x \sin x + \frac{4}{5}\int \cos^3 x dx$$

$$
\begin{aligned}
&= \frac{1}{5}\cos^4 x\sin x + \frac{4}{5}\left[\frac{1}{3}\cos^2 x\sin x + \frac{2}{3}\int \cos x dx\right] \\
&= \frac{1}{5}\cos^4 x\sin x + \frac{4}{15}\cos^2 x\sin x + \frac{8}{15}\sin x + C.
\end{aligned}
$$

Example 2.33 *Find* $\int \tan^6 x dx$.

Solution

$$
\begin{aligned}
\int \tan^6 x dx &= \int \tan^4 x\tan^2 x dx \\
&= \int \tan^4 x(\sec^2 x - 1)dx \\
&= \int (\tan^4 x\sec^2 x - \tan^4 x)dx \\
&= \int (\tan^4 x\sec^2 x - (\sec^2 x - 1)\tan^2 x)dx \\
&= \int (\tan^4 x\sec^2 x - \tan^2 x\sec^2 x + \tan^2 x)dx \\
&= \int (\tan^4 x\sec^2 x - \tan^2 x\sec^2 x + \sec^2 x - 1)dx \\
&= \int \tan^4 x\sec^2 x dx - \int \tan^2 x\sec^2 x dx + \int \sec^2 x dx - \int dx \\
&= \frac{1}{5}\tan^5 x - \frac{1}{3}\tan^3 x + \tan x - x + C
\end{aligned}
$$

Activity 2.9 *Find*

1. $\int \tan^5 x\sec^4 x dx$
2. $\int \tan^5 x\sec^3 x dx$

Hint: *Write*

$$\int \tan^5 x\sec^4 x dx = \int \tan^5 x\sec^2 x\sec^2 x dx$$

and

$$\int \tan^5 x\sec^3 x dx = \int (\sec^2 x - 1)^2\sec^2 x\sec x\tan x dx.$$

Example 2.34 *Establish the following:*

1. $\int \tan^n x dx = \frac{1}{n-1}\tan^{n-1} x - \int \tan^{n-2} x dx, \quad n \geq 2$ *and*

2. $\displaystyle\int \sec^n x dx = \frac{1}{n-1}\sec^{n-2} x \tan x + \frac{n-2}{n-1}\int \sec^{n-2} x dx, \quad n \geq 2.$

Solution

1. Exercise.
2. Write $\displaystyle\int \sec^n x dx = \int \sec^{n-2} x \sec^2 x dx$ and let $f(x) = \sec^{n-2} x, \quad g'(x) = \sec^2 x$. In this way we have

$$f'(x) = (n-2)\sec^{n-2} x \tan x$$

and $g(x) = \tan x$. Hence, integration by parts gives

$$\begin{aligned}
\int \sec^n x dx &= \sec^{n-2} x \tan x - (n-2)\int \sec^{n-2} x \tan^2 x dx \\
&= \sec^{n-2} x \tan x - (n-2)\int \sec^{n-2} x(\sec^2 x - 1)dx \\
&= \sec^{n-2} x \tan x - (n-2)\int \sec^n x dx + (n-2)\int \sec^{n-2} x dx \\
\Longrightarrow (n-1)\int \sec^n x dx &= \sec^{n-2} x \tan x + (n-2)\int \sec^{n-2} x dx \\
\Longrightarrow \int \sec^n x dx &= \frac{1}{n-1}\sec^{n-2} x \tan x + \frac{n-2}{n-1}\int \sec^{n-2} x dx, \quad n \geq 2.
\end{aligned}$$

Example 2.35 *Using the established reduction formula above find* $\displaystyle\int \sec^6 x dx$.

Solution

$$\begin{aligned}
\int \sec^6 x dx &= \frac{1}{5}\sec^4 x \tan x + \frac{4}{5}\int \sec^4 x dx \\
&= \frac{1}{5}\sec^4 x \tan x + \frac{4}{5}\left[\frac{1}{3}\sec^2 x \tan x + \frac{2}{3}\int \sec^2 x dx\right] \\
&= \frac{1}{5}\sec^4 x \tan x + \frac{4}{15}\sec^2 x \tan x + \frac{8}{15}\tan x + C.
\end{aligned}$$

Activity 2.10 *Show that the solution obtained above is equivalent to*

$$\frac{1}{5}\tan^5 x + \frac{2}{3}\tan^3 x + \tan x + C$$

as obtained earlier.

2.6.7 Hyperbolic Functions

As with trigonometric functions, your knowledge of derivatives of hyperbolic functions will help in evaluating integrals involving hyperbolic functions.

Example 2.36 *Find* $\int x^2 \cosh(1-x^3)dx$.

Solution
$$\int x^2 \cosh(1-x^3)dx = -\frac{1}{3}\int(-3x^2)\cosh(1-x^3)dx = -\frac{1}{3}\sinh(1-x^3)+C.$$

Example 2.37 *Find* $\int \frac{\coth(e^x)\text{cosech}(e^x)}{e^{-x}}dx$.

Solution
$$\int \frac{\coth(e^x)\text{cosech}(e^x)}{e^{-x}}dx = \int e^x \coth(e^x)\text{cosech}(e^x) = -\text{cosech}(e^x)+C$$

Activity 2.11 *Determine the following integrals:*
(i) $\int e^{2x}\text{cosech}(e^{2x})dx$ *(ii)* $\int \tanh^2 4x dx$ *(iii)* $\int \frac{\text{sech}(\cos x)\tanh(\cos x)}{\text{cosec}x}dx$
(iv) $\int e^{-x}\sinh 2x dx$ *(v)* $\int \frac{\cosh\sqrt{x}}{\sqrt{x}}dx$ *(vi)* $\int \text{sech}^2(3+2x)dx$ *(vii)* $\int \coth\left(\frac{2}{3}x\right)dx$

2.6.8 Inverse Functions and The technique of completing the square

Integrals that result in the inverse trigonometric functions $\tan^{-1}$, $\sin^{-1}$ or $\cos^{-1}$ have been dealt with earlier in this unit using the technique of substitution. Such forms and others alike can be evaluated easily by using standard integrals from the table of integration formulae.

Example 2.38 *Find* $\int \frac{1}{x^2+9}dx$

Solution
$$\int \frac{1}{x^2+9}dx = \int \frac{1}{(3)^2+(x)^2}dx = \frac{1}{3}\arctan\left(\frac{x}{3}\right)+C$$

Example 2.39 *Determine* $\int \frac{1}{\sqrt{2x^2+9}}dx$

Solution
$$\int \frac{1}{\sqrt{2x^2+9}}dx = \int \frac{1}{\sqrt{(\sqrt{2}x)^2+(3)^2}}dx = \frac{1}{\sqrt{2}}\int \frac{\sqrt{2}}{\sqrt{(\sqrt{2}x)^2+(3)^2}}dx$$
$$= \frac{1}{\sqrt{2}}\text{arcsinh}\left(\frac{\sqrt{2}x}{3}\right)+C$$

Example 2.40 *Solve the integral* $\displaystyle\int \frac{\cos x}{\sin^2 x - 1} dx$

Solution
$$\int \frac{\cos x}{\sin^2 x - 1} dx = \int \frac{\cos x}{(\sin x)^2 - 1} dx = -\text{arccoth}(\sin x) + C$$

Example 2.41 *Evaluate the integral* $\displaystyle\int \frac{1}{\sqrt{3 + 6x - x^2}} dx$

Solution
We notice that, since the denominator does not factorise, it is convenient to first complete the square in the denominator. The resultant form will be easier to handle than the former. We write $3 + 6x - x^2 = 12 - (x-3)^2$, and thus
$$\int \frac{1}{\sqrt{3 + 6x - x^2}} dx = \int \frac{1}{\sqrt{12 - (x-3)^2}} dx =$$
$$\int \frac{1}{\sqrt{(\sqrt{12})^2 - (x-3)^2}} dx = \sin^{-1}\left(\frac{x-3}{\sqrt{12}}\right) + C$$

Example 2.42 *Find* $\displaystyle\int \frac{dx}{x^2 - 4x + 1}$.

Write
$$x^2 - 4x + 1 = x^2 - 4x + 4 + (1 - 4) = (x-2)^2 - 3.$$
We have completed the square. Now, let $t = x - 2, \Longrightarrow dt = dx, \quad x = t + 2$. So
$$\begin{aligned}
\int \frac{dx}{x^2 - 4x + 1} &= \int \frac{dx}{(x-2)^2 - 3} \\
&= \int \frac{dt}{t^2 - 3} \\
&= \int \frac{dt}{(t - \sqrt{3})(t + \sqrt{3})} \\
&= \frac{1}{2\sqrt{3}} \int \frac{dt}{t - \sqrt{3}} - \frac{1}{2\sqrt{3}} \int \frac{dt}{t + \sqrt{3}} \\
&= \frac{1}{2\sqrt{3}} \ln|t - \sqrt{3}| - \frac{1}{2\sqrt{3}} \ln|t + \sqrt{3}| + C \\
&= \frac{1}{2\sqrt{3}} \ln\left|\frac{x - 2 - \sqrt{3}}{x - 2 + \sqrt{3}}\right| + C
\end{aligned}$$

Alternative solution:
$$\int \frac{dx}{x^2 - 4x + 1} = \int \frac{dx}{(x-2)^2 - 3} = -\frac{1}{\sqrt{3}} \coth^{-1}\left(\frac{x-2}{\sqrt{3}}\right) + C$$

Example 2.43 *Find* $\displaystyle\int \frac{x - 2}{x^2 + x + 2} dx$.

Solution

$$\begin{aligned}
\int \frac{x-2}{x^2+x+2}dx &= \int \frac{x}{x^2+x+2}dx - \int \frac{2dx}{x^2+x+2} \\
&= \frac{1}{2}\int \frac{2x+1-1}{x^2+x+2}dx - \int \frac{2dx}{x^2+x+2} \\
&= \frac{1}{2}\int \frac{2x+1}{x^2+x+2}dx - \frac{1}{2}\int \frac{dx}{x^2+x+2} - \int \frac{2dx}{x^2+x+2} \\
&= \frac{1}{2}\ln|x^2+x+2| - \frac{5}{2}\int \frac{dx}{x^2+x+2} \\
&= \frac{1}{2}\ln|x^2+x+2| - \frac{5}{2}\int \frac{dx}{(x+\frac{1}{2})^2+\frac{7}{4}} \\
&= \frac{1}{2}\ln|x^2+x+2| - \frac{5}{2}\int \frac{\frac{\sqrt{7}}{2}\sec^2\theta d\theta}{\frac{7}{4}(\tan^2\theta+1)}, \quad \text{let} \quad x+\frac{1}{2} = \frac{\sqrt{7}}{2}\tan\theta \\
&= \frac{1}{2}\ln|x^2+x+2| - \frac{5}{2}\int \frac{2}{\sqrt{7}}d\theta \\
&= \frac{1}{2}\ln|x^2+x+2| - \frac{5}{\sqrt{7}}\theta + C \\
&= \frac{1}{2}\ln|x^2+x+2| - \frac{5}{\sqrt{7}}\tan^{-1}\left(\frac{2x+1}{\sqrt{7}}\right) + C.
\end{aligned}$$

Activity 2.12 *Complete the square:*
(i) x^2+6x+4 *(ii)* $x^2-10x+18$ *(iii)* $5+4x-2x^2$ *(iv)* $4-4x-x^2$

Activity 2.13 *Determine the the following integrals:*
(i) $\int \frac{e^x}{1-e^{2x}}dx$ *(ii)* $\int \frac{1}{\sqrt{11x^2-5}}dx$ *(iii)* $\int \frac{dx}{\sqrt{3x^2+16}}$
(iv) $\int \frac{dx}{\sqrt{9-4x^2}}$ *(v)* $\int \frac{1}{x^2+2x+5}dx$ *(vi)* $\int \frac{8}{5-4x-x^2}dx$

2.6.9 integrals of the form $\int f'(x)\sqrt{a^2-[f(x)]^2}dx$, $\int f'(x)\sqrt{a^2+[f(x)]^2}dx$, $\int f'(x)\sqrt{[f(x)]^2-a^2}dx$,

We illustrate the method by solving $\int f'(x)\sqrt{a^2-[f(x)]^2}dx$. We use the substitution $f(x) = a\sin\theta \implies f'(x) = a\cos\theta d\theta$. In this way the integral becomes
$\int f'(x)\sqrt{a^2-[f(x)]^2}dx = \int \sqrt{a^2-(a\sin\theta)^2}(a\cos\theta)d\theta = a\int \sqrt{1-\sin^2\theta}(a\cos\theta)d\theta =$
$a\int \sqrt{\cos^2\theta}(a\cos\theta)d\theta = a^2\int \cos^2\theta d\theta = \frac{a^2}{2}\int(1+\cos 2\theta)d\theta$
$= \frac{a^2}{2}\left(\theta+\frac{1}{2}\sin 2\theta\right) + C = \frac{a^2}{2}(\theta+\sin\theta\cos\theta) + C$

$$= \frac{a^2}{2}\left(\sin^{-1}\left(\frac{f(x)}{a}\right) + \frac{f(x)}{a}\frac{\sqrt{a^2-[f(x)]^2}}{a}\right) + C$$
$$= \frac{a^2}{2}\sin^{-1}\left(\frac{f(x)}{a}\right) + \frac{f(x)}{2}\sqrt{a^2-[f(x)]^2} + C$$

Solutions can however be written down following standard forms form the table of integrals.

Example 2.44 *Determine* $\int \sqrt{25-x^2}dx$

Solution

$$\int \sqrt{25-x^2}dx = \int \sqrt{(5)^2+(x)^2}dx = \frac{25}{2}\sin^{-1}\frac{x}{5} + \frac{x}{2}\sqrt{25-x^2} + C$$

Example 2.45 *Find* $\int \sqrt{10-4x^2}dx$

Solution

$$\int \sqrt{10-4x^2}dx = \int \sqrt{(\sqrt{10})^2+(2x)^2}dx = \frac{10}{2}\sin^{-1}\frac{2x}{\sqrt{10}} + \frac{2x}{2}\sqrt{10-4x^2} + C$$
$$= 5\sin^{-1}\frac{2x}{\sqrt{10}} + x\sqrt{10-4x^2} + C$$

Example 2.46 *Solve the integral* $\int \sqrt{x^2-8x-9}dx$

Solution

$$\int \sqrt{x^2-8x-9}dx = \int \sqrt{(x-4)^2-25}dx$$
$$= \frac{x-4}{2}\sqrt{x^2-8x-9} - \frac{25}{2}\cosh^{-1}\left(\frac{x-4}{5}\right) + C$$

Activity 2.14 *Evaluate the integrals:*
(i) $\int \sqrt{9x^2-3}dx$ *(ii)* $\int \sqrt{x^2-4x-1}dx$ *(iii)* $\int \sqrt{6+x-x^2}dx$
(iv) $\int \sqrt{5-16x^2}dx$

2.6.10 The form $\int \frac{a\cos x + b\sin x}{A\cos x + B\sin x + C}dx$

This integral can be evaluated by writing the numerator in the form

$$p(A\cos x + B\sin x + C) + q\frac{d}{dx}(A\cos x + B\sin x + C) + r$$

where p, q, r are constants to be determined.

Example 2.47 *Find* $\displaystyle\int \frac{1+\cos x-3\sin x}{2+2\cos x-\sin x}dx.$

Solution
Let

$$1+\cos x-3\sin x \equiv p(2+2\cos x-\sin x)+q(-2\sin x-\cos x)+r.$$

In order to find the values of p, q and r, the following system of equations arise:

$$\begin{aligned} 1 &= 2p-q \\ -3 &= -p-2q \\ 1 &= 2p+r \end{aligned}$$

Solving this system simultaneously we get $p=1, \quad q=1, \quad r=-1$. Thus,

$$\begin{aligned} \int \frac{1+\cos x-3\sin x}{2+2\cos x-\sin x}dx &= \int \left(1+\frac{-2\sin x-\cos x}{2+2\cos x-\sin x}-\frac{1}{2+2\cos x-\sin x}\right)dx \\ &= x+\ln|2+2\cos x-\sin x|-\int \frac{dx}{2+2\cos x-\sin x}. \end{aligned}$$

Now, the last integral can be evaluated using the t-formula as discussed earlier on.
Let $t=\tan\frac{x}{2}, \Longrightarrow \sin x=\dfrac{2t}{1+t^2}$, $\cos x=\dfrac{1-t^2}{1+t^2}$ and $dx=\dfrac{2dt}{1+t^2}$. Hence

$$\begin{aligned} \int \frac{dx}{2+2\cos x-\sin x} &= \int \frac{1}{2+2\left(\frac{1-t^2}{1+t^2}\right)-\left(\frac{2t}{1+t^2}\right)}\cdot\frac{2dt}{1+t^2} \\ &= \int \frac{2dt}{2+2t^2+2-2t^2-2t} \\ &= \int \frac{2dt}{4-2t} \\ &= \int \frac{dt}{2-t} \\ &= -\ln|2-t|+C \\ &= -\ln\left|2-\tan\frac{x}{2}\right|+C \end{aligned}$$

Finally,

$$\int \frac{1+\cos x-3\sin x}{2+2\cos x-\sin x}dx = x+\ln|2+2\cos x-\sin x|+\ln\left|2-\tan\frac{x}{2}\right|+C$$

Activity 2.15 *Find* $\displaystyle\int \frac{2+4\cos x+7\sin x}{10-4\cos x+3\sin x}dx.$

Exercises

1. Find the integrals.
   (a) $\int \frac{1}{x^2-1}dx$ (b) $\int \frac{dx}{3x^2-x}$ (c) $\int \frac{5-x}{2x^2+x-1}dx$
   (d) $\int \frac{x^2+12x+12}{x^3-4x}dx$ (e) $\int \frac{4x^2-1}{2x(x^2+2x+1)}dx$ (f) $\int \frac{x^4}{(x-1)^3}dx$
   (g) $\int \frac{3x}{x^2-6x+9}dx$ (h) $\int \frac{x^3-2x^2+1}{x^2-3x}dx$

2. Evaluate the given integral:
   (a) $\int \sin^3 x dx$ (b) $\int \sin^2 3x dx$ (c) $\int \cos^4 x \sin^3 x dx$ (d) $\int \tan^3 x dx$
   (e) $\int \sin 2x \cos 3x dx$ (f) $\int \cos^2 4x dx$ (g) $\int \sin^2 x \cos^4 x dx$ (h) $\int \tan^3 x \sec^3 x dx$
   (i) $\int \sin 5x \sin 2x dx$ (j) $\int_0^{\frac{1}{2}} \cos(\pi x)\cos(\frac{\pi}{2}x)dx$ (k) $\int \sin^6 x dx$

3. Calculate the indefinite integral:
   (a) $\int \frac{1}{x^2+6x-8}dx$ (b) $\int \frac{1}{x^2+4x-5}dx$ (c) $\int \frac{1}{2x^2-4x-6}dx$

4. If $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x dx$, prove that for $n \geq 2$, $I_n = n\left(\frac{\pi}{2}\right)^{n-1} - n(n-1)I_{n-2}$.

5. Evaluate the following
   (a) $\int \frac{\cos x + \sin x}{\sin x - \cos x}dx$ (b) $\int \frac{2\sin x + 9\cos x}{\sin x + 2\cos x}dx$ (c) $\int \frac{\sin x}{\sin x + \cos x}dx$
   (d) $\int \frac{\cos x + 8\sin x - 2}{3\sin x + 2\cos x - 1}dx$

2.6.11 The V-formula

The result $\int_0^a f(x)dx = \int_0^a f(a-u)du$, which follows by putting $x = a-u$ in the first integral is called the V-formula.

Example 2.48 *Find* $I = \int_0^{\frac{\pi}{2}} \ln(\sin x)dx$.

Solution
Let $y = \frac{\pi}{2} - x$, $\Longrightarrow dy = -dx$ and $\sin(\frac{\pi}{2} - y) = \cos y$. Thus

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2}} \ln(\sin x)dx \\ &= -\int_{\frac{\pi}{2}}^0 \ln\sin(\frac{\pi}{2} - y)dy \end{aligned}$$

$$
\begin{aligned}
&= \int_0^{\frac{\pi}{2}} \ln\sin(\frac{\pi}{2} - y)dy \\
&= \int_0^{\frac{\pi}{2}} \ln\cos y dy \\
&= \int_0^{\frac{\pi}{2}} \ln\cos x dx \\
I + I &= \int_0^{\frac{\pi}{2}} (\ln\sin x + \ln\cos x)dx \\
2I &= \int_0^{\frac{\pi}{2}} \ln(\sin x\cos x)dx, \quad \frac{\sin 2x}{2} = \sin x\cos x \\
&= \int_0^{\pi} 2\ln\left(\frac{\sin 2x}{2}\right)dx \\
&= \int_0^{\frac{\pi}{2}} (\ln\sin 2x - \ln 2)dx \\
&= \int_0^{\frac{\pi}{2}} \ln\sin 2x dx - \frac{\pi}{2}\ln 2 \\
2I &= \frac{1}{2}\int_0^{\pi} \ln\sin v dv - \frac{\pi}{2}\ln 2, \quad 2x = v \\
&= \frac{1}{2}\left[\int_0^{\frac{\pi}{2}} \ln\sin v dv + \int_{\frac{\pi}{2}}^{\pi}\right] - \frac{\pi}{2}\ln 2 \\
&= \frac{1}{2}I + \frac{1}{2}I - \frac{\pi}{2}\ln 2 \\
\Longrightarrow I &= -\frac{\pi}{2}\ln .
\end{aligned}
$$

Example 2.49 *Prove that* $I = \int_0^{\frac{\pi}{2}} \frac{\sin^m x}{\sin^m x + \cos^m x}dx = \frac{\pi}{4}, \quad \forall \quad m.$

Solution
Let $x = \frac{\pi}{2} - y$, then $dx = -dy$ and $\sin x = \sin(\frac{\pi}{2} - y) = \cos y$, $\cos x = \cos(\frac{\pi}{2} - y) = \sin y$. Therefore we have

$$
\begin{aligned}
I &= \int_0^{\frac{\pi}{2}} \frac{\sin^m x}{\sin^m x + \cos^m x}dx \\
&= \int_0^{\frac{\pi}{2}} \frac{\cos^m y}{\cos^m y + \sin^m y}dy \\
&= \int_0^{\frac{\pi}{2}} \frac{\cos^m x}{\cos^m x + \sin^m x}dx \\
2I &= \int_0^{\frac{\pi}{2}} \frac{\sin^m x}{\sin^m x + \cos^m x}dx + \int_0^{\frac{\pi}{2}} \frac{\cos^m x}{\cos^m x + \sin^m x}dx
\end{aligned}
$$

$$\begin{aligned}
&= \int_0^{\frac{\pi}{2}} \frac{\sin^m x + \cos^m x}{\sin^m x + \cos^m x} dx \\
&= \int_0^{\frac{\pi}{2}} dx \\
&= \frac{\pi}{2} \\
\Longrightarrow I &= \frac{\pi}{4}.
\end{aligned}$$

Activity 2.16 *1. Prove that* $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \frac{\pi}{4}$.

2. Prove that $\displaystyle\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi^2}{4}$.

Unit 3
Applications of integration

3.1 What this Unit is all About

In this Unit,we discuss mean values and room-mean-square (RMS) values, how to find the area between two curves and how to find the length of a given smooth curve using integrals. We also discuss polar coordinates and the length of a curve represented in polar coordinates or by parametric equations. We conclude the unit by discussing volumes of solids of revolution.

By the end of this unit, you should be able to:

1. Calculate mean values and root-mean-square (RMS)values.
2. Find the area between two curves using integrals.
3. Find the length of a given smooth curve.
4. Convert equations from cartesian form to polar form and vice-versa.
5. Sketch polar graphs.
6. Find the length of a curve represented in polar form.
7. Find volumes of solids of revolution.

Introduction
Integration can be applied to many real life situations. Many things we need to know and that are useful in everyday life can be calculated using integrals. Examples of such things are areas between curves, the volumes and surface areas of solids, the length of curves, the forces against floodgates and many more. We want to explore some of these situations.

As we will still be integrating, the reader is reminded to recall the various techniques of integration from unit 2.

3.2 Mean Values and RMS values

3.2.1 Mean values

Mean is a term which refers to average. Average values of quantities such as current, voltage, force and other engineering quantities are important in everyday life. This section is concerned with calculating the average value of a function.

Figure 3.1

Let the function $y = f(x)$ be defined on $[a, b]$ as in the figure. Suppose M is the mean value of this function between a and b, then M is such that
$M(b-a) =$ shaded area A. This gives
$M = \dfrac{\text{shaded area A}}{b-a} = \dfrac{\int_a^b f(x)dx}{b-a} = \dfrac{1}{b-a}\displaystyle\int_a^b f(x)dx$. We denote M by $\bar{y}$.

Example 3.1 *Find the mean value of $y = x^2 + 3$ between $x = 1$ and $x = 3$.*

Solution
$\bar{y} = \dfrac{1}{b-a}\displaystyle\int_a^b y dx = \dfrac{1}{3-1}\int_1^3 (x^2+3)dx = \dfrac{1}{2}\left[\dfrac{x^3}{3} + 3x\right]_1^3 = \dfrac{1}{2}\left(18 - \dfrac{10}{3}\right) = \dfrac{44}{6} = 3,33$

Example 3.2 *Evaluate the mean value of the voltage, v, over the interval $0 \le t \le 10$ where*

$$v = 10(1 - e^{-0.1t})$$

Solution
$\bar{v} = \dfrac{1}{10}\displaystyle\int_0^{10} 10(1-e^{-0.1t})dt = \int_0^{10}(1-e^{-0.1t})dt = \left[t + 10e^{-0.1t}\right]_0^{10} = \dfrac{10}{e}$

3.2.2 Root-mean-square values (RMS)

The root-mean-square, RMS, value of $y = f(x)$ between $x = a$ and $x = b$ is given by RMS $= \sqrt{\text{mean value of } y^2}$, that is $(\text{RMS})^2 =$ mean value of y^2. Thus $(\text{RMS})^2 = \dfrac{1}{b-a}\displaystyle\int_a^b y^2 dx$ and therefore RMS $= \sqrt{\dfrac{1}{b-a}\displaystyle\int_a^b y^2 dx}$.

Example 3.3 *Calculate the RMS value of $y = 20\sin 20\pi t$ between $t = 0$ and $t = \dfrac{1}{10}$.*

Solution
$(\text{RMS})^2 = \dfrac{1}{\frac{1}{10}} \displaystyle\int_0^{\frac{1}{10}} 400 \sin^2 20\pi t dt = 4000 \int_0^{\frac{1}{10}} \sin^2 20\pi t dt$
$= 2000 \displaystyle\int_0^{\frac{1}{10}} (1 - \cos 40\pi t) dt = 2000 \left[t - \frac{1}{40\pi} \sin 40\pi t \right]_0^{\frac{1}{10}} = 2000 \cdot \frac{1}{10} = 200.$ Thus RMS$=\sqrt{200}$.

Activity 3.1 *1. Find the mean value of each of the following:*
(i) $y = 3\sin 5t + 2\cos 3t$ *between* $t = 0$ *and* $t = \pi$
(ii) $i = I\cos t$ *between* $t = 0$ *and* $t = 2\pi$, I *is a constant.*
(iii) $y = 3x^2 + 4x + 1$ *between* $x = -1$ *and* $x = 2$.

2. Calculate the RMS value of each of the following:
(i) $v = 10(1 - e^{-0.1t})$ *over the interval* $0 \le t \le 10$
(ii) $i = I\cos t$ *over* $0 \le t \le 10$, I *is a constant*
(iii) $y = 400\cos 200\pi t$ *between* $t = 0$ *and* $t = \dfrac{1}{100}$.
(iv) $y = 10\cosh \frac{x}{10}$ *between* $x = -1$ *and* $x = 2$.

3.3 Area between curves

In this section, we show how to find the areas of regions in the coordinate plane by integrating the functions that define the region's boundaries.

3.3.1 The basic formula

Suppose we want to find the area of a region that is bounded above by the curve $y = f(x)$, below by $y = g(x)$, and on the left and right by the lines $x = a$ and $x = b$ (See figure). The region might by chance have a shape whose area we could find with geometry, but if f and g are arbitrary continuous functions we usually have to find the area with an integral.

Figure 3.2

To see what the integral should be, we first approximate the region with n vertical rectangles based on a partition

$$P = \{x_0, x_1, \cdots, x_n\}$$

of $[a, b]$. The area of the kth rectangle is

$$\Delta A_k = \text{height} \times \text{width} = [f(c_k) - g(c_k)]\Delta x_k$$

We then approximate the area of the region by adding the areas of the n rectangles:

$$A = \sum_{k=1}^{n} \Delta A_k = \sum_{k=1}^{n} [f(c_k) - g(c_k)]\Delta x_k$$

Now, as $\|P\| \longrightarrow 0$ the sums on the right approach the limit

$$\int_a^b [f(c_k) - g(c_k)]dx$$

because f and g are continuous. We take the area to be the value of this integral. That is,

$$A = \lim_{\|P\| \longrightarrow 0} \sum_{k=1}^{n} [f(c_k) - g(c_k)]\Delta x_k = \int_a^b [f(c_k) - g(c_k)]dx$$

Definition 3.1 *If f and g are continuous functions with $f(x) \geq g(x)$ throughout $[a, b]$, then the* **area** *of the region between the curves $y = f(x)$ and $y = g(x)$ from a to b is given by*

$$A = \int_a^b [f(x) - g(x)]dx$$

General Procedure to find the area between two curves

1. Graph the curves and draw a representative rectangle. This reveals which curve is the upper curve and which curve is the lower curve. It also helps to find the limits of integration if you do not already know them.
2. Find the limits of integration.
3. Write a formula for $f(x) - g(x)$. Simplify it if you can.
4. Integrate $[f(x) - g(x)]$ from a to b. The number you get is the area.

Example 3.4 *Find the area between $y = \cos x$ and $y = -\sin x$ from $x = 0$ to $x = \frac{\pi}{2}$.*

Solution

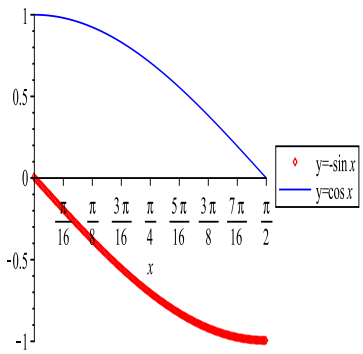

Figure 3.3

Here $f(x) = \cos x$ and $g(x) = -\sin x$ and $f(x) - g(x) = \cos x + \sin x$.
$a = 0, \quad b = \frac{\pi}{2}$. Hence

$$A = \int_0^{\frac{\pi}{2}} (\cos x + \sin x)dx = [\sin x - \cos x]_0^{\frac{\pi}{2}} = (1 - 0) - (0 - 1) = 2$$

The area of the region is 2.
When a region is determined by curves that intersect, the intersection points give the limits of integration.

Example 3.5 *Find the area of the region enclosed by the parabola $y = 2 - x^2$ and the line $y = -x$.*

Solution

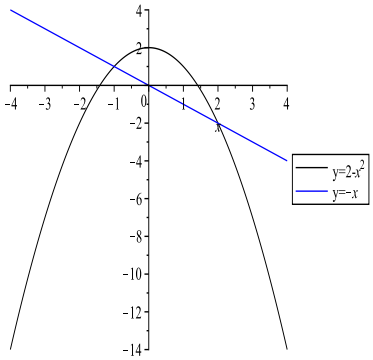

Figure 3.4

The upper curve is $f(x) = 2 - x^2$ and the lower curve is $g(x) = -x$. The x-coordinates of the intersection points are the limits of integration.
We find the limits of integration with respect to x by solving the equations $y = 2 - x^2$ and $y = -x$ simultaneously for x:

$$\begin{aligned} 2 - x^2 &= -x \\ x^2 - x - 2 &= 0 \\ (x+1)(x-2) & \\ &= 0 \end{aligned}$$

This gives $x = -1, \quad x = 2$. Thus the region runs from $x = -1$ to $x = 2$. Hence $a = -1, \quad b = 2$. $f(x) - g(x) = (2 - x^2) - (-x) = 2 + x - x^2$

$$\begin{aligned}
\int_a^b [f(x) - g(x)]dx &= \int_{-1}^{2} (2 + x - x^2)dx \\
&= \left[2x + \frac{x^2}{2} - \frac{x^3}{3}\right]_{-1}^{2} \\
&= (4 + \frac{4}{2} - \frac{8}{3}) - (-2 + \frac{1}{2} + \frac{1}{3}) \\
&= \frac{9}{2}.
\end{aligned}$$

Activity 3.2 *Find the area of the region bounded by* $y = 2x^2$ *and* $y = x^4 - 2x^2$. *Ans:* $\frac{64}{15}$

If a region's bounding curves are described by functions of y, the approximating rectangles are horizontal and the basic formula is in terms of y instead of x.

Example 3.6 *Find the area of the region in the first quadrant that is bounded above by* $y = \sqrt{x}$ *and below by the* x*-axis and the line* $y = x - 2$.

Solution

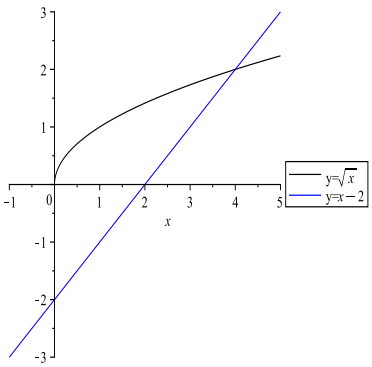

Figure 3.5

The region's right hand boundary is the line $x = y + 2$, so $f(y) = y + 2$. The left hand boundary is the curve $x = y^2$, so $g(y) = y^2$. The lower limit of integration is $y = 0$. We find the upper limit of integration by solving the equations $x = y + 2$ and $x = y^2$ simultaneously for y:

$$\begin{aligned}
y + 2 &= y^2 \\
y^2 - y - 2 &= 0 \\
(y + 1)(y - 2) &= 0
\end{aligned}$$

This gives $y = -1,\ \ y = 2$ and the upper limit of integration is $b = 2$. $f(y) - g(y) = 2 + y - y^2$. Hence

$$\begin{aligned}\int_a^b [f(y) - g(y)]dy &= \int_0^2 [2 + y - y^2]dy \\ &= \left[2y + \frac{y^2}{2} - \frac{y^3}{3}\right]_0^2 \\ &= 4 + \frac{4}{2} - \frac{8}{3} \\ &= \frac{10}{3}.\end{aligned}$$

The area of the region is $\frac{10}{3}$.

Activity 3.3 *Calculate the area of the region bounded by the curves $x = y^2$ and $x - y = 2$.*

Exercises

Sketch the region bounded by the curves and find its area.

1. $4x = 4y - y^2,\ 4x - y = 0$
2. $x + y^2 - 4 = 0,\ x + y = 2$
3. $x = y^2,\ x = 3 - 2y^2$
4. $x + y = 2y^2,\ y = x^3$
5. $x + y - y^3 = 0,\ x - y + y^2 = 0$
6. $8x = y^3,\ 8x = 2y^3 + y^2 - 2y$
7. $y = \cos x,\ y = \sec^2 x,\ x \in [\frac{-\pi}{4}, \frac{\pi}{4}]$
8. $y = \sin^2 x,\ y = \tan^2 x,\ x \in [\frac{-\pi}{4}, \frac{\pi}{4}]$

3.4 Curve length

3.4.1 The basic length formula

Suppose the curve whose length we want to find is the graph of the function $y = f(x)$ from $x = a$ to $x = b$. We partition $[a, b]$ as before and connect the corresponding points on the curve with the line segments (see figure). The line segments taken together form a polygonal path that approximates the curve.

Figure 3.6

If we can find a formula for the sum of the lengths of these line segments, we will have a formula for approximating the length of the curve.
The length of a typical line segment PQ is

$$\sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}.$$

The length of the curve is therefore approximated by the sum

$$\sum_{k=1}^{n} \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}.$$

Now, suppose that f has a derivative that is continuous at every point of $[a, b]$. Then, by the Mean Value Theorem, there is a point $(c_k, f(c_k))$ on the curve between PQ where the tangent is parallel to the segment PQ. At this point

$$f'(c_k) = \frac{\Delta y_k)}{\Delta x_k)}, \Longrightarrow \Delta y_k = f'(c_k)\Delta x_k.$$

We now substitute for Δy_k into the formula to get

$$\sum_{k=1}^{n} \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2} = \sum_{k=1}^{n} \sqrt{(\Delta x_k)^2 + (f'(c_k))\Delta x_k)^2} = \sum_{k=1}^{n} \sqrt{1 + (f'(c_k))^2}\Delta x_k.$$

Because $\sqrt{1 + (f'(c_k))^2}\Delta x_k$ is continuous on $[a, b]$, the limit of the sums on the right as the norm of the partition goes to zero is $\displaystyle\int_a^b \sqrt{1 + (f'(x))^2}dx$.

Definition 3.2 *If f is smooth on $[a, b]$, the length of the curve $y = f(x)$ from a to b is the number*

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx.$$

Example 3.7 *Find the length of the curve*

$$y = \frac{4\sqrt{2}}{3}x^{\frac{3}{2}} - 1, \quad 0 \leq x \leq 1.$$

Solution
Here $\dfrac{dy}{dx} = 2\sqrt{2}x^{\frac{1}{2}}, \implies \left(\dfrac{dy}{dx}\right)^2 = (2\sqrt{2}x^{\frac{1}{2}})^2 = 8x$. So the length of the curve from $x = 0$ to $x = 1$ is

$$L = \int_0^1 \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = \int_0^1 \sqrt{1+8x}dx = \frac{2}{3} \cdot \frac{1}{8}(1+8x)^{\frac{3}{2}}]_0^1 = \frac{13}{6}.$$

At a point on a curve where $\dfrac{dy}{dx}$ fails to exist, $\dfrac{dx}{dy}$ may exist and we may be able to find the curve's length by expressing x as a function of y and applying the following analogue formula:
Formula for the length of a smooth curve $x = g(y)$ **from** $y = c$ **to** $y = d$.

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$$

Example 3.8 *Find the length of the curve* $y = \left(\frac{x}{2}\right)^{\frac{2}{3}}$ *from* $x = 0$ *to* $x = 2$.

Solution
The derivative

$$\frac{dy}{dx} = \frac{2}{3}\left(\frac{x}{2}\right)^{\frac{-1}{3}} \cdot \frac{1}{2} = \frac{1}{3}\left(\frac{2}{x}\right)^{\frac{1}{3}}$$

is not defined at $x = 0$, thus we need to adopt the alternative approach to solve this problem.
We rewrite the equation expressing x in terms of y:

$$y = \left(\frac{x}{2}\right)^{\frac{2}{3}} \quad \implies x = 2y^{\frac{3}{2}}.$$

From this we see that the curve whose length we want is the graph of $x = 2y^{\frac{3}{2}}$ from $y = 0$ to $y = 1$. Now

$$\frac{dx}{dy} = 3y^{\frac{1}{2}}$$

This derivative is continuous on $[0, 1]$. Thus

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy = \int_0^1 \sqrt{1+9y}dy = \frac{1}{9}\cdot\frac{2}{3}(1+9y)^{\frac{3}{2}}]_0^1 = \frac{2}{27}(10\sqrt{10}-1) \approx 2.27.$$

Exercises

1. Find the length of the curves.

   (a) $y = \frac{1}{3}(x^2+2)^{\frac{3}{2}}$ from $x = 0$ to $x = 3$.

   (b) $y = x^{\frac{3}{2}}$ from $x = 0$ to $x = 4$.

   (c) $x = \frac{y^3}{3} + \frac{1}{4y}$ from $y = 1$ to $y = 3$.

   (d) $x = \frac{y^{\frac{3}{2}}}{3} - y^{\frac{1}{2}}$ from $y = 1$ to $y = 9$.

2. Find a curve through the point $(0, 1)$ whose length from $x = 0$ to $x = 1$ is

$$L = \int_0^1 \sqrt{1+e^{2x}}dx.$$

3. Find a curve through the point $(1, 0)$ whose length from $x = 1$ to $x = 2$ is

$$L = \int_1^2 \sqrt{1+\frac{1}{x^2}}dx.$$

3.5 Polar coordinates

In this section we study polar coordinates and their relation to cartesian coordinates. Our ultimate aim will be to find the length of a curve represented in polar form. While a point in the plane has just one pair of cartesian coordinates, it has infinitely many pairs of polar coordinates. As we will discover, this has interesting consequences for graphing.

3.5.1 Definition of polar coordinates

Refer to the figure. We fix an **origin** O, the pole and an **initial ray** from O. Then each point P can be located by assigning to it a **polar coordinate pair** (r, θ) where r is the directed distance from O to P and θ is the directed angle from the initial ray to ray OP.
Now $r = \sqrt{x^2+y^2}$ and $\frac{x}{r} = \cos\theta$ and $\frac{y}{r} = \sin\theta$ or $x = r\cos\theta, \quad y = r\sin\theta$.
As in trigonometry, θ is positive when measured anticlockwise and negative when measured clockwise. The angle associated with a given point is not unique. For instance, the point 2 units from the origin along the ray $\theta = \frac{\pi}{6}$ has polar coordinates $(2, \frac{\pi}{6})$ as well as $(2, \frac{-11\pi}{6})$. In the same spirit the point $(2, \frac{7\pi}{6})$ is the same as $(-2, \frac{\pi}{6})$.

Example 3.9 *The equation $r = 1$ is an equation for the circle of radius 1 centred at the origin. So is the equation $r = -1$.*

An interesting exercise is to rewrite an equation given in cartesian form in polar form and vice versa. We achieve this by using the transformations

$$x = r\cos\theta$$

and

$$y = r\sin\theta$$

as stated earlier on. We give some examples in the table below.

Polar equation	Cartesian equivalent
$r\cos\theta = 2$	$x = 2$
$r^2\cos\theta\sin\theta = 4$	$xy = 4$
$r^2\cos^2\theta - r^2\sin^2\theta = 1$	$x^2 - y^2 = 1$
$r = 1 + 2r\cos\theta$	$y^2 - 3x^2 - 4x - 1 = 0$
$r = 1 - \cos\theta$	$x^4 + y^4 + 2x^2y^2 + 2x^3 + 2xy^2 - y^2 = 0$

Activity 3.4 *Verify the equivalence of the equations in the table above.*

3.5.2 Polar graphs

The reader is reminded that it is not our intention in this module to explore polar graphs in detail. We include this section for purposes of completeness only. Interested readers are referred to the references listed at the end of the module for further explorations as regards this one section.
We, by the use of examples, describe techniques for graphing equations in polar coordinates.

Example 3.10 *Graph the curve $r = 1 - \cos\theta$*

Solution
The curve is symmetric about the x-axis because
(r, θ) on the graph

$$\begin{aligned} \Rrightarrow r &= 1 - \cos\theta \\ \Rrightarrow r &= 1 - \cos(-\theta). \end{aligned}$$

$\Rrightarrow (r, -\theta)$ is on the graph.
As θ increases from 0 to π, $\cos\theta$ decreases from 1 to -1, and $r = 1 - \cos\theta$ increases from a minimum value of 0 to a maximum value of 2. As θ continues on from π to 2π, $\cos\theta$ increases from -1 back to 1 and r decreases from 2 back to 0. The curve starts to repeat when $\theta = 2\pi$ because cosine has period 2π. The curve leaves the origin with slope $\tan 0 = 0$ and returns to the origin with slope

$\tan 2\pi = 0$.
We make a table of values from $\theta = 0$ to $\theta = \pi$, plot the points, draw a smooth curve through them with a horizontal tangent at the origin, and reflect the curve across the x-axis to complete the graph.

θ	$r = 1 - \cos\theta$
0	0
$\frac{\pi}{3}$	$\frac{1}{2}$
$\frac{\pi}{2}$	1
$\frac{2\pi}{3}$	$\frac{3}{2}$
π	2

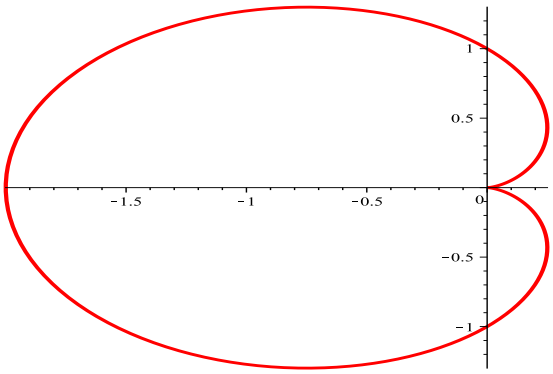

Figure 3.7

Example 3.11 *Graph the curve* $r^2 = 4\cos\theta$.

Solution
Although $\cos\theta$ has period 2π, the equation $r^2 = 4\cos\theta$ requires that $\cos\theta \geq 0$, so we get the entire graph by running θ from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$. The curve is symmetric about the x-axis because
(r, θ) on the graph

$$\begin{aligned} \Rrightarrow r^2 &= 4\cos\theta \\ \Rrightarrow r^2 &= 4\cos(-\theta). \end{aligned}$$

$\Rrightarrow (r, -\theta)$ is on the graph.
The curve is also symmetric about the origin because
(r, θ) on the graph

$$\begin{aligned} \Rrightarrow r^2 &= 4\cos\theta \\ \Rrightarrow (-r)^2 &= 4\cos\theta. \end{aligned}$$

$\Rrightarrow (-r, \theta)$ is on the graph.
Together these two symmetries imply symmetry about the y-axis.

θ	$\cos\theta$	$r = \pm 2\sqrt{\cos\theta}$
0	1	± 2
$\pm\frac{\pi}{6}$	$\frac{\sqrt{3}}{2}$	± 1.9
$\pm\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}$	± 1.7
$\pm\frac{\pi}{3}$	$\frac{1}{2}$	± 1.4
$\pm\frac{\pi}{2}$	0	0

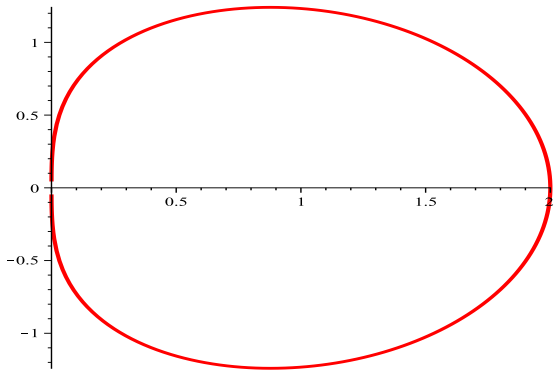

Figure 3.8

Activity 3.5 *Graph the curve* $r = 1 + \cos\frac{\theta}{2}$.

3.5.3 Polar coordinates and length of a curve

We can obtain a polar coordinate formula for the length of a curve $r = f(\theta)$, $\alpha \le \theta \le \beta$, by parametrizing the curve as

$$x = r\cos\theta = f(\theta)\cos\theta, \quad y = r\sin\theta = f(\theta)\sin\theta, \quad \alpha \le \theta \le \beta.$$

The parametric length formula then gives the length as

$$L = \int_\alpha^\beta \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}\, d\theta.$$

After substituting for x and y, this equation becomes

$$L = \int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta.$$

Theorem 3.1 *If $r = f(\theta)$ has a continuous first derivative for $\alpha \le \theta \le \beta$ and if the point $P(r, \theta)$ traces the curve $r = f(\theta)$ exactly once as θ runs from α to β, then the length of the curve is*

$$L = \int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta.$$

Example 3.12 *Find the length of the curve* $r = 1 - \cos\theta$.

Solution
The sketch of the curve reveals the limits of integration (see the sketch in the previous section). The point $P(r, \theta)$ traces the curve once, counterclockwise as θ runs from 0 to 2π, so these are the values we take for α and β. With

$$r = 1 - \cos\theta, \quad \frac{dr}{d\theta} = \sin\theta$$

and

$$\begin{aligned} r^2 + \left(\frac{dr}{d\theta}\right)^2 &= (1 - \cos\theta)^2 + (\sin\theta)^2 \\ &= 1 - 2\cos\theta + \cos^2\theta + \sin^2\theta \\ &= 2 - 2\cos\theta. \end{aligned}$$

Thus

$$\begin{aligned} L &= \int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta \\ &= \int_0^{2\pi} \sqrt{2 - 2\cos\theta}\, d\theta \\ &= \int_0^{2\pi} \sqrt{4\sin^2\frac{\theta}{2}}\, d\theta \\ &= \int_0^{2\pi} 2\left|\sin\frac{\theta}{2}\right| d\theta \\ &= \int_0^{2\pi} 2\sin\frac{\theta}{2}\, d\theta, \quad \sin\frac{\theta}{2} \geq 0 \ \text{ for } \ 0 \leq \theta \leq 2\pi \\ &= \left[-4\cos\frac{\theta}{2}\right]_0^{2\pi} \\ &= 4 + 4 \\ &= 8. \end{aligned}$$

Exercises
Find the lengths of the curves

1. The spiral $r = \theta^2, \quad 0 \leq \theta \leq \sqrt{5}$
2. The spiral $r = \dfrac{e^\theta}{\sqrt{2}}, \quad 0 \leq \theta \leq \pi$
3. The cardioid $r = 1 + \cos\theta$
4. The curve $r = a\sin^2\left(\frac{\theta}{2}\right), \quad 0 \leq \theta \leq \pi, \quad a > 0$
5. The curve $r = \sqrt{1 + \sin 2\theta}, \quad 0 \leq \theta \leq \pi\sqrt{2}$

3.6 Volumes of Revolution

If a plane figure bounded by the curve $y = f(x)$, the x-axis and the ordinates at $x = x_1$ and $x = x_2$ is rotated through a complete revolution about the x-axis, a solid of revolution is generated. The nature or type of the solid generated is dependent on the shape of the plane figure rotated. Consider an element of volume obtained by cutting a strip of thickness Δx perpendicular to the x-axis at distance x from the origin. The strip will be a cylinder of radius $y = f(x)$ and height Δx.

Figure 3.9

The element of volume is thus given by

$$\Delta V = \pi y^2 \Delta x.$$

Summing the volumes of all such cylinder elements from x_1 to x_2 we obtain

$$V = \sum \pi y^2 \Delta x.$$

Taking the limit as $\Delta x \longrightarrow 0$, we obtain

$$V = \pi \int_{x_1}^{x_2} y^2 dx.$$

If, on the other hand, the same area is rotated about the y-axis, the solid formed will be a shell and it's volume is given by

$$V = 2\pi \int_{x_1}^{x_2} xy dx.$$

In general, the volume of a solid generated by revolving the area bounded by the graph of $y = f(x)$ from above and the graph of $y = g(x)$ from below from $x = x_1$ to $x = x_2$ about the x-axis is given by

$$V = \pi \int_{x_1}^{x_2} \left((f(x))^2 - (g(x))^2\right) dx$$

If this same area is rotated about the y-axis, the volume will be given by

$$V = 2\pi \int_{x_1}^{x_2} x\left(f(x) - g(x)\right) dx.$$

If, instead, we rotate the area bounded by the curve $x = f(y)$ and the ordinates $y = y_1$ and $y = y_2$ about the y-axis and the x-axis respectively, the corresponding volume formulae are deduced to be

$$V = \pi \int_{y_1}^{y_2} x^2 dy$$

and

$$V = 2\pi \int_{y_1}^{y_2} xy dy$$

respectively. The corresponding more general formulae involving 2 curves are

$$V = \pi \int_{y_1}^{y_2} \left((f(y))^2 - (g(y))^2\right) dy$$

and

$$V = 2\pi \int_{y_1}^{y_2} y\left(f(y) - g(y)\right) dy.$$

Example 3.13 *Find the volume of the solid generated when the plane figure bounded by $y = \cos 2x$, the x-axis and the ordinates $x = 0$ and $x = \dfrac{\pi}{4}$ is rotated about the x-axis.*

Solution

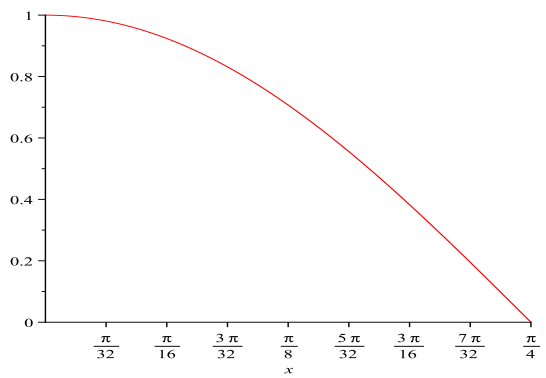

Figure 3.10

$$V = \pi \int_{x_1}^{x_2} y^2 dx = \pi \int_0^{\frac{\pi}{4}} \cos^2 2x dx = \frac{\pi}{2} \int_0^{\frac{\pi}{4}} (1 + \cos 4x) dx$$
$$= \frac{\pi}{2}\left[x + \frac{1}{4}\sin 4x\right]_0^{\frac{\pi}{4}} = \frac{\pi}{2} \cdot \frac{\pi}{4} = \frac{\pi^2}{8}.$$

Example 3.14 *Find the volume of the solid obtained by rotating about the y-axis the region bounded by $y = 2x^2 - x^3$ and $y = 0$.*

Solution

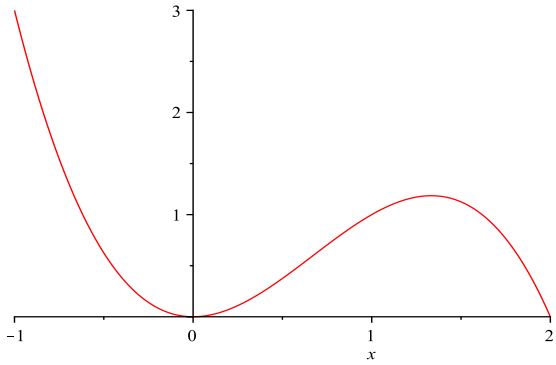

Figure 3.11

$$V = 2\pi \int_{x_1}^{x_2} xydx = 2\pi \int_0^2 x(2x^2 - x^3)dx = 2\pi \int_0^2 (2x^3 - x^4)dx = 2\pi \left[\frac{1}{2}x^4 - \frac{1}{5}x^5\right]_0^2 = 2\pi\left(8 - \frac{32}{5}\right) = \frac{16}{5}\pi.$$

Example 3.15 *Find the volume of the solid obtained by rotating about the y-axis the region between $y = x$ and $y = x^2$.*

Solution

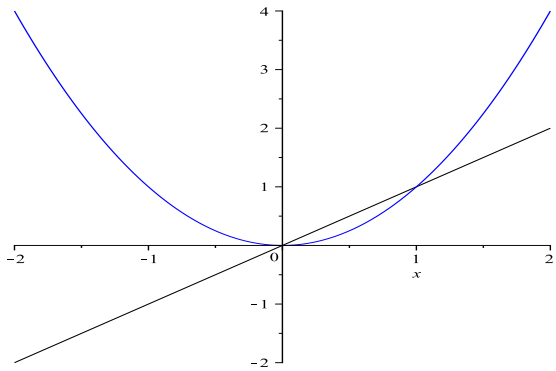

Figure 3.12

$$V = 2\pi \int_{x_1}^{x_2} x\left(f(x) - g(x)\right)dx = 2\pi \int_0^1 x(x - x^2)dx = 2\pi \int_0^1 (x^2 - x^3)dx = 2\pi \left[\frac{x^3}{3} - \frac{x^4}{4}\right]_0^1 = \frac{\pi}{6}.$$

Activity 3.6 *1. Find the volume of the solid obtained by rotating the region bounded by the given curves about the specified axis. Sketch the region, the solid, and a typical disk/strip or washer.*

(a) $y = 2 - \frac{1}{2}x,\ y = 0,\ x = 1,\ x = 2$: *about the x-axis.*

(b) $y = 1 - x^2,\ y = 0$; *about the x-axis.*

(c) $x = 2\sqrt{y},\ x = 0,\ y = 9$; *about the y-axis*

(d) $y = \ln x,\ y = 1,\ y = 2,\ x = 0$; *about the y-axis*

(e) $y = x^3,\ y = x,\ x \geq 0$; *about the x-axis*

(f) $y^2 = x,\ x = 2y$; *about the y-axis*

2. Find the volume generated by rotating the region bounded by the given curves about the y-axis. Sketch the region and a typical shell.

(a) $y = \frac{1}{x},\ y = 0,\ x = 1,\ x = 2$

(b) $y = x^2,\ y = 0,\ x = 1$

(c) $y = e^{-x^2},\ y = 0,\ x = 0,\ x = 1$

(d) $y = 3 + 2x - x^2,\ x + y = 3$

Unit 4
Ordinary differential equations

4.1 What this Unit is all About

In this Unit, we discuss first order separable equations as well as first order linear equations.We also discuss equations reducible to variable separable type and homogeneous ordinary differential equations. We conclude the unit by discussing second order linear constant coefficient homogeneous ordinary differential equations. We will be using the expressions ordinary differential equation, differential equations, equations and o.d.e's interchangeably.

By the end of this unit, you should be able to:

1. Explain what a solution of a differential equation is.
2. Classify differential equations according to type, order and linearity.
3. Solve separable variable first order o.d.e's.
4. Solve first order linear o.d.e's.
5. Transform to variable separable form.
6. Determine whether an o.d.e is homogeneous or not.
7. Solve homogeneous o.d.e's
8. solve second order linear constant coefficient homogeneous ordinary differential equations

Introduction

You have already acquired knowledge of various methods of integration. You are urged to review these methods and get ready to use them to solve simple ordinary differential equations.

In this unit, we introduce the standard form of a first order differential equation:

$$P(x,y)dx + Q(x,y)dy = 0 \tag{4.1}$$

In the domain in which $Q(x,y)$ is not zero, equation (1.1) can equivalently be written as;

$$\frac{dy}{dx} = f(x,y) \tag{4.2}$$

where

$$f(x,y) = -\frac{P(x,y)}{Q(x,y)} \tag{4.3}$$

We will discuss some methods which allow us to solve some of the equations of this form and those equations which reduce to the first order form.

4.2 Basic concepts and definitions

Definition 4.1 *Any equation of the form:*

$$F(x, y, y', y'', \cdots, y^{(n)}) = 0,$$

where F is a real valued function of the $n+1$ variables $x, y, y', y'', \cdots, y^{(n)}$ is called an nth **order ordinary differential equation** *where x is the independent variable and the variables $y, y', y'', \cdots, y^{(n)}$ depend on x.*
Any equation of the form:

$$F(x, y, y') = 0,$$

where F is a real valued function of three variables x, y and y' is called a **first order ordinary differential equation** *where x is the independent variable and y and y' depend on x.*

The following are examples of ordinary differential equations:

Example 4.1 *1.* $\dfrac{dy}{dt} = ky$, *k is a constant.*

2. $\dfrac{dy}{dx} = \dfrac{xy - y^2}{x^2 - xy}, \quad x^2 - xy \neq 0$

3. $x\sqrt{1 + (y')^2} = 0$

4. $x^4 \dfrac{dy}{dx} + \dfrac{d^5 y}{dx^5} = y + x^7$

One can rewrite a first order equation in the form:

$$F\left(x, y\frac{dy}{dx}\right) = 0,$$

which is often rearranged into the form:

$$\frac{dy}{dx} = f(x, y).$$

Definition 4.2 *The* **order** *of a differential equation is that of the highest derivative occurring in the differential equation.*

Example 4.2

$$y' + ay^2 = bx^2$$

is a first order differential equation, while

$$(y''')^2 - 2xy''' + y(y')^3 = 0$$

is a third order equation.

Definition 4.3 *A* **solution** *of a first order differential equation is a function $y = \phi(x)$, with first derivatives defined such that when substituted into the o.d.e it yields an identity on the domain of definition of $\phi(x)$.*
The **general solution** *of a first order o.d.e is one with the arbitrary constant that describes all possible solutions of the equation, while a* **particular solution** *is determined by certain initial conditions, and is free of any arbitrary constants.*

Exercises

1. Determine the order of each of the following o.d.e's.
   (a) $x^4\dfrac{dy}{dx} + \dfrac{d^5y}{dx^5} = y + x^7$
   (b) $\dfrac{d^3y}{dx^3} + x\dfrac{dy}{dx} - y = x^6$
   (c) $[x + (y')^2]^{\frac{3}{2}} = y''$
   (d) $y''' + \cos y'' + x^2y' = x^3$
   (e) $(x')^2x''' = x^4x'' + t^5x'$
2. Check whether the given function $y = \phi(x)$ is a solution of the given o.d.e.
   (a) $\phi(x) = \dfrac{1-x}{1+x}$; $(x^2+1)y' + y^2 + 1 = 0$
   (b) $\phi(x) = e^{3x}$; $\dfrac{d^3y}{dx^3} - 9\dfrac{d^2y}{dx^2} = 0$
   (c) $\phi(x) = xe^{3x}$; $\dfrac{d^2y}{dx^2} - 9\dfrac{dy}{dx} = 6e^{3x}$
   (d) $\phi(x) = \ln(-x),\ x < 0$; $xy' = 1$

4.3 Types of first order differential equations

4.3.1 Equations with separable variables

We say that the first order differential equation of the form (3.2) is one of **separable variables** if the function $f(x, y)$ can be expressed as a product of a function

which solely depends on the variable x, $g(x)$, and another function which solely depends on the variable y, $h(y)$, such that the differential equation then assumes the form:

$$\frac{dy}{dx} = g(x) \cdot h(y) \tag{4.4}$$

$$\frac{dy}{h(y)} = g(x)dx, \quad h(y) \neq 0 \tag{4.5}$$

If the differential equation can be presented in the form (3.1) where

$$P(x, y) = P_1(x) \cdot P_2(y)$$

and

$$Q(x, y) = Q_1(x) \cdot Q_2(y)$$

then we also consider such a differential equation to be of variables separable type. At all points (x, y) where the functions $P(x, y)$ and $Q(x, y)$ are defined, the equation can still be written in the form

$$M(x)dx + N(y)dy = 0 \tag{4.6}$$

with

$$M(x) = \frac{P_1(x)}{Q_1(x)}, \quad N(y) = \frac{Q_2(y)}{P_2(y)}, \quad Q_1(x) \neq 0, \; P_2(y) \neq 0$$

The variables in equations (3.5) and (3.6) are said to be separable where (3.5) is the separated form, hence the name **separable variables**. Now, suppose that the function $g(x)$ is continuous in an interval I_1 and that the function $h(y)$ is also continuous and never zero in the interval I_2. Then the solution of (3.5) is given by

$$\int \frac{dy}{h(y)} = \int g(x)dx + C$$

where C is an arbitrary constant. Note that it is not necessary to introduce a constant of integration for the first integral since this will be combined with the one from the second integral to produce one constant C.

Let $H(y) = \displaystyle\int \frac{dy}{h(y)}$ and let $G(x) = \displaystyle\int g(x)dx$, then

$$H(y) = G(x) + C \tag{4.7}$$

is the general solution of (3.5). The differential equation of the form

$$\frac{dy}{dx} = g(x) \tag{4.8}$$

is a particular case of $\dfrac{dy}{dx} = g(x)h(y)$ where $h(y) = 1$ and g is continuous in the interval I. Similarly if $g(x) = 1$ in $\dfrac{dy}{dx} = g(x)h(y)$, we obtain

$$\frac{dy}{dx} = h(y) \tag{4.9}$$

The class of equations (3.8) and (3.9) is called **autonomous differential equations**. Autonomous equations are such that the right side of the equations does not depend explicitly on the independent variable x. In Physics, the independent variable is usually the time component t.

Example 4.3 *Show that the equation* $\dfrac{dy}{dx} = \dfrac{x^2}{1-y^2}$ *is separable and then find an equation for the integral curves.*

Solution
If we rewrite the equation as

$$-x^2 + (1-y^2)\frac{dy}{dx} = 0,$$

then it has the form

$$M(x) + N(y)\frac{dy}{dx} = 0$$

and therefore separable. Solving:

$$\begin{aligned}(1-y^2)dy &= x^2dx \\ y - \frac{y^3}{3} &= \frac{x^3}{3} + C\end{aligned}$$

Example 4.4 *Show that the following differential equation is separable and find a solution that satisfies the given initial condition.*

$$y' = \frac{xy-y}{1+y}, \quad y(2) = 1$$

Solution

$$\begin{aligned}y'(y+1) &= xy - y \\ y'(y+1) &= y(x-1) \\ 1 - x + y'\left(1 + \frac{1}{y}\right) &= 0 \\ (1-x)dx + \left(1 + \frac{1}{y}\right)dy &= 0\end{aligned}$$

$$\int (1-x)dx + \int \left(1 + \frac{1}{y}\right)dy = C$$

$$x - \frac{1}{2}x^2 + y + \ln|y| = C$$

Now, $y(2) = 1, \Longrightarrow 2 - 2 + 1 + 0 = C, \Longrightarrow C = 1$ Hence

$$x - \frac{1}{2}x^2 + y + \ln|y| = 1$$

Activity 4.1 *Show that the differential equation*

$$y' = xe^{y-x}$$

is separable and find the general solution.

Example 4.5 *Life scientists have used the equation*

$$\frac{dy}{dt} = ky(m-y),$$

called the logistic equation to model the spread of a disease or information through a population. If m denotes the total number of people in the population and $y(t)$ is the number of infected people at time t, then the o.d.e. states that the rate of growth of the disease is proportional to the product of the number of people who are infected and the number who are not. i.e. $\frac{dy}{dt} \propto y(m-y)$. Now

$$\begin{aligned}
\frac{dy}{y(m-y)} &= kdt \\
\Longrightarrow \int kdt - \int \frac{dy}{y(m-y)} &= C \\
kt - \int \left(\frac{\frac{1}{m}}{y} + \frac{\frac{1}{m}}{m-y} \right) dy &= C \\
\frac{1}{m} \ln|y| - \frac{1}{m} \ln|m-y| &= kt + C \\
\frac{1}{m} \ln \left| \frac{y}{m-y} \right| &= kt + C \\
\frac{y}{m-y} &= C_2 e^{mkt} \\
\Longrightarrow y &= \frac{mC_2 e^{mkt}}{1 + C_2 e^{mkt}} \\
\Longrightarrow y &= \frac{C_2 m}{C_2 + e^{-mkt}}
\end{aligned}$$

Suppose we are given that $y(0) = R < m$, then we can find C_2:
$R = \frac{C_2 m}{1 + C_2}, \quad \Longrightarrow C_2 = \frac{R}{m-R}$, *so*

$$y(t) = \frac{mR}{R + (m-R)e^{-mkt}}$$

Observe that as $t \longrightarrow \infty$, $y(t) \longrightarrow m$ which means assuming no cure is found, eventually all the people will be infected.

Example 4.6 *A rumour spreads through a population of* 5000 *people at a rate proportional to the product of the number of people who have heard it and a*

number who have not. Suppose that 100 *people initiate the rumour and that a total of* 500 *people know the rumour after two days. How long will it take for half the people to hear the rumour?*

Solution

Here $m = 5000$ and $R = y(0) = 100$, so $y(t) = \dfrac{100(5000)}{100 + 4900e^{-5000kt}} = \dfrac{5000}{1 + 49e^{-5000kt}}$.
Now, we are given that $y(2) = 500$. We determine k using this.

$$500 = \frac{5000}{1 + 49e^{-10000k}}, \quad \Longrightarrow k = 0.00017$$

Therefore

$$y(t) = \frac{5000}{1 + 49e^{-0.85t}}$$

Now, to determine how long it will take for half the population to hear the rumour, we solve

$$2500 = \frac{5000}{1 + 49e^{-0.85t}}$$

which gives $t = 4.58$. Thus, it will take slightly more than $4\frac{1}{2}$ days for half the population to hear the rumour.

Exercises

1. Solve the differential equation.
   (a) $y' = y\sin(2x + 3)$ (b) $y' = (x^2 + 1)(y^2 + y)$
   (c) $y' = (xy)^3$ (d) $y' = 3x(1 + y^2)$
   (e) $y' = \dfrac{\sin\frac{1}{x}}{x^2 y\cos y}$ (f) $y' = \dfrac{y^2 + 1}{y + xy}$
   (g) $y' = xe^{y-x}$ (h) $y' = xy^2 - x - y^2 + 1$
   (i) $(y\ln x)y' = \dfrac{(y+1)^2}{x}$ (j) $e^y \sin 2x dx + \cos x(e^{2y} - y)dy = 0.$

2. Solve the initial value problems.
   (a) $y' = x\sqrt{\frac{1-y^2}{1-x^2}}$, $y(0) = 0$. (b) $\dfrac{dy}{dx} = \dfrac{e^{x-y}}{1 + e^x}$, $y(1) = 0$.
   (c) $\dfrac{dy}{dx} = \dfrac{x^2 y - y}{y + 1}$, $y(3) = 1$. (d) $x^2\dfrac{dy}{dx} = y - xy$, $y(-1) = -1$
   (e) $(xy^2 + y^2 + x + 1)dx + (y - 1)dy = 0$, $y(2) = 0$
   (f) $\cos y dx + (1 + e^{-x})\sin y dy = 0$, $y(0) = \dfrac{\pi}{4}$.

4.3.2 Equations reducible to variables separable type

These are some equations which can be transformed by certain substitutions or transformations into the variables separable form.

Example 4.7 *The differential equation*

$$\frac{dy}{dx} = x + y$$

can be transformed to one of variables separable by the substitution $u = x + y$, *noting therefore that* $y = u - x$ *so that the o.d.e takes the form*

$$\frac{du}{dx} = 1 + u.$$

All first order o.d.e.'s of the form

$$\frac{dy}{dx} = F(a_1x + a_2y + a_3),$$

where a_1, a_2 and a_3 are constants can be reduced to variables separable type by introducing the substitution

$$z = a_1x + a_2y + a_3$$

Now, $\dfrac{dz}{dx} = a_1 + a_2\dfrac{dy}{dx}$ and substituting for z into

$$\frac{dy}{dx} = F(a_1x + a_2y + a_3)$$

we obtain

$$\frac{dz}{dx} = a_1 + a_2F(z).$$

Separating the variables we obtain

$$\frac{dz}{a_1 + a_2F(z)} = dx$$

and on integration we obtain the general solution

$$x = \int \frac{dz}{a_1 + a_2F(z)} + C.$$

We can have the case where $a_1 = 0$; which implies that the differential equation

$$\frac{dy}{dx} = F(a_2y + a_3)$$

is already of the variables separable type with the general solution given by

$$x = \int \frac{dz}{F(a_2y + a_3)} + C.$$

We may also have the case where $a_2 = 0$ and similarly we obtain the general solution after separating the variables;

$$y = \int F(a_1x + a_3)dx + C.$$

Example 4.8 *Prove that the differential equation* $\dfrac{dy}{dx} = (x+y)^2$ *can be reduced to one of variables separable type by the substitution* $v = x + y$. *Obtain the solution if* $y(0) = 0$.

Solution
The substitution $v = x + y$ allows us to obtain

$$\frac{dv}{dx} = 1 + \frac{dy}{dx}$$

or

$$\frac{dy}{dx} = \frac{dv}{dx} - 1$$

after we differentiate with respect to x, and the differential equation takes the form

$$\frac{dv}{dx} - 1 = v^2$$

or

$$\frac{dv}{dx} = v^2 + 1$$

which is separable. Its general solution is

$$v = \tan(x + C)$$

which becomes

$$y = \tan(x + C) - x$$

after back substitution. Now, $y(0) = 0$ gives the particular solution

$$y = \tan x - x.$$

Activity 4.2 *Use the substitution* $xy = t$ *to solve* $(x^2y^2 + 1)dx + 2x^2dy = 0$.

4.3.3 Homogeneous equations

Definition 4.4 *The differential equation* $\dfrac{dy}{dx} = f(x, y)$ *is called a first order homogeneous equation if it can be written in the form:*

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right), \quad x \neq 0. \tag{4.10}$$

Differential equations like $y' = \dfrac{y}{x}$; $y' = \log x - \log y$; $y' = \dfrac{y}{x} + \sin\dfrac{y}{x}$ and $y' = \dfrac{y-x}{y+x}$ are all homogeneous. However, the equation $y' = \dfrac{y}{x} + \dfrac{x^2}{y}$ is not homogeneous. The form (3.10) suggests the introduction of a new variable by the substitution

$v = \dfrac{y}{x}$ that is, $y = vx$. This substitution, first used by the Mathematician Leibniz in 1961, transforms the differential equation (3.10) into one of variables separable.

$$\frac{dy}{dx} = x\frac{dv}{dx} + v = F(v),$$

or

$$\frac{dv}{dx} = \frac{F(v) - v}{x}, \quad x \neq 0.$$

We separate variables and then integrate;

$$\int \frac{dv}{F(v) - v} = \int \frac{dx}{x}, \; x \neq 0, \quad F(v) \neq v, \Longrightarrow G(v) = \ln|x| + C.$$

for which

$$G\left(\frac{y}{x}\right) = \ln|x| + C.$$

Definition 4.5 *A function $f(x, y)$ is said to be* **a homogeneous function of degree r** *with respect to the variables x and y if*

$$\forall\ t \in \mathbb{R}, \quad f(tx, ty) = t^r f(x, y)$$

If the differential equation (3.10) is expressible in the form

$$P(x, y)dx + Q(x, y)dy = 0,$$

where $P(x, y)$ and $Q(x, y)$ are homogeneous functions of the same degree, then it is a homogeneous differential equation.

Example 4.9 *Solve* $y' = \dfrac{y - x}{y + x}$.

Solution
The differential equation is homogeneous given that:

$$f(x, y) = \frac{y - x}{y + x} = \frac{1 - \frac{y}{x}}{1 + \frac{y}{x}} = F\left(\frac{y}{x}\right)$$

is a homogeneous function of degree $r = 0$. To determine its solution we use the substitution $y = vx$. Thus

$$\frac{dy}{dx} = v + x\frac{dv}{dx} = \frac{1 - v}{1 + v}$$

which is variable separable. Separating the variables we obtain:

$$\frac{(1 + v)dv}{1 - 2v - v^2} = \frac{dx}{x}, \quad \text{with} \quad x \neq 0$$

The solution to this equation is $x^2 - 2xy - y^2 = C$

Example 4.10 *Solve* $y' = \dfrac{y^2 + x^2}{yx}$, $xy \neq 0$.

Solution
Rewrite the equation in the form $\dfrac{dy}{dx} = \dfrac{x}{y} + \dfrac{y}{x}$, with $x, y \neq 0$. It is clear that this equation is homogeneous. Let $y = vx$ to obtain

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

which reduces to

$$v + x\frac{dv}{dx} = \frac{1}{v} + v, \quad v \neq 0,$$

or

$$\frac{dv}{dx} = \frac{1}{vx}, \ vx \neq 0.$$

Separating variables and integrating we obtain

$$\frac{1}{2}v^2 - \ln|x| = C, \ x \neq 0.$$

Since $y = vx$, the general solution of this equation is given by

$$y^2 = Cx^2 + x^2 \ln x^2.$$

Activity 4.3 *Solve* $(xe^{\frac{y}{x}} + y)dx - xdy = 0, \quad x \neq 0$.

Exercises

1. Use the substitution $xy = t$ to solve $(1 + x^2y^2)y + (xy - 1)^2xy' = 0$.

2. Use the substitution $xy = t$ to solve $(x^2y^3 + y + x - 2)dx + (x^3y^2 + x)dy = 0$.

3. Use the substitution $y = tx$ to solve
$(x^6 - 2x^5 + 2x^4 - y^3 + 4x^2y)dx + (xy^2 - 4x^3)dy = 0$.

4. Use the substitution $x \ln y = t$ to solve
$(xy + 2xy \ln^2 y + y \ln y)dx + (2x^2 \ln y + x)dy$

5. Solve the differential equations

   (a) $xy' = y + \sqrt{y^2 - x^2}$

   (b) $2xy'(x^2 + y^2) = y(y^2 + 2x^2)$

   (c) $xy' = \sqrt{y^2 - x^2}$

4.3.4 Linear differential equations

A first order differential equation is linear if it has the form

$$a_1(x)y' + a_0(x)y = f(x) \tag{4.11}$$

The characteristic element for linearity for this equation is in y and y'. Linear differential equations are important given the frequency in which they occur in applications.
The linear equation (3.11) requires an **integrating factor** of the form $v(x) = e^{\int P(x)dx}$ where

$$P(x) = \frac{a_0(x)}{a_1(x)}$$

for which the general solution becomes

$$y = e^{-\int P(x)dx} \int Q(x)e^{\int P(x)dx}dx + Ce^{-\int P(x)dx}$$

Example 4.11 *Solve* $y' + \dfrac{1}{x}y = 2, \quad x \neq 0$

Solution
From the general form, $P(x) = \dfrac{1}{x}$ so that $v(x) = e^{\int \frac{1}{x}dx} = e^{\ln x} = x$. We multiply the differential equation by the integrating factor to get:

$$x\left(y' + \frac{1}{x}y\right) = 2x$$

or

$$xy' + y = 2x$$

which is then written as

$$\frac{d}{dx}(xy) = 2x$$
$$\Longrightarrow yx = x^2 + C$$

or

$$y = x + Cx^{-1}, \quad x \neq 0.$$

Example 4.12 *Solve* $y' + y = x$.

Solution
Here $P(x) = 1$, so that $v(x) = e^x$ and multiplying the differential equation by this integrating factor and integrating we obtain

$$e^x y = (x-1)e^x + C$$

which simplifies to

$$y = Ce^{-x} + x - 1.$$

Activity 4.4 *1. Find the general solution of the linear differential equation*

$$y' + ay = b$$

where a, b are constants with $a \neq 0$.

2. Find the general solution of

$$y' + 2xy = x$$

and find the particular solution that satisfies $y(0) = 2$.

3. Solve the initial value problem

$$xy' - 2y = 3x^4, \quad y(-1) = 2.$$

Example 4.13 *Newton's Law of Cooling states that the rate of change of the temperature T of an object is proportional to the difference between T and the (constant) temperature τ of the surrounding medium, called the ambient temperature. Assuming that the temperature of the object is a differentiable function, the Mathematical formulation of Newton's Law is*

$$\frac{dT}{dt} \propto T - \tau$$

$$\Longrightarrow \frac{dT}{dt} = -k(T - \tau), k > 0 \quad \text{constant}$$

$$\Longrightarrow \frac{dT}{dt} + kT = k\tau$$

which is a first order linear equation with $P(x) = k$ and $f(x) = k\tau$. We solve this equation.

$$\begin{aligned} v(t) &= e^{\int k dt} \\ &= e^{kt} \\ e^{kt}\frac{dT}{dt} + e^{kt}kT &= k\tau e^{kt} \\ \Longrightarrow \frac{d}{dt}(e^{kt}T) &= k\tau e^{kt} \\ \Longrightarrow e^{kt}T &= \tau e^{kt} + C \\ \Longrightarrow T &= \tau + Ce^{-kt} \end{aligned}$$

An example to apply Newton's Law of cooling is the following:

Example 4.14 *A cup of coffee is served to you at $185°$ in a room where the temperature is $65°$. Two minutes later, the temperature of the coffee has dropped to $155°$. How many more minutes would you expect to wait for the coffee to cool to $105°$?*

Solution
Here $\tau = 65$ and $T(0) = 185$. So by Newton's Law of cooling

$$T = \tau + Ce^{-kt}$$

we find C:

$$185 = 65 + C, \Longrightarrow C = 120.$$

Thus

$$T = 65 + 120e^{-kt}$$

Now, using $T(2) = 155$, we determine k.

$$155 = 65 + 120e^{-2k}$$

This gives $k = 0.144$. Hence

$$T = 65 + 120e^{-0.144t}$$

Finally

$$105 = 65 + 120e^{-0.144t}$$

gives $t \approx 7.63$ minutes. Thus you should wait for another 7.63 minutes.
Exercises

1. Find the general solution of the following differential equations.
   (a) $y' - 2y = 1$ (b) $xy' - 2y = -x$
   (c) $2y' + 5y = 2$ (d) $y' - y = -2e^{-x}$
   (e) $y' - 2y = 1 - 2x$ (f) $xy' + 2y = \dfrac{\cos x}{x}$
   (g) $xy' - y = 2x \ln x$ (h) $y' + \dfrac{2}{x+1}y = 0$
   (i) $y' + \dfrac{2}{x+1}y = (x+1)^{\frac{5}{2}}$.

2. Find the particular solution determined by the given conditions.
   (a) $y' + y = x, \quad y(0) = 1.$ (b) $xy' + 2y = xe^{-x}, \quad y(1) = -1.$
   (c) $y' + y = \dfrac{1}{1+e^x}, \quad y(0) = e.$ (d) $xy' - 2y = x^3e^x, \quad y(1) = 0$

3. An object falling from rest in air is subject not only to the gravitational force but also to air resistance. Assume that the air resistance is proportional to the velocity and acts in a direction opposite to the motion. Then the velocity of the object at time t satisfies

   $$v'(t) = 32 - kv(t), \quad k > 0 \text{ constant}, \quad v(0) = 0.$$

   Here we are measuring distance in metres and the positive direction is down.
   (a) Find $v(t)$.
   (b) Show that $v(t)$ cannot exceed $\dfrac{32}{k}$ and that $v(t) \to \dfrac{32}{k}$ as $t \to \infty$.
   (c) Sketch the graph of $v(t)$.

4. Suppose that a certain population P has a known birth rate $\frac{dB}{dt}$ and a known death rate $\frac{dD}{dt}$. Then the rate of change of P is given by

$$\frac{dP}{dt} = \frac{dB}{dt} - \frac{dD}{dt}.$$

(a) Assume that $\frac{dB}{dt} = aP$ and $\frac{dD}{dt} = bP$, where a and b are constants. Find $P(t)$ if $P(0) = P_0 > 0$.
(b) Analyze the cases (i) $a > b$, (ii) $a = b$, (iii) $a < b$.
Find $\lim_{t\to\infty} P(t)$ in each case.

4.4 Second order linear differential equations with constant coefficients

The differential equations we consider in this section take the general form:

$$a_n \frac{d^n y}{dx^n} + a_{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_2\frac{d^2y}{dx^2} + a_1\frac{dy}{dx} + a_0 y = f(x),$$

where the coefficients $a_i, i = 0, 1, \cdots, n$ are real constants and $a_n \neq 0$. If $f(x) = 0$, the equation is said to be homogeneous. Solutions of such equations are of exponential form.

4.4.1 Homogeneous equations of first order

$$a\frac{dy}{dx} + by = 0$$

To solve this equation, write $\frac{dy}{dx} = -\frac{b}{a}y$ and we see that this is a separable equation. Separating the variables and integrating both sides:
$\int \frac{dy}{y} = -\int \frac{b}{a}dx \Longrightarrow \ \ln y = -\frac{b}{a}x + C \Longrightarrow \ y = C_1 e^{-\frac{b}{a}x}$

4.4.2 Homogeneous equations of second order

$$a\frac{d^2y}{dx^2} + d\frac{dy}{dx} + cy = 0$$

If we try a solution of the form $y = e^{mx}$, then $y' = me^{mx}$ and $y'' = m^2e^{mx}$. Substituting into the equation we get

$$am^2e^{mx} + bme^{mx} + ce^{mx} = 0$$

$$\implies \quad e^{mx}(am^2 + bm + c) = 0$$
$$\implies \quad am^2 + bm + c = 0.$$

The last equation is called the **auxiliary equation** or the **characteristic equation** for the second order ode. Solving this equation leads to 3 cases of solution:
Case 1: Distinct real roots:
If the roots are m_1 and m_2, $m_1 \neq m_2$, then we get two solutions $y_1 = e^{m_1 x}$ and $y_2 = e^{m_2 x}$. The general solution is thus

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}.$$

Case 2: Repeated real roots:
When $m_1 = m_2$, we obtain only one exponential solution $y_1 = e^{m_1 x}$. A second solution of the equation is $y_2 = e^{m_1 x} \int \frac{e^{2m_1 x}}{e^{2m_1 x}} dx = x e^{m_1 x}$. The general solution then becomes

$$y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}$$

Case 3: Complex conjugate roots:
If m_1 and m_2 are complex conjugates, then we can write $m_1 = \alpha + j\beta, \quad m_2 = \alpha - j\beta$ where $\alpha, \beta > 0$ are real and $j^2 = -1$. In this way we have

$$y = c_1 e^{(\alpha + j\beta)x} + c_2 e^{(\alpha - j\beta)x}.$$

However this solution is not practically useful as it is in terms of j. We need to get rid of the j. Using Euler's formula, we write $e^{j\beta x} = \cos\beta x + j\sin\beta x$ and $e^{-j\beta x} = \cos\beta x - j\sin\beta x$. Adding and subtracting the two equations results in

$$e^{j\beta x} + e^{-j\beta x} = 2\cos\beta x$$

and

$$e^{j\beta x} - e^{-j\beta x} = 2j\sin\beta x$$

Since $y = c_1 e^{(\alpha + j\beta)x} + c_2 e^{(\alpha - j\beta)x}$ is a solution, the choices $c_1 = c_2 = 1$ and $c_1 = 1, \quad c_2 = -1$ give, in turn, 2 solutions:
$y_1 = e^{(\alpha + j\beta)x} + e^{(\alpha - j\beta)x} = e^{\alpha x}\left(e^{j\beta x} + e^{-j\beta x}\right) = 2e^{\alpha x}\cos\beta x$
and $y_2 = e^{(\alpha + j\beta)x} - e^{(\alpha - j\beta)x} = e^{\alpha x}\left(e^{j\beta x} - e^{-j\beta x}\right) = 2je^{\alpha x}\sin\beta x$
We have thus, without loss of generality, shown that the real functions $e^{\alpha x}\cos\beta x$ and $e^{\alpha x}\sin\beta x$ are solutions of the second order equation under this case 3. Consequently, the general solution is

$$y = Ae^{\alpha x}\cos\beta x + Be^{\alpha x}\sin\beta x = e^{\alpha x}(A\cos\beta x + B\sin\beta x).$$

Example 4.15 *Solve the following differential equations:*

1. $2\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 3y = 0$

2. $\dfrac{d^2y}{dx^2} - 10\dfrac{dy}{dx} + 25y = 0$
3. $y'' + y' + y = 0$

Solution
The solution process involve giving the auxiliary equations, the roots and the corresponding general solutions.

1. $2m^2 - 5m - 3 = 0 \Longrightarrow (2m+1)(m-3) = 0, \Longrightarrow \quad m_1 = -\dfrac{1}{2}, \quad m_2 = 3$ and $y = c_1 e^{-\frac{x}{2}} + c_2 e^{3x}$
2. $m^2 - 10m + 25 = 0 \Longrightarrow \quad (m-5)^2 = 0 \Longrightarrow \quad m_1 = m_2 = 5$ and $y = Ae^{5x} + Bxe^{5x}$
3. $m^2 + m + 1 = 0 \Longrightarrow m_1 = -\dfrac{1}{2} + \dfrac{\sqrt{3}}{2}j, \quad m_2 = -\dfrac{1}{2} - \dfrac{\sqrt{3}}{2}j$ and $y = e^{-\frac{x}{2}}\left(c_1 \cos\dfrac{\sqrt{3}}{2}x + c_2 \sin\dfrac{\sqrt{3}}{2}x\right)$.

Activity 4.5 *1. Find the general solution of the given differential equation:*

(a) $y'' - y' - 6y = 0$
(b) $\dfrac{d^2y}{dx^2} + 8\dfrac{dy}{dx} + 16y = 0$
(c) $y'' + 3y' - 5y = 0$
(d) $12y'' - 5y' - 2y = 0$
(e) $\dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + 2y = 0$
(f) $y'' + 9y = 0$
(g) $4y'' + y' = 0$

2. Solve the given differential equation subject to the indicated initial conditions:

(a) $y'' + 16y = 0, \quad y(0) = 2, \quad y'(0) = -2$
(b) $y'' - y = 0, \quad y(0) = y'(0) = 1$
(c) $y'' + 6y' + 5y = 0, \quad y(0) = 0, \quad y'(0) = 3$
(d) $y'' - 8y' + 17y = 0, \quad y(0) = 4, \quad y'(0) = -1$

Unit 5
Matrix Theory

5.1 What this Unit is all About

In this unit we discuss the basic algebra of matrices namely matrix addition and subtraction, matrix multiplication and division. We also discuss matrix determinants and inverses.

By the end of this unit, you should be able to:

1. Define a matrix.
2. Perform matrix addition and subtraction.
3. Multiply matrices.
4. State and understand axioms of matrix multiplication.
5. Find the determinant of a matrix.
6. Find the inverse of a matrix.

5.2 Addition, Scalar Multiplication, Multiplication

5.2.1 Introduction and terminology

• A **matrix** is an array of numbers arranged into rows and columns. Rows are horizontal and columns are vertical. Rows are numbered starting from the top and counting downwards. Columns are numbered starting from the left and moving to the right.
• If a matrix has s rows and t columns, its **dimension** is $s \times t$ read as "s by t".
• A matrix with only one row (i.e. dimension $1 \times t$) is called a **row vector** while a matrix with only one column is called a **column vector**. (dimension $s \times 1$).
• The $(i, j)^{th}$ position in a matrix is the intersection of the ith row and the jth column and the number appearing in that position is called the $(i, j)^{th}$ **entry** of the matrix. The $(i, j)^{th}$ entry of a matrix is denoted by the symbol a_{ij} if the matrix is labelled by A. The first subscript indicates the row it comes from while the second indicates the column.

• The general 2×3 matrix is denoted by $A = [a_{ij}]_{2\times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$ The 2nd row is $\begin{bmatrix} a_{21} & a_{22} & a_{23} \end{bmatrix}$. Note that 1st subscript is always 2. The 2nd column is $\begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix}$. Second subscript always 2.

Below we give some matrices that will be used to illustrate concepts.

$$A = \begin{bmatrix} -2 & -4 & 0 & -1 \\ 1 & -3 & 2 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 4 & 0 \\ 6 & -9 \\ 1 & 0 \\ 6 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix}$$

$$E = \begin{bmatrix} 1 & 8 & 1 & 4 \\ 2 & 0 & 6 & -2 \end{bmatrix} \quad F = \begin{bmatrix} 3 & 2 & 1 \\ 4 & 3 & 2 \\ 5 & 4 & 3 \end{bmatrix}$$

Definition 5.1 *Addition/Sum*
Let $A = [a_{ij}]_{s\times t}$ and $B = [b_{ij}]_{s\times t}$ be two matrices. The **sum** *$A + B$ of A and B is the $s \times t$ matrix $A + B = C = [c_{ij}]_{s\times t}$, where $\forall\ i, j \quad c_{ij} = a_{ij} + b_{ij}$*

Points that must be noted carefully are:
• The sum of two matrices is only defined for two matrices <u>of the same dimension.</u>
• The dimension of the sum is the same as the dimension of the two summands.
• The $(i, j)^{th}$ entry of the sum of two matrices is the sum of the corresponding entries in the two matrices, i.e. the sum of the $(i, j)^{th}$ entries of A and B.
Examples from above given matrices:

Example 5.1 $A+E = \begin{bmatrix} -2 & -4 & 0 & -1 \\ 1 & -3 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 8 & 1 & 4 \\ 2 & 0 & 6 & -2 \end{bmatrix} = \begin{bmatrix} -1 & 4 & 1 & 3 \\ 4 & -3 & 8 & 1 \end{bmatrix} = E + A$

$$D + F = \begin{bmatrix} 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix} + \begin{bmatrix} 3 & 2 & 1 \\ 4 & 3 & 2 \\ 5 & 4 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 4 & 5 \\ 5 & 6 & 11 \\ 6 & 8 & 19 \end{bmatrix} = F + D$$

Most of the rules of number addition apply to matrix addition:

1. <u>Commutative law:</u> $A + B = B + A$
2. <u>Associative law:</u> $A + (B + C) = (A + B) + C$
3. For all positive integers s, t, there is an $s \times t$ matrix Z which obeys the rule $A + Z = Z + A = A \quad \forall \quad s \times t$ matrices A. It is easy to see that every entry of Z is the number 0, so Z is usually denoted by O and is called the <u>zero matrix.</u> Note that there is one zero matrix for each possible dimension.
4. Every matrix $A = [a_{ij}]_{s\times t}$ has a negative matrix $B = [b_{ij}]_{s\times t}$ satisfying the equations $A + B = B + A = O$. It is usual to write $-A$ rather than B and it is obvious that $\forall\ \ i, j \ \ b_{ij} = -a_{ij}$.

Definition 5.2 *If k is a scalar, i.e. $k \in \mathbb{R}$ or $k \in \mathbb{C}$ and $A = [a_{ij}]_{s\times t}$, then* **Scalar Multiplication** *is defined by $B = kA = [b_{ij}]_{s\times t}$ with $b_{ij} = ka_{ij}$.*

Points to note are:
• A and kA are of the same dimension.
• The matrix kA is obtained by multiplying each entry of A by k.

Example 5.2 $3C = 3\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}$

• Basic rules of multiplication by a scalar:
For all scalars h, k and matrix A, the following are true.

1. $k(A + B) = kA + kB$.
2. $(h + k)A = hA + kA$.
3. $hA = Ah$.
4. $(hk)A = h(kA)$.
5. $0A = kO = O, \quad 1A = A1 = A, \quad (-1)A = -A$

Definition 5.3 *If A is any $s \times t$ matrix, then the* **transpose of A**, *denoted A^T is defined to be the $t \times s$ matrix that results from interchanging the rows and columns of A. That is, the first column of A^T is the first row of A, the second column of A^T is the second row of A, and so forth. The $(i, j)^{th}$ entry of A^T is the $(j, i)^{th}$ entry of A.*

Example 5.3 $F = \begin{bmatrix} 3 & 2 & 1 \\ 4 & 3 & 2 \\ 5 & 4 & 3 \end{bmatrix}$, $F^T = \begin{bmatrix} 3 & 4 & 5 \\ 2 & 3 & 4 \\ 1 & 2 & 3 \end{bmatrix}$

Example 5.4 *Let* $A = \begin{bmatrix} 1 & 4 & -1 & 2 \\ 3 & 1 & 2 & 4 \\ 5 & -5 & 10 & 8 \end{bmatrix}$ $B = \begin{bmatrix} 2 & 1 & -1 & -2 \\ 3 & -2 & -2 & 1 \\ 5 & 6 & -2 & 9 \end{bmatrix}$
$C = \begin{bmatrix} 1 & 2 & -3 \\ -2 & -5 & 7 \\ 2 & 2 & -4 \\ 1 & 3 & 2 \end{bmatrix}$. *Calculate (i)* $A + B$ *(ii)* $A^T + C$ *(iii)* $A^T + C^T$
(iv) $6B - 2A$ *(v)* $3B^T + 7C$ *(vi)* $7B + 3C$.

Solution
(i) $A+B = \begin{bmatrix} 1 & 4 & -1 & 2 \\ 3 & 1 & 2 & 4 \\ 5 & -5 & 10 & 8 \end{bmatrix} + \begin{bmatrix} 2 & 1 & -1 & -2 \\ 3 & -2 & -2 & 1 \\ 5 & 6 & -2 & 9 \end{bmatrix} = \begin{bmatrix} 3 & 5 & -2 & 0 \\ 6 & -1 & 0 & 5 \\ 10 & 1 & 8 & 17 \end{bmatrix}$

(ii) $A^T + C = \begin{bmatrix} 1 & 3 & 5 \\ 4 & 1 & -5 \\ -1 & 2 & 10 \\ 2 & 4 & 8 \end{bmatrix} + \begin{bmatrix} 1 & 2 & -3 \\ -2 & -5 & 7 \\ 2 & 2 & -4 \\ 1 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 5 & 2 \\ 2 & -4 & 2 \\ 1 & 4 & 6 \\ 3 & 7 & 10 \end{bmatrix}$

(iii) A^T is 4×3, C^T is 3×4, so the matrices are not conformable for addition and $A^T + C^T$ is undefined.

(iv) $6B - 2A = \begin{bmatrix} 12 & 6 & -6 & -12 \\ 18 & -12 & -12 & 6 \\ 30 & 36 & -12 & 54 \end{bmatrix} - \begin{bmatrix} 2 & 8 & -2 & 4 \\ 6 & 2 & 4 & 8 \\ 10 & -10 & 20 & 16 \end{bmatrix} = \begin{bmatrix} 10 & -2 & -4 & -16 \\ 12 & -14 & -16 & -2 \\ 20 & 46 & -32 & 38 \end{bmatrix}$.

(v) $3B^T + 7C = \begin{bmatrix} 6 & 9 & 15 \\ 3 & -6 & 18 \\ -3 & -6 & -6 \\ -6 & 3 & 27 \end{bmatrix} + \begin{bmatrix} 7 & 14 & -21 \\ -14 & -35 & 49 \\ 14 & 14 & -28 \\ 7 & 21 & 14 \end{bmatrix} = \begin{bmatrix} 13 & 23 & -6 \\ -11 & -41 & 67 \\ 11 & 8 & -34 \\ 1 & 24 & 41 \end{bmatrix}$.

(vi) $7B$ is 3×4, $3C$ is 4×3, so $7B + 3C$ is undefined.

5.2.2 Matrix Multiplication and Division

Definition 5.4 *Let $A = [a_{ij}]_{s\times t}$ and $B = [b_{ij}]_{p\times q}$ be two matrices.*
The **PRODUCT** *AB of A and B, in that order, is defined only if $p = t$ in which case it is the $s \times q$ matrix.*

$$AB = C = [c_{ij}]_{s\times q} \text{ with } c_{ij} = \sum_{k=1}^{t} a_{ik}b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{it}b_{tj}$$

Points to note:

1. The product of two matrices is only defined when the number of columns in the first matrix is the same as the number of rows in the second.

2. $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, $B = \begin{bmatrix} u & v \\ w & x \\ y & z \end{bmatrix}$ are 2×3 and 3×2 matrices respectively.
   They are conformable for multiplication and the product AB is of dimension 2×2 while BA is of dimension 3×3.
   $AB = \begin{bmatrix} 1u + 2w + 3y & 1v + 2x + 3z \\ 4u + 5w + 6y & 4v + 5x + 6z \end{bmatrix}$, $BA = \begin{bmatrix} 1u + 4v & 2u + 5v & 3u + 6v \\ 1w + 4x & 2w + 5x & 3w + 6x \\ 1y + 4z & 2y + 5z & 3y + 6z \end{bmatrix}$

3. We observe that:
   entry $c_{11} = a_{11}b_{11} + a_{12}b_{21}$
   entry $c_{12} = a_{11}b_{12} + a_{12}b_{22}$
   entry $c_{21} = a_{21}b_{11} + a_{22}b_{21}$
   entry $c_{22} = a_{21}b_{12} + a_{22}b_{22}$

   $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$.
   It is easy to generalize from this simple illustration.

4. The dimension of the product AB is the number of rows of A by the number of columns of B.

5. **Axiom M1** Associative Law of Matrix Multiplication
   Matrix multiplication is associative. i.e. For all matrices A, B, C for which the products are defined, $(AB)C = A(BC)$.

6. **Axiom M2** Commutative Law
   Item 2 means that matrix multiplication is highly non-commutative in that there are many cases where $AB \neq BA$, since if A is $s \times t$ and B is $t \times q$ with $s \neq q$ then AB is defined but BA is not defined.
   The dimension of the product compounds the problem of non-commutativity of matrix multiplication. For, consider the case where the products $AB,\ BA$ are both defined. This only happens if B is $t \times s$ when A is $s \times t$ (i.e.$p = t,\ q = s$). Under these conditions AB is $s \times s$ and BA is $t \times t$, so that if $s \neq t$, AB and BA are of different dimensions and so are not equal.

   Note that if AB and BA are both defined, then they are both square and the products AB, BA can only be equal if $p = t = s = q$, in which case A, B are both square and of the same dimension. That is $AB = BA \Longrightarrow A, B$ are both square of dimension $s \times s$. However even this is not sufficient (which is the same as saying that the reverse implication is not true) as shown in the example below:

   Example 5.5 *$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ are square and of the same dimension but $AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, while $BA = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. So $AB \neq BA$. However $AB \neq BA$ for all $s \times s$ matrices is not a true statement either i.e. A, B both $s \times s$ does not imply $AB \neq BA$.*

   Example 5.6 *Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $B = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$. Then $AB = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$ while $BA = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix} = AB$.*

   Summary
   Matrix multiplication is not commutative. That is the statement "for all matrices A, B for which the products are defined $AB = BA$" is false. This, however, does not mean that the statement "For all matrices A, B for which the products are defined $AB \neq BA$" is true.

7. **Axiom M3** Existence of an identity for matrix multiplication

Definition 5.5 • *A* <u>*diagonal matrix*</u> *is a square matrix with every entry off the main diagonal* 0.
• *A diagonal matrix is a* <u>*scalar matrix*</u> *if all the diagonal entries are equal. e.g.* $B = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$ *is a scalar matrix.*
• *The* $n \times n$ <u>*identity matrix,*</u> I_n*, is defined as the scalar matrix of dimension* $n \times n$ *whose diagonal entries are all* 1. $I_n = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \end{bmatrix}_{n \times n}$.
For each $s \times s$ *matrix* A, $I_s A = A I_s = A$.

8. **Axiom M4** <u>Distributive Law</u>
Matrix multiplication distributes from the right and from the left over matrix addition. i.e. $(A+B)C = BC + AC$; $A(B+C) + AB + AC$.
$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 4 & 3 \\ 5 & 6 \end{bmatrix}, C = \begin{bmatrix} -2 & 2 \\ -1 & 3 \\ 5 & -4 \end{bmatrix}$
$A(B+C) = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \cdot \begin{bmatrix} -1 & 4 \\ 3 & 6 \\ 10 & 2 \end{bmatrix} = \begin{bmatrix} 35 & 22 \\ 71 & 58 \end{bmatrix}$
$AB + AC = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 4 & 3 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \cdot \begin{bmatrix} -2 & 2 \\ -1 & 3 \\ 5 & -4 \end{bmatrix}$
$= \begin{bmatrix} 24 & 26 \\ 54 & 59 \end{bmatrix} + \begin{bmatrix} 11 & -4 \\ 17 & -1 \end{bmatrix} = \begin{bmatrix} 35 & 22 \\ 71 & 58 \end{bmatrix}$ which verifies the distributive law but does not prove it.

9. Matrix multiplication and multiplication by a scalar obey a law which looks similar to an associative law: $(ab)C = a(bC)$; $A(bc) = (Ab)c$.

Example 5.7 $(2 \cdot 3) \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = 6 \cdot \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 6 & 12 & 18 \\ 24 & 30 & 36 \end{bmatrix}$ *while* $2\left(3 \cdot \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}\right) = 2 \cdot \begin{bmatrix} 3 & 6 & 9 \\ 12 & 15 & 18 \end{bmatrix} = \begin{bmatrix} 6 & 12 & 18 \\ 24 & 30 & 36 \end{bmatrix}$

10. There are matrices A, B, neither of which is a zero matrix, whose product is a zero matrix. See (6) above. Thus in matrix multiplication $AB = O$ does not imply $A = O$ or $B = O$.

Example 5.8 *Let* $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 4 \\ 1 & 5 \\ 2 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.
State which of the following do not define matrices and evaluate the others. AB, AC, BC, $AC + C, B^T C, A + B$.

Solution

AB is 3×3 by 3×2 so is defined and is 3×2.

$$AB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 1 & 5 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 1 & 5 \\ 2 & 1 \end{bmatrix} = B.$$

AC is 3×3 by 3×1 so is defined and is 3×1.

$$AC = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = C$$

BC is 3×2 by 3×1 and thus is not defined.

$AC + C$ is 3×1 plus 3×1 so is defined and is 3×1.

$$AC + C = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 2C.$$

$B^T C$ is 2×3 by 3×1 so is defined and is 2×1.

$$B^T C = \begin{bmatrix} 1 & 1 & 2 \\ 4 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \end{bmatrix}.$$

$A + B$ is 3×3 plus 3×2 so is undefined.

Activity 5.1 *Let* $U = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 0 & -2 \end{bmatrix}, V = \begin{bmatrix} 2 & 1 & -2 & 3 \\ -2 & 2 & 1 & -3 \\ -1 & 0 & -1 & 0 \end{bmatrix}, W = \begin{bmatrix} 3 & -1 & 0 & 5 \\ 1 & -4 & 7 & 3 \\ 0 & -2 & 1 & 2 \end{bmatrix}.$
Calculate $V - W$*, then* $U(V - W)$*. Calculate* UV *and* UW*. Is* $U(V - W) = UV - UW$*? Compare* UV *and* UW*. Discuss the problems that these matrices highlight for matrix multiplication.*

Activity 5.2 *1. Let* $B = \begin{bmatrix} 2 & 1 & -1 & -2 \\ 3 & -2 & -2 & 1 \\ 5 & 6 & -2 & 9 \end{bmatrix}$,

$$C = \begin{bmatrix} 1 & 2 & -3 \\ -2 & -5 & 7 \\ 2 & 2 & -4 \\ 1 & 3 & 2 \end{bmatrix}.$$

With the matrices B *and* C^T

(a) Verify the commutative law of addition, i.e. $B + C^T = C^T + B$.

(b) Verify the law $a(B + C^T) = aB + aC^T$ *for any real number* a.

2. Let $G = \begin{bmatrix} -6 & -3 & 5 \\ -3 & -2 & 0 \\ 5 & 0 & 3 \end{bmatrix}$,

$H = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$, *and* $K = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ -1 & -2 & 1 \end{bmatrix}$. *Calculate (i)* $G+H$ *(ii)* H^T+

K^T *(iii)* $3K - 6G$ *(iv)* $3K^T + 2G$ *(v)* $2K^T + 3H^T$ *(vi)* $2K - 3G^T$.

3. *(a) Let* $X = \begin{bmatrix} 1 & 3 & 0 & 4 \\ 2 & 1 & 1 & 0 \end{bmatrix}$ *and* $Y = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$. *Calculate* XY *and* YX *and note that the two products are different because they have different dimensions.*

   (b) Let $X = \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix}$ *and* $Y = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$. *Calculate* XY *and* YX *and note that the two are different because they have different dimensions.*

   (c) Let $X = \begin{bmatrix} 3 & 1 & -2 \\ 2 & -2 & 0 \\ -1 & 1 & 2 \end{bmatrix}$ *and* $Y = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$. *Calculate* XY *and* YX *and note whether they are equal to one another or not. Calculate* XY^T *and* Y^TX *and note whether they are equal or not. What does this last pair of examples tell you about the commutative law of matrix multiplication?*

5.3 Matrix determinants and inverses

5.3.1 Determinants

Definition 5.6 *The* **determinant** *of the* 2×2 *matrix* $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ *written as* $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ *is given by* $ad - bc$.

3×3 **Matrix**

Definition 5.7 *The* $(i, j)^{th}$ *Minor of the matrix* A, $M_{ij}(A)$ *is the determinant of the matrix obtained by deleting the* i^{th} *row and the* j^{th} *column of* A.

If $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ *then* $M_{11}(A) = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$, $M_{12}(A) = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$,

$M_{13}(A) = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$, $M_{21}(A) = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$, $M_{22}(A) = \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$,

$$M_{23}(A) = \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}, \; M_{31}(A) = \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix},$$
$$M_{32}(A) = \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}, \; M_{33}(A) = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}.$$

Definition 5.8 *The $(i,j)^{th}$ Cofactor of the matrix A, $C_{ij}(A)$ is the signed minor $C_{ij}(A) = (-1)^{i+j}M_{ij}(A)$.*

Definition 5.9 *The determinant of A, $|A|$ is defined by the Laplace expansion by any row or any column. As an example the following is the expansion by the first row:*

$$\begin{aligned} |A| &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \\ &= a_{11}C_{11}(A) + a_{12}C_{12}(A) + a_{13}C_{13}(A) \\ &= a_{11}(-1)^2M_{11}(A) + a_{12}(-1)^3M_{12}(A) + a_{13}(-1)^4M_{13}(A) \\ &= a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}. \end{aligned}$$

The following scheme can be adopted: $\left|\begin{array}{ccc|cc} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} \end{array}\right.$

This expansion gives $|A| = a_{11}a_{22}a_{33}+a_{12}a_{23}a_{31}+a_{13}a_{21}a_{32}-a_{11}a_{23}a_{32}-a_{12}a_{21}a_{33}-a_{13}a_{22}a_{31}$ as before.

To generalize for an $s \times s$ matrix:

For $A = [a_{ij}]_{s\times s}$, the Laplace expansion of $|A|$ by the ith row is given by $|A| = a_{i1}C_{i1}(A) + a_{i2}C_{i2}(A) + \cdots + a_{is}C_{is}(A) = \sum_{k=1}^{s} a_{ik}C_{ik}(A)$ while that by the jth column is given by $|A| = a_{1j}C_{1j}(A) + a_{2j}C_{2j}(A) + \cdots + a_{sj}C_{sj}(A) = \sum_{k=1}^{s} a_{kj}C_{kj}(A)$. Note that the value of the determinant does not depend on the row or column used in the Laplace expansion.

Example 5.9 *Consider the matrices* $E = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -3 & -1 \\ 2 & -1 & -2 \end{bmatrix}, F = \begin{bmatrix} 2 & -3 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$.

Calculate $|E|$ and $|F|$, the former by expanding by row 1, the later by expanding by a row that gives the simplest expansion. Calculate $|F|$ using the special method for 3×3 determines.

$$\begin{aligned}
|E| &= \begin{vmatrix} 1 & 0 & 2 \\ 0 & -3 & -1 \\ 2 & -1 & -2 \end{vmatrix} \\
&= 1(-1)^{1+1}\begin{vmatrix} -3 & -1 \\ -1 & -2 \end{vmatrix} + 0(-1)^{1+2}\begin{vmatrix} 0 & -1 \\ 2 & -2 \end{vmatrix} + 2(-1)^{1+3}\begin{vmatrix} 0 & -3 \\ 2 & -1 \end{vmatrix} \\
&= 6 - 1 + 12 \\
&= 17.
\end{aligned}$$

As was seen in the last calculation, the more zeros there are in a row the simpler it is to expand by that row. So we expand $|F|$ by row 2.

$$\begin{aligned}
|F| &= \begin{vmatrix} 2 & -3 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} \\
&= 0(-1)^{2+1}\begin{vmatrix} -3 & -1 \\ 0 & 1 \end{vmatrix} + 1(-1)^{2+2}\begin{vmatrix} 2 & -1 \\ -1 & 1 \end{vmatrix} + 0(-1)^{2+3}\begin{vmatrix} 2 & -3 \\ -1 & 0 \end{vmatrix} \\
&= 0 + 1 + 0 \\
&= 1.
\end{aligned}$$

Using the special method

$$\begin{aligned}
|F| &= \left|\begin{array}{ccc|cc} 2 & -3 & -1 & 2 & -3 \\ 0 & 1 & 0 & 0 & 1 \\ -1 & 0 & 1 & -1 & 0 \end{array}\right. \\
&= 2(1)(1) + (-3)(0)(-1) + (-1)(0)(0) - (-1)(1)(-1) - (0)(0)(2) - (1)(0)(-3) \\
&= 2 + 0 + 0 - 1 - 0 - 0 \\
&= 1.
\end{aligned}$$

Activity 5.3 *Obtain $|E|$ using the special method.*

5.3.2 Rules for determinants

1. If A has a zero row, $|A| = 0$. Proof is by expanding by that row.
2. If B is obtained from A by interchanging two adjacent rows, then $|B| = -|A|$.
3. If B is obtained from A by interchanging any two different rows, then $|B| = -|A|$.
4. If a matrix has two identical rows, then its determinant is zero.
5. If B is obtained from A by multiplying a row by a scalar, then $|B| = k|A|$.

6. If one row of a matrix is a multiple of another, then the determinant of that matrix is zero.

7. If B is obtained from A by the elementary row operation $R_i \longrightarrow R_i + kR_j$, then $|B| = |A|$.

8. If A is equivalent to B, then $|B|$ is a non-zero multiple of $|A|$.

9. $|kA| = k^s|A|$ if A is $s \times s$.

10. $|A^T| = |A|$,so all the above results hold with row replaced by column.

5.3.3 The inverse of a matrix

Definition 5.10 *If A is a square matrix, and if a matrix B of the same size can be found such that $AB = BA = I$,then A is said to be* **invertible** *and B is called the* **inverse** *of A. If no such matrix B can be found, then A is said to be* **singular**.

Example 5.10 *The matrix* $B = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$ *is an inverse of* $A = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}$ *since* $AB = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$. $BA = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$.

Theorem 5.1 *An invertible matrix has exactly one inverse. i.e. if B and C are both inverses of the matrix A, then $B = C$.*

Proof
B inverse of $A \Longrightarrow BA = I$. Multiplying both sides by C: $(BA)C = C$. But $(BA)C = B(AC) = BI = B$. So $B = C$. △

Notation
If A is invertible, then its inverse will be denoted by the symbol A^{-1}. Thus $AA^{-1} = I = A^{-1}A$.

Theorem 5.2 *The matrix* $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ *is invertible if* $ad-bc \neq 0$, *in which case the inverse is given by the formula* $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix}$

Activity 5.4 *Verify that* $AA^{-1} = I_2, \quad A^{-1}A = I_2$.

5.3.4 Finding the inverse by elementary row operations

To find the inverse of an invertible matrix A, we must find a sequence of elementary row operations that reduces A to the identity and then perform this same sequence of operations on I_n to obtain A^{-1}. That is, to find the inverse of an $s \times s$ matrix A, a solution to the equations $AX = I_s$ has to be found. To do this a matrix $[A|I_s]$, the **combined augmented matrix**, is formed and Gauss-Jordan elimination is performed to reduce it to $[I_s|B]$. Then $B = A^{-1}$. If the process fails to reach I_s, no inverse exists.

Example 5.11 *Let* $E = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -3 & -1 \\ 2 & -1 & -2 \end{bmatrix}$, $F = \begin{bmatrix} 2 & -3 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$. *Using Gauss-Jordan elimination, find the inverses of* E *and* F*, and calculate* $E^{-1}F^{-1}$ *and* $F^{-1}E^{-1}$.

Solution

Form $[E|I_3]$ and row reduce it to $[I_3|E^{-1}]$.

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -3 & -1 & 0 & 1 & 0 \\ 2 & -1 & -2 & 0 & 0 & 1 \end{array}\right]$$

$$R_3 \longrightarrow R_3 - 2R_1 \quad \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -3 & -1 & 0 & 1 & 0 \\ 0 & -1 & -6 & -2 & 0 & 1 \end{array}\right]$$

At this stage $|E| = 17$.

$$\begin{array}{c} R_2 \longrightarrow \frac{-1}{3}R_2 \\ R_3 \longrightarrow R_3 + R_2 \end{array} \quad \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & \frac{1}{3} & 0 & \frac{-1}{3} & 0 \\ 0 & 0 & \frac{-17}{3} & -2 & \frac{-1}{3} & 1 \end{array}\right]$$

$$R_3 \longrightarrow \frac{-3}{17}R_3 \quad \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & \frac{1}{3} & 0 & \frac{-1}{3} & 0 \\ 0 & 0 & 1 & \frac{6}{17} & \frac{1}{17} & \frac{-3}{17} \end{array}\right]$$

$$\begin{array}{c} R_1 \longrightarrow R_1 - 2R_3 \\ R_2 \longrightarrow R_2 - \frac{1}{3}R_3 \end{array} \quad \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{5}{17} & \frac{-2}{17} & \frac{6}{17} \\ 0 & 1 & 0 & \frac{-2}{17} & \frac{-6}{17} & \frac{1}{17} \\ 0 & 0 & 1 & \frac{6}{17} & \frac{1}{17} & \frac{-3}{17} \end{array}\right]$$

So $E^{-1} = \begin{bmatrix} \frac{5}{17} & \frac{-2}{17} & \frac{6}{17} \\ \frac{-2}{17} & \frac{-6}{17} & \frac{1}{17} \\ \frac{6}{17} & \frac{1}{17} & \frac{-3}{17} \end{bmatrix}$.

Now the inverse of F:

$$\left[\begin{array}{ccc|ccc} 2 & -3 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ -1 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$ It is easy to see that $|F| = 1$.

$$\begin{array}{c} R_1 \longrightarrow \frac{1}{2}R_1 \\ R_3 \longrightarrow R_3 + R_1 \end{array} \left[\begin{array}{ccc|ccc} 1 & \frac{-3}{2} & \frac{-1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & \frac{-3}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 1 \end{array}\right]$$

$$\begin{array}{c} R_1 \longrightarrow R_1 + \frac{3}{2}R_2 \\ R_3 \longrightarrow R_3 + \frac{3}{2}R_2 \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{-1}{2} & \frac{1}{2} & \frac{3}{2} & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & \frac{3}{2} & 1 \end{array}\right]$$

$$\begin{array}{c} R_3 \longrightarrow 2R_3 \\ R_1 \longrightarrow R_1 + \frac{1}{2}R_3 \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 3 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 3 & 2 \end{array}\right]$$

and $F^{-1} = \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 0 \\ 1 & 3 & 2 \end{bmatrix}$

Checking: $E^{-1}E = \begin{bmatrix} \frac{5}{17} & \frac{-2}{17} & \frac{6}{17} \\ \frac{-2}{17} & \frac{-6}{17} & \frac{1}{17} \\ \frac{6}{17} & \frac{1}{17} & \frac{-3}{17} \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & -3 & -1 \\ 2 & -1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$

$$F^{-1}F = \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 0 \\ 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & -3 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$E^{-1}F^{-1} = \frac{1}{17} \begin{bmatrix} 5 & -2 & 6 \\ -2 & -6 & 1 \\ 6 & 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 0 \\ 1 & 3 & 2 \end{bmatrix} = \frac{1}{17} \begin{bmatrix} 11 & 31 & 17 \\ -1 & -9 & 0 \\ 3 & 10 & 0 \end{bmatrix}$$

$$F^{-1}E^{-1} = \frac{1}{17} \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 0 \\ 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} 5 & -2 & 6 \\ -2 & -6 & 1 \\ 6 & 1 & -3 \end{bmatrix} = \frac{1}{17} \begin{bmatrix} 5 & -19 & 6 \\ -2 & -6 & 1 \\ 11 & -18 & 3 \end{bmatrix}$$

Activity 5.5 *Calculate* $(EF)(F^{-1}E^{-1})$ *using the matrices obtained above. Find* $(FE)^{-1}$ *using Gauss-Jordan elimination and check that it is the same as* $E^{-1}F^{-1}$

Example 5.12 *Find* $\begin{bmatrix} 1 & 4 & 6 \\ 3 & 6 & 48 \\ -2 & -12 & 8 \end{bmatrix}^{-1}$

Solution

$$[B|I_3] = \left[\begin{array}{ccc|ccc} 1 & 4 & 6 & 1 & 0 & 0 \\ 3 & 6 & 48 & 0 & 1 & 0 \\ -2 & -12 & 8 & 0 & 0 & 1 \end{array}\right]$$

$$\begin{array}{c} R_2 \longrightarrow R_2 - 3R_1 \\ R_3 \longrightarrow R_3 + 2R_1 \end{array} \left[\begin{array}{ccc|ccc} 1 & 4 & 6 & 1 & 0 & 0 \\ 0 & -6 & 30 & -3 & 1 & 0 \\ 0 & -4 & 20 & 2 & 0 & 1 \end{array}\right]$$

At this stage we see that the determinant of the matrix is 0.

$$\begin{array}{c} R_2 \longrightarrow \frac{-1}{6}R_2 \\ R_3 \longrightarrow R_3 + 4R_2 \end{array} \left[\begin{array}{ccc|ccc} 1 & 4 & 6 & 1 & 0 & 0 \\ 0 & 1 & -5 & \frac{1}{2} & \frac{-1}{6} & 0 \\ 0 & 0 & 0 & 4 & \frac{-4}{6} & 1 \end{array}\right]$$

The last row is equivalent to the equations

$$0x_{11} + 0x_{21} + 0x_{31} = 4$$

$$\begin{aligned} 0x_{12} + 0x_{22} + 0x_{32} &= \frac{-4}{6} \\ 0x_{13} + 0x_{23} + 0x_{33} &= 1 \end{aligned}$$

These equations have no solution and so there is no inverse. In fact, we could have stopped earlier on when we found out that the determinant is 0 because if a matrix has determinant equal to 0 that matrix has no inverse.

Activity 5.6 *Find the inverse of* $\begin{bmatrix} 1 & -3 & 2 \\ -2 & 7 & -4 \\ 3 & -4 & 6 \end{bmatrix}$

5.3.5 Rules for inverses

1. If A and B are invertible matrices of the same size, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

2. If A is an invertible matrix, then

   (a) A^{-1} is invertible and $(A^{-1})^{-1} = A$.

   (b) A^n is invertible and $(A^n)^{-1} = (A^{-1})^n, \quad n = 0, 1, 2, \cdots$.

   (c) for any non-zero scalar k, the matrix kA is invertible and $(kA)^{-1} = \frac{1}{k}A^{-1}$.

3. If A is an invertible matrix, then A^T is also invertible and $(A^T)^{-1} = (A^{-1})^T$.

5.3.6 Adjoint and inverse matrices

Definition 5.11 *The* **adjoint** *of a matrix A, written $AdjA$, is the transposed matrix of cofactors of A. Thus, if A is a 3×3 matrix then* $AdjA = \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix}$ *and in general if A is an $n \times n$ matrix then* $AdjA = \begin{bmatrix} C_{11} & C_{21} & \dots & C_{n1} \\ C_{12} & C_{22} & \dots & C_{n2} \\ \vdots & \vdots & \vdots & \vdots \\ C_{1n} & C_{2n} & \dots & C_{nn} \end{bmatrix}$.

Remark 5.1 *As seen earlier, the matrix A is non-singular if $|A| \neq 0$. In this case the matrix A will be invertible. On the other hand the matrix A is said to be singular if $|A| = 0$ and in this case A is not invertible.*

The above definition provides us with an alternative way of finding the inverse of a given invertible matrix.

Definition 5.12 *If A is non-singular we define the inverse of A by* $A^{-1} = \dfrac{1}{|A|} adj A$.

Example 5.13 *Consider the matrix* $E = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -3 & -1 \\ 2 & -1 & -2 \end{bmatrix}$ *as discussed earlier. We want to use the adjoint method to find the inverse of this matrix.*

As obtained earlier $|E| = 17$. The matrix of cofactors is $\begin{bmatrix} 5 & -2 & 6 \\ -2 & -6 & 1 \\ 6 & 1 & -3 \end{bmatrix}$. Thus $AdjE = \begin{bmatrix} 5 & -2 & 6 \\ -2 & -6 & 1 \\ 6 & 1 & -3 \end{bmatrix}$.

Consequently $E^{-1} = \dfrac{1}{|E|} AdjE = \dfrac{1}{17} \begin{bmatrix} 5 & -2 & 6 \\ -2 & -6 & 1 \\ 6 & 1 & -3 \end{bmatrix} = \begin{bmatrix} \frac{5}{17} & \frac{-2}{17} & \frac{6}{17} \\ \frac{-2}{17} & \frac{-6}{17} & \frac{1}{17} \\ \frac{6}{17} & \frac{1}{17} & \frac{-3}{17} \end{bmatrix}$ as found earlier.

Activity 5.7 *1. Evaluate the determinants of the following matrices.* $A = \begin{bmatrix} 3 & 1 \\ 4 & -2 \end{bmatrix}$,

$B = \begin{bmatrix} 1 & 2 & 3 \\ -4 & 5 & 6 \\ 7 & -8 & 9 \end{bmatrix}$,

$C = \begin{bmatrix} 3 & 0 & 0 \\ 2 & -1 & 5 \\ 1 & 9 & -4 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 1 \\ 1 & -1 & 1 & 0 \\ -1 & 1 & 0 & 1 \end{bmatrix}$,

2. Find all values of λ for which $|A| = 0$, where $A = \begin{bmatrix} \lambda - 2 & 1 \\ -5 & \lambda + 4 \end{bmatrix}$,

$A = \begin{bmatrix} \lambda - 4 & 0 & 0 \\ 0 & \lambda & 2 \\ 0 & 3 & \lambda - 1 \end{bmatrix}$,

3. In each section of the question you should try to find the inverse of the given matrix by Gauss - Jordan elimination techniques. After reducing the matrix to echelon form or reduced echelon form write down the value of the determinant of the matrix. Check your arithmetic by expanding each determinant.

(a) $\begin{bmatrix} 1 & -2 & -5 \\ 2 & -3 & -8 \\ -1 & 1 & 2 \end{bmatrix}$, *(b)* $\begin{bmatrix} 1 & -3 & 2 \\ -2 & 7 & -4 \\ 3 & -4 & 6 \end{bmatrix}$, *(c)* $\begin{bmatrix} 1 & 0 & -2 & 3 \\ -1 & 2 & 6 & -6 \\ 0 & 4 & 2 & 3 \\ 2 & -6 & -14 & 12 \end{bmatrix}$,

(d) $\begin{bmatrix} 1 & -1 & -1 & 2 \\ -2 & 3 & 1 & 2 \\ 5 & -2 & -8 & 7 \\ 1 & -3 & 1 & 6 \end{bmatrix}$, *(e)* $\begin{bmatrix} -3 & -6 & 3 \\ -1 & -2 & 3 \\ 6 & 8 & -2 \end{bmatrix}$, *(f)* $\begin{bmatrix} 2 & 2 & 0 \\ -3 & -2 & 2 \\ 4 & 2 & -8 \end{bmatrix}$,

(g) $\begin{bmatrix} 1 & -2 & 5 & 1 \\ -1 & 3 & -2 & -3 \\ -1 & 1 & -8 & 1 \\ 2 & 2 & 7 & 6 \end{bmatrix}$, *(h)* $\begin{bmatrix} 5 & 2 & 3 & -1 \\ 2 & 2 & -1 & 1 \\ 1 & -4 & -5 & 0 \\ -6 & 2 & -4 & -4 \end{bmatrix}$, *(i)* $\begin{bmatrix} 3 & 2 & -2 \\ -8 & -4 & 6 \\ 2 & 2 & -2 \end{bmatrix}$,

(j) $\begin{bmatrix} -12 & 30 & -12 \\ -8 & 20 & -8 \\ -1 & 2 & -1 \end{bmatrix}$, *(k)* $\begin{bmatrix} 1 & -1 & 3 & 5 \\ -1 & 0 & -1 & -2 \\ 1 & 1 & 0 & 14 \\ -1 & -1 & 5 & 0 \end{bmatrix}$, *(l)* $\begin{bmatrix} 1 & -1 & 1 & -1 \\ -1 & 0 & 0 & -1 \\ -1 & 1 & -1 & 1 \\ 0 & -1 & -1 & 0 \end{bmatrix}$,

4. *For each* 3×3 *matrix in Question* 3 *above, find the inverse using the adjoint method.*

Unit 6
Systems of simultaneous linear equations

6.1 What this Unit is all About

In this Unit, we discuss Gauss elimination and Gauss - Jordan elimination techniques of solving systems of simultaneous linear equations. We also discuss the method of solution by use of the inverse matrix as well as by use of Cramer's rule.

By the end of this unit, you should be able to:

1. Define echelon form.
2. Define rank of a matrix.
3.Solve systems of simultaneous linear equations by the inverse matrix method.
4. Solve systems of simultaneous linear equations by Gauss elimination.
5. Solve systems of simultaneous linear equations by Gauss - Jordan elimination.
6. Distinguish between Gauss elimination and Gauss - Jordan elimination.
7. Solve systems of simultaneous linear equations by Cramer's rule.

6.2 Linear equations

Definition 6.1 *A* **linear equation** *in the n variables $x_1, x_2, \cdots, x_n$ is an equation that can be expressed in the form $a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$ where $a_1, a_2, \cdots, a_n$ and b are real constants and $x_1, x_2, \cdots, x_n$ are the variables or* **unknowns**.

Example 6.1 *The following equations are linear:*
- $x_1 + 3y = 7$
- $y = \frac{1}{2}x + 3z + 1$
- $x_1 - 2x_2 - 3x_3 + x_4 = 7$

Note that in these examples there are no products of variables, no roots of variables, no powers greater than 1, no trigonometric terms, no logs and no exponentials.

In the same spirit we note that the equations:
- $x + 3\sqrt{y} = 5$
- $3x + 2y - z + xz = 4$
- $y = \sin x$

are not linear. (Why?)

Our objective is to solve systems of simultaneous linear equations. The method of solution we are going to use this time around is the use of 'Elementary row operations'.

6.3 Notation and terminology

We adopt the **matrix notation** and in the foregoing we outline a summary of definitions of the various items involved.
• A **matrix** is an array of numbers arranged into rows and columns. Rows are horizontal and columns are vertical. Rows are numbered starting from the top and counting downwards. Columns are numbered starting from the left and moving to the right.
• If a matrix has s rows and t columns, its **dimension** is $s \times t$ read as "s by t".
• A matrix with only one row (i.e. dimension $1 \times t$) is called a **row vector** while a matrix with only one column is called a **column vector**. (dimension $s \times 1$).
• The $(i, j)^{th}$ position in a matrix is the intersection of the ith row and the jth column and the number appearing in that position is called the $(i, j)^{th}$ **entry** of the matrix. The $(i, j)^{th}$ entry of a matrix is denoted by the symbol a_{ij} if the matrix is labelled by A. The first subscript indicates the row it comes from while the second indicates the column.
• The general 2×3 matrix is denoted by $A = [a_{ij}]_{2\times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$ The 2nd row is $\begin{bmatrix} a_{21} & a_{22} & a_{23} \end{bmatrix}$. Note that 1st subscript is always 2. The 2nd column is $\begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix}$. Second subscript always 2.

Consider the following system of s simultaneous linear equations in t variables (unknowns) $\{x_1, x_2, \cdots, x_t\}$:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1t}x_t &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2t}x_t &= b_2 \\ \vdots \quad \vdots \qquad \vdots & \\ a_{s1}x_1 + a_{s2}x_2 + \cdots + a_{st}x_t &= b_s \end{aligned}$$

• The coefficient matrix of the system is $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1t} \\ a_{21} & a_{22} & \dots & a_{2t} \\ \vdots & \vdots & \dots & \vdots \\ a_{s1} & a_{s2} & \dots & a_{st} \end{bmatrix}$. Note that the dimension of this matrix is $s \times t$. s is the number of equations in the system and t is the number of variables.

It must be noted that a_{ij}, b_i are constants and are considered "known" while the x_j's are unknowns and/or variables and the problem is always to find the x_j's in terms of the a_{ij}, b_i.

The vector of unknowns and the vector of constants respectively are

$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_t \end{bmatrix}$, $\underline{c} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_s \end{bmatrix}$.

The system of equations is then written as a matrix equation

$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1t} \\ a_{21} & a_{22} & \dots & a_{2t} \\ \vdots & \vdots & \dots & \vdots \\ a_{s1} & a_{s2} & \dots & a_{st} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_t \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_s \end{bmatrix}$ which is written in the form $A\underline{x} = \underline{b}$ or $AX = B$ with $\underline{x} = X$ being a $t \times 1$ column vector and $\underline{b} = B$ being an $s \times 1$ column vector.

• The **augmented matrix** is obtained by augmenting the coefficient matrix with the vector of constants: $\left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1t} & b_1 \\ a_{21} & a_{22} & \dots & a_{2t} & b_2 \\ \vdots & \vdots & \dots & \vdots & \vdots \\ a_{s1} & a_{s2} & \dots & a_{st} & b_s \end{array}\right]$ denoted briefly by $[A|b]$.

6.4 Elementary row operations

1. Multiplication of a row by a nonzero scalar.
   Notation: $R_i \longrightarrow kR_i$, $k \in \mathbb{R} - \{0\}$. This operation when performed on an augmented matrix of a system of equations changes that matrix to the augmented matrix of a system obtained from the original system by multiplying through the ith equation by k. Since the later operation has no effect on the solutions to the system, every solution of the original system is a solution of the new system and vice versa.

2. Replacement of a given row by the sum of that row and a constant (scalar) multiple of another (distinct) row.
   Notation: $R_i \longrightarrow R_i + kR_j$, $i \neq j$, $k \in \mathbb{R}$. If this operation is applied

to the augmented matrix of a given system of equations, the augmented matrix of the system in which k times the jth equation is added to the ith equation, is obtained.

3. Interchange of two rows.
   Notation: $R_i \longleftrightarrow R_j, \quad i \neq j$. When applied to the augmented matrix of a system of simultaneous linear equations this operation has the effect of interchanging the positions of the two equations.

Definition 6.2 • *Two matrices are equivalent if one can be obtained from the other by a sequence of elementary row operations.*
• *Two systems of equations are equivalent if every solution to the first is a solution to the second and vice-versa. It is a fact that two systems of equations are equivalent if and only if their augmented matrices are equivalent.*

Definition 6.3 *(Echelon form)*
A matrix is in echelon form if

1. *each zero row occurs below all non-zero rows. A row is a zero row if all its entries are* 0*. If one or more entries are different from* 0*, that row is a non-zero row.*

2. *the first non-zero entry in a non-zero row (called the leading entry) is* 1*.*

3. *the leading entry in a non-zero row is to the left of the leading entry in the next non-zero row and to the right of the leading entry in the preceding non-zero row. Thus all the entries in the column below a leading entry are zero.*

Definition 6.4 *The* <u>*rank*</u> *of a matrix is the number of non-zero rows in an equivalent echelon form.*

Definition 6.5 *A system of equations is* <u>*consistent*</u> *if it has at least one solution and is* <u>*inconsistent*</u> *if it has no solution.*

6.5 Gaussian Elimination

This is a systematic procedure for solving systems of simultaneous linear equations. The procedure is based on the idea of reducing the augmented matrix of a system to another augmented matrix that is simple enough that the solution of the system can be found by inspection.

Example 6.2 *Solve the system*

$$\begin{aligned} 3x_1 - x_2 + x_3 &= -1 \\ 9x_1 - 2x_2 + x_3 &= -9 \\ 3x_1 + x_2 - 2x_3 &= -9 \end{aligned}$$

Solution

The augmented matrix of the system is $\left[\begin{array}{ccc|c} 3 & -1 & 1 & -1 \\ 9 & -2 & 1 & -9 \\ 3 & 1 & -2 & -9 \end{array}\right]$.

Apply the row operations:

$$\begin{array}{c} R_1 \longrightarrow \frac{1}{3}R_1 \\ R_2 \longrightarrow R_2 - 9R_1 \\ R_3 \longrightarrow R_3 - 3R_1 \end{array} \quad \left[\begin{array}{ccc|c} 1 & -\frac{1}{3} & \frac{1}{3} & -\frac{1}{3} \\ 0 & 1 & -2 & -6 \\ 0 & 2 & -3 & -8 \end{array}\right].$$

$$R_3 \longrightarrow R_3 - 2R_2 \quad \left[\begin{array}{ccc|c} 1 & \frac{-1}{3} & \frac{1}{3} & \frac{-1}{3} \\ 0 & 1 & -2 & -6 \\ 0 & 0 & 1 & 4 \end{array}\right].$$ The resulting matrix is in echelon form.

<u>Back substitution</u>:
Third row gives $x_3 = 4$. Substitution using second row (equation) gives $x_2 = 2x_3 - 6 = 2(4) - 6 = 2$. Substitution using first (equation) row gives $x_1 = \frac{1}{3}x_2 - \frac{1}{3}x_3 - \frac{1}{3} = \frac{1}{3}(2) - \frac{1}{3}(4) - \frac{1}{3} = -1$.

Checking

$$\begin{aligned} 3(-1) - 2 + 4 &= -1 \\ 9(-1) - 2(2) + 4 &= -9 \\ 3(-1) + 2 - 2(4) &= -9 \end{aligned}$$

which agrees with the equations.

Example 6.3 *Solve*

$$\begin{aligned} 5x_2 &= 4 \\ 2x_1 + 4x_2 &= 3 \\ x_1 + 3x_2 &= 6 \end{aligned}$$

Solution

The augmented matrix is $\left[\begin{array}{cc|c} 0 & 5 & 4 \\ 2 & 4 & 3 \\ 1 & 3 & 6 \end{array}\right]$.Apply row operations.

$$\begin{array}{c} R_1 \longleftrightarrow R_3 \\ R_2 \longrightarrow R_2 - 2R_1 \end{array} \quad \left[\begin{array}{cc|c} 1 & 3 & 6 \\ 0 & -2 & -9 \\ 0 & 5 & 4 \end{array}\right].$$

$$\begin{array}{c} R_2 \longrightarrow \frac{-1}{2}R_2 \\ R_3 \longrightarrow R_3 - 5R_2 \end{array} \quad \left[\begin{array}{cc|c} 1 & 3 & 6 \\ 0 & 1 & \frac{9}{2} \\ 0 & 0 & -\frac{37}{2} \end{array}\right].$$

The last row corresponds to the equation $0x_1 + 0x_2 = -\frac{37}{2}$ which clearly has no solution. The original system is <u>INCONSISTENT</u>.

Example 6.4 *Solve the system*

$$\begin{aligned} 2x_1 + x_2 + 2x_3 + x_4 &= 5 \\ 4x_1 + 3x_2 + 7x_3 + 3x_4 &= 8 \\ -8x_1 - x_2 - x_3 + 3x_4 &= 4 \\ 6x_1 + x_2 + 2x_3 - x_4 &= 1 \end{aligned}$$

Solution

The augmented matrix of the system is $\left[\begin{array}{cccc|c} 2 & 1 & 2 & 1 & 5 \\ 4 & 3 & 7 & 3 & 8 \\ -8 & -1 & -1 & 3 & 4 \\ 6 & 1 & 2 & -1 & 1 \end{array}\right]$. The first group of row operations is

Group 1 $\begin{array}{c} R_1 \longrightarrow \frac{1}{2}R_1 \\ R_2 \longrightarrow R_2 - 4R_1 \\ R_3 \longrightarrow R_3 + 8R_1 \\ R_4 \longrightarrow R_4 - 6R_1 \end{array} \quad \left[\begin{array}{cccc|c} 1 & \frac{1}{2} & 1 & \frac{1}{2} & \frac{5}{2} \\ 0 & 1 & 3 & 1 & -2 \\ 0 & 3 & 7 & 7 & 24 \\ 0 & -2 & -4 & -4 & -14 \end{array}\right]$

Group 2

$\begin{array}{c} R_3 \longrightarrow R_3 - 3R_2 \\ R_4 \longrightarrow R_4 + 2R_2 \end{array} \quad \left[\begin{array}{cccc|c} 1 & \frac{1}{2} & 1 & \frac{1}{2} & \frac{5}{2} \\ 0 & 1 & 3 & 1 & -2 \\ 0 & 0 & -2 & 4 & 30 \\ 0 & 0 & 2 & -2 & -18 \end{array}\right]$

Group 3

$\begin{array}{c} R_3 \longrightarrow \frac{-1}{2}R_3 \\ R_4 \longrightarrow R_4 - 2R_3 \end{array} \quad \left[\begin{array}{cccc|c} 1 & \frac{1}{2} & 1 & \frac{1}{2} & \frac{5}{2} \\ 0 & 1 & 3 & 1 & -2 \\ 0 & 0 & 1 & -2 & -15 \\ 0 & 0 & 0 & 2 & 12 \end{array}\right]$

Group 4

$R_4 \longrightarrow \frac{1}{2}R_4 \quad \left[\begin{array}{cccc|c} 1 & \frac{1}{2} & 1 & \frac{1}{2} & \frac{5}{2} \\ 0 & 1 & 3 & 1 & -2 \\ 0 & 0 & 1 & -2 & -15 \\ 0 & 0 & 0 & 1 & 6 \end{array}\right]$

Back substitution gives $x_4 = 6, \quad x_3 = -3, \quad x_2 = 1, \quad x_1 = 2.$

6.6 Gauss - Jordan Elimination

Example 6.5 *Solve*

$$x - y + z - w = 2$$

$$\begin{aligned} x - z + 2w &= 0 \\ -x + 2y - 2z + 7w &= -7 \\ 2x - y - z &= 3 \end{aligned}$$

Solution

The augmented matrix of the system is $\left[\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 2 \\ 1 & 0 & -1 & 2 & 0 \\ -1 & 2 & -2 & 7 & -7 \\ 2 & -1 & -1 & 0 & 3 \end{array}\right]$.

We reduce each column to a unit vector as follows:

$$\begin{array}{c} R_2 \longrightarrow R_2 - R_1 \\ R_3 \longrightarrow R_3 + R_1 \\ R_4 \longrightarrow R_4 - 2R_1 \end{array} \left[\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 2 \\ 0 & 1 & -2 & 3 & -2 \\ 0 & 1 & -1 & 6 & -5 \\ 0 & 1 & -3 & 2 & -1 \end{array}\right]$$

$$\begin{array}{c} R_1 \longrightarrow R_1 + R_2 \\ R_3 \longrightarrow R_3 - R_2 \\ R_4 \longrightarrow R_4 - R_2 \end{array} \left[\begin{array}{cccc|c} 1 & 0 & -1 & 2 & 0 \\ 0 & 1 & -2 & 3 & -2 \\ 0 & 0 & 1 & 3 & -3 \\ 0 & 0 & -1 & -1 & 1 \end{array}\right]$$

$$\begin{array}{c} R_1 \longrightarrow R_1 + R_3 \\ R_2 \longrightarrow R_2 + 2R_3 \\ R_4 \longrightarrow R_4 + R_3 \end{array} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 5 & -3 \\ 0 & 1 & 0 & 9 & -8 \\ 0 & 0 & 1 & 3 & -3 \\ 0 & 0 & 0 & 2 & -2 \end{array}\right]$$

$$\begin{array}{c} R_4 \longrightarrow \frac{1}{2}R_4 \\ R_1 \longrightarrow R_1 - 5R_4 \\ R_2 \longrightarrow R_2 - 9R_4 \\ R_3 \longrightarrow R_3 - 3R_4 \end{array} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{array}\right].$$

This gives $x = 2,\ y = 1,\ z = 0,\ w = -1$.

Example 6.6

$$\begin{aligned} x + z + 2w &= 0 \\ 3x + y + z + 7w &= 0 \\ 2x - y + 4z + 3w &= 0 \\ x + 2y - 2z + 8w &= 0 \end{aligned}$$

Solution

Since the last column of the augmented matrix remains 0 throughout, the solution is obtained by reducing the coefficient matrix.

The coefficient matrix is $\begin{bmatrix} 1 & 0 & 1 & 2 \\ 3 & 1 & 1 & 7 \\ 2 & -1 & 4 & 3 \\ 1 & 2 & -2 & 8 \end{bmatrix}$

$$\begin{array}{c} R_2 \longrightarrow R_2 - 3R_1 \\ R_3 \longrightarrow R_3 - 2R_1 \\ R_4 \longrightarrow R_4 - R_1 \end{array} \quad \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & -2 & 1 \\ 0 & -1 & 2 & -1 \\ 0 & 2 & -3 & 6 \end{bmatrix}$$

$$\begin{array}{c} R_3 \longrightarrow R_3 + R_2 \\ R_4 \longrightarrow R_4 - 2R_2 \end{array} \quad \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 4 \end{bmatrix}$$

$$\begin{array}{c} R_3 \longleftrightarrow R_4 \\ R_1 \longrightarrow R_1 - R_3 \\ R_2 \longrightarrow R_2 + 2R_3 \end{array} \quad \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 9 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Observe that the rank of the coefficient matrix is 3. Thus, there is 1 degree of freedom, i.e. there is one arbitrary variable.

The general solution is $x = 2w, \quad y = -9w, \quad z = -4w$ or $\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 2 \\ -9 \\ -4 \\ 1 \end{bmatrix} w$.

w is arbitrary. Particular solution for $w = 1$ is $\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 2 \\ -9 \\ -4 \\ 1 \end{bmatrix}$. If $w = t$, say, then the solution is $\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 2 \\ -9 \\ -4 \\ 1 \end{bmatrix} t$.

Activity 6.1 *Solve*

$$\begin{aligned} a - 2b + 2c &= 0 \\ 4a - b + 5c + 4d &= 0 \\ 2a + 3b + c + 4d &= 0 \\ 7b - 3c + 4d &= 0 \end{aligned}$$

by Gauss Elimination.

Activity 6.2 *1. (a) Write down the augmented matrix corresponding to the following system of equations*

$$\begin{aligned} x_1 + 2x_2 - 5x_3 + x_4 &= 3 \\ 2x_1 - 3x_2 + 4x_3 - 2x_4 &= 2 \\ 4x_1 + x_2 - 6x_3 + 3x_4 &= 11 \\ 5x_1 + 9x_2 - 20x_3 + x_4 &= 10 \end{aligned}$$

(b) *Solve the system in (a) by performing elementary row operations on the augumented matrix. Check your solution by substituting into the original equations.*
Use the following listed groups of operations:
*Group*1*:*

$$\begin{aligned} R_2 &\rightarrow R_2 - 2R_1 \\ R_3 &\rightarrow R_3 - 4R_1 \\ R_4 &\rightarrow R_4 - 5R_1 \end{aligned}$$

*Group*2*:*

$$\begin{aligned} R_2 &\rightarrow -\frac{1}{7}R_2 \\ R_3 &\rightarrow R_3 + 7R_2 \\ R_4 &\rightarrow R_4 + R_2 \end{aligned}$$

*Group*3*:*

$$\begin{aligned} R_3 &\rightarrow \frac{1}{3}R_3 \\ R_4 &\rightarrow \frac{1}{3}R_4 \\ R_3 &\leftrightarrow R_1 \end{aligned}$$

2. *Each of the following matrices is the augmented matrix of a system of equations. Write down the system that the matrix represents and use Gauss Elimination on the augmented matrix to obtain the solution of the system.*

(a)

$$\left[\begin{array}{ccc|c} 5 & 0 & 2 & -1 \\ -2 & 3 & 8 & 3 \\ 0 & 6 & 5 & -8 \\ 0 & 0 & 4 & 5 \end{array}\right] \tag{6.1}$$

(b)

$$\left[\begin{array}{cccc|c} 1 & 4 & 8 & 2 & 2 \\ 2 & 5 & 3 & 8 & 9 \end{array}\right] \tag{6.2}$$

3. *Use Gaussian Elimination to solve the following systems of equations. (***NB***: Find the augmented matrix, row reduce to echelon form and back substitute.)*

(a)

$$\begin{aligned} -x_1 - x_2 + 2x_3 &= - \quad 5 \\ -3x_1 - x_2 + 7x_3 &= \quad -22 \\ x_1 - 3x_2 - x_3 &= \quad 10 \end{aligned}$$

(b)

$$\begin{aligned} -3x_1 - 6x_2 + 9x_3 &= 0 \\ x_1 + 4x_2 + x_3 &= 6 \\ 2x_1 + 8x_2 + 3x_3 &= 13 \end{aligned}$$

(c)

$$\begin{aligned} 2x_1 - x_2 + 2x_3 + 3x_4 &= 3 \\ -4x_1 + 6x_2 - 3x_3 - 6x_4 &= 2 \\ 6x_1 - x_2 + 8x_3 + 5x_4 &= 9 \\ 4x_1 - 2x_2 + 6x_3 + 12x_4 &= 12 \end{aligned}$$

4. *Solve, using Gauss elimination on the augmented matrix,*

$$\begin{aligned} x_1 + 2x_2 - x_4 &= -1 \\ x_2 + 2x_4 &= 8 \\ 3x_4 - 3x_1 - x_3 &= 10 \\ 7x_1 + 4x_3 &= 6 \end{aligned}$$

5. *Solve the following using Gauss - Jordan elimination.*

(a)

$$\begin{aligned} x - 2y + z - w &= 2 \\ -x + 4y - 2z + 3w &= -1 \\ 2x + z + 3w &= 9 \\ -2x + 6y + 5w &= 2 \end{aligned}$$

(b)

$$\begin{aligned} 2x - y + z &= 8 \\ x + y + 2z &= 4 \\ 3x - 3y - z &= 10 \end{aligned}$$

(c)

$$\begin{aligned} u - v + 2w + 2x + y &= 3 \\ 2u + v + 5w + 2x + 2y &= 6 \\ -u + 4v - 6x + y &= -2 \\ -2u - 4v - 4w - x + y &= -3 \\ 2u + 4v + 4w + 7x - y &= 9 \end{aligned}$$

6. *Solve the following systems of homogeneous equations, alternately using the Gauss - Jordan elimination method and the Gauss elimination method.*

   (a)

$$\begin{aligned} a + 2b - c + 3d &= 0 \\ 4a + 3b - c + 10d &= 0 \\ 2a - b + c + 4d &= 0 \\ 3a + b + 7d &= 0 \end{aligned}$$

   (b)

$$\begin{aligned} x + 3y + z + 2w &= 0 \\ x + 2y + 4z &= 0 \\ x + 5y - 5z + 6w &= 0 \\ 2x + 3y + 11z - 4w &= 0 \end{aligned}$$

   (c)

$$\begin{aligned} 2a + b - c - 2d &= 0 \\ 3a - 2b - 2c + d &= 0 \\ 5a + 6b - 2c - 9d &= 0 \end{aligned}$$

   (d)

$$\begin{aligned} 3x - 2y + z &= 0 \\ x + 4y + 2z &= 0 \\ 7x + 4z &= 0 \end{aligned}$$

6.7 Cramer's Rule

Determinants can be applied to solving a system of n linear equations in n unknowns. This method is very useful as it enables us to solve for one unknown without having to solve for the others.

Consider a system of two linear equations with two unknowns:

$$\begin{aligned} ax + by &= h \\ cx + dy &= k. \end{aligned}$$

Multiplying the first equation by d and the second equation by b gives:

$$\begin{aligned} adx + bdy &= dh \\ cbx + bdy &= bk. \end{aligned}$$

This allows us to eliminate the variable y by subtracting the second equation from the first equation or vice versa. In the process we get

$$x = \frac{hd - bk}{ad - cb}.$$

We observe the following equivalence:

$$x = \frac{hd - bk}{ad - cb} = \frac{\begin{vmatrix} h & b \\ k & d \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}},$$

where we require $\begin{vmatrix} a & b \\ c & d \end{vmatrix} \neq 0$. It can similarly be shown that

$$y = \frac{ak - ch}{ad - cb} = \frac{\begin{vmatrix} a & h \\ c & k \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}}.$$

Let us adopt the following notation: $\Delta = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$, $\Delta_x = \begin{vmatrix} h & b \\ k & d \end{vmatrix}$, $\Delta_y = \begin{vmatrix} a & h \\ c & k \end{vmatrix}$. We are now in a position to state Cramer's rule for solving two linear equations in two variables.

Cramer's rule for solving a 2×2 linear system:

The unique solution to the linear system $a_1x + b_1y = k_1, \quad a_2x + b_2y = k_2$ is

$$x = \frac{\Delta_x}{\Delta} = \frac{\begin{vmatrix} k_1 & b_1 \\ k_2 & b_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}} \text{ and } y = \frac{\Delta_y}{\Delta} = \frac{\begin{vmatrix} a_1 & k_1 \\ a_2 & k_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

provided that $\Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \neq 0$ and $k_1,\ k_2 \neq 0$.

If $k_1,\ k_2 = 0$ we have a homogeneous system and Cramer's rule cannot be applied as the determinants become 0. If $\Delta = 0$ the equations in the system are either inconsistent or dependent.

Example 6.7 *Solve the following system*

$$\begin{aligned} 2x + y + 5 &= 0 \\ 3y + x &= 6 \end{aligned}$$

We begin by rewriting the system in an appropriate form:

$$\begin{aligned} 2x + y &= -5 \\ x + 3y &= 6 \end{aligned}$$

The determinant of the coefficient matrix is $\Delta = \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} = 6 - 1 = 5$. *Since* $\Delta \neq 0$, *a unique solution exists. Thus we have* $x = \dfrac{\begin{vmatrix} -5 & 1 \\ 6 & 3 \end{vmatrix}}{5} = \dfrac{-15-6}{5} = \dfrac{-21}{5}$ *and* $y = \dfrac{\begin{vmatrix} 2 & -5 \\ 1 & 6 \end{vmatrix}}{5} = \dfrac{12-(-5)}{5} = \dfrac{17}{5}$.

One can always check the solution by substituting back into the original equations.

We are now in a position to generalize the rule to a system of n linear equations in n unknowns.

Cramer's Rule

Let a system of n linear equations in n unknowns be given by

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= k_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= k_2 \\ \vdots \quad \vdots \qquad \vdots & \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= k_n \end{aligned}$$

If the determinant of the coefficients $\Delta = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \dots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix}$ is not equal to zero, the system has a unique solution. The solution is

$$x_1 = \frac{\Delta_{x_1}}{\Delta}, \quad x_2 = \frac{\Delta_{x_2}}{\Delta}, \cdots, x_n = \frac{\Delta_{x_n}}{\Delta}$$

, where Δ_{x_m}, the numerator of x_m, is the determinant obtained by replacing the mth column of Δ by the column of the constants $k_1, \cdots, k_n$.

Example 6.8 *Solve the system*

$$\begin{aligned} x + 2y - z &= -3 \\ 3x + y + z &= 4 \\ x - y + 2z &= 6 \end{aligned}$$

$\Delta = \begin{vmatrix} 1 & 2 & -1 \\ 3 & 1 & 1 \\ 1 & -1 & 2 \end{vmatrix} = -3$, $\Delta_x = \begin{vmatrix} -3 & 2 & -1 \\ 4 & 1 & 1 \\ 6 & -1 & 2 \end{vmatrix} = -3$, $\Delta_y = \begin{vmatrix} 1 & -3 & -1 \\ 3 & 4 & 1 \\ 1 & 6 & 2 \end{vmatrix} =$ 3 *and* $\Delta_z = \begin{vmatrix} 1 & 2 & -3 \\ 3 & 1 & 4 \\ 1 & -1 & 6 \end{vmatrix} = -6$. *Thus,*

$$x = \frac{\Delta_x}{\Delta} = \frac{-3}{-3} = 1, \quad y = \frac{\Delta_y}{\Delta} = \frac{3}{-3} = -1, \quad z = \frac{\Delta_z}{\Delta} = \frac{-6}{-3} = 2$$

6.8 The inverse matrix method

Consider, again, the system of equations in matrix notation

$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1t} \\ a_{21} & a_{22} & \dots & a_{2t} \\ \vdots & \vdots & \dots & \vdots \\ a_{s1} & a_{s2} & \dots & a_{st} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_t \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_s \end{bmatrix}$ which is written in the form $A\underline{x} = \underline{b}$ or $AX = B$ with $\underline{x} = X$ being a $t \times 1$ column vector and $\underline{b} = B$ being an $s \times 1$ column vector.

Solving this system for X can be achieved easily by multiplying both sides of $AX = B$ by the inverse of the matrix A, A^{-1}, i.e. $A^{-1}AX = A^{-1}B$. This gives $IX = A^{-1}B \Longrightarrow \quad X = A^{-1}B$.

Example 6.9 *Use the inverse matrix method to solve the system:*

$$\begin{aligned} x + 2z &= 1 \\ -3y - z &= 2 \\ 2x - y - 2z &= -1 \end{aligned}$$

solution

$\begin{bmatrix} 1 & 0 & 2 \\ 0 & -3-1 & \\ 2 & -1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$

$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -3 & -1 \\ 2 & -1 & -2 \end{bmatrix}$ and $A^{-1} = \frac{1}{17} \begin{bmatrix} 5 & -2 & 6 \\ -2 & -6 & 1 \\ 6 & 1 & -3 \end{bmatrix}$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{17} \begin{bmatrix} 5 & -2 & 6 \\ -2 & -6 & 1 \\ 6 & 1 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -\frac{5}{17} \\ -\frac{15}{17} \\ \frac{11}{17} \end{bmatrix}$

and $x = -\frac{5}{17}$, $y = -\frac{15}{17}$, $z = \frac{11}{17}$

Activity 6.3 *1. Solve the following systems of equations (i) by the inverse matrix method and (ii) by Cramer's rule:*

(a)

$$\begin{aligned} 2x + y &= 7 \\ 3x - 2y &= -7 \end{aligned}$$

(b)

$$\begin{aligned} 3x &= 3 - y \\ 2x &= 13 + 3y \end{aligned}$$

(c)

$$\begin{aligned} 2,5x + 3,8y &= 9,3 \\ 0,5x + 0,76y &= -2,44 \end{aligned}$$

(d)

$$\begin{aligned} x + 3y - z &= 7 \\ 5x - 7y + z &= 3 \\ 2x - y - 2z &= 0 \end{aligned}$$

(e)

$$\begin{aligned} 5x + 3y - 2z &= 5 \\ 3x - 4y + 3z &= 13 \\ x + 6y - 4z &= -8 \end{aligned}$$

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